

COMPUTER VISION ALGORITHMS FOR MICROSCOPY IMAGE SEGMENTATION

by

Khaled Alrfou

A Dissertation Submitted in
Partial Fulfillment of the
Requirements for the Degree

Doctor of Philosophy
in Engineering

at

The University of Wisconsin-Milwaukee

August 2024

ABSTRACT

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Khaled Alrfou

The University of Wisconsin-Milwaukee, 2024
Under the Supervision of Dr.Tian Zhao

In the fields of medical and materials sciences, accurate and robust segmentation of microscopy images is of great importance. The manual analysis methods used in the past face limitations, from subjectivity introduced by user judgments to inability to scale for processing large image datasets. Automated segmentation with traditional computer vision methods, such as image thresholding and morphology operations, offers speed and repeatability but is challenging to implement and not robust to changes in imaging or sample conditions. Integrated computer vision techniques have emerged as promising solutions to address these challenges where Convolutional Neural Network (CNN) and Vision Transformer (ViT) are two widely used techniques. Deep learning algorithms have demonstrated significant potential in improving microscopy image analysis in the areas of medical and materials sciences. They have been employed in medical applications for disease diagnosis, medical imaging analysis, and drug discovery by analyzing images such as X-rays, CT scans, and MRI scans. Materials science has used these algorithms for material characterization, defect detection, and quality control, analyzing materials such as metals, ceramics, and polymers.

Segmentation is a critical task in microscopy image analysis, as it involves identifying and separating different objects or regions of interest within an image. Segmentation can be used for tasks such as cell counting, measuring cell size and shape, and identifying specific structures or features within a tissue or organ. Deep learning models have shown great potential for improving the accuracy and efficiency of segmentation tasks in microscopy image

analysis. However, training deep learning models for segmentation can be challenging, as it requires a large amount of labeled data and can be computationally intensive. Additionally, microscopy images differ significantly from natural images due to variable signal-to-noise ratios, rich information content, and complex structures. Acquiring microscopy images can be a difficult and expensive process, and they often require expert knowledge to label and annotate. To overcome these challenges, transfer learning can be used, where models trained on large-scale datasets can be fine-tuned for specific microscopy analysis tasks when confronted with limited training data. Nevertheless, transfer learning based on natural images tend to yield sub-optimal results when applied to microscopy image segmentation due to the mismatch of domain information in microscopy images. Also, popular segmentation networks use CNN, which often struggles to capture long-range dependencies. This deficiency was solved in the Transformer-based segmentation models though they are limited in capturing low-level features.

To address these challenges, this thesis developed several deep learning models for microscopy image analysis, which use transfer learning based on large sets of microscopy images to provide accurate, robust, and generalizable approaches for microscopy image segmentation. Such advancements have the potential to improve the analysis of medical and material science images, impacting disease diagnosis, material characterization, and various critical research areas. One of the key steps is to pre-train CNN and Transformer models on large in-domain datasets to learn high-level microscopy features that are not typically present in natural images. Experiments show that such models outperform those pre-trained on natural image data like ImageNet. Practical and effective strategies emerge when hybrid approaches are employed, combining the strengths of CNNs and Transformers. These approaches allow for the exploitation of the unique advantages inherent in each architectural design. Through the fusion of domain expertise, the power of pre-training, and the combining of diverse architectural approaches offer a path toward enhanced precision and efficiency in microscopy image segmentation.

In this thesis, we make the following contributions:

1. We surveyed the areas of Computer Vision (CV) research in microscopy image analysis covering topics such as image classification and image segmentation.
2. We designed a robust CV pipeline for the automation of area fraction measurement in ductile iron microstructures with significant improvement in accuracy and efficiency.
3. We developed a new U-shaped hybrid segmentation network called CS-UNet that combines the strengths of CNNs and Transformers for image segmentation. Experimental results show that CS-UNet outperforms the state-of-the-art CNN-based, Transformer-based, and hybrid models. We also collected a set of microscopy images, MicroLite, for the purpose of pre-training segmentation encoders. Models pre-trained on MicroLite have superior performance than models pre-trained on natural images.
4. While CS-UNet has superior segmentation accuracy than the state-of-the-art models, it is computational intensive due to the combination of both CNN and Transformer encoders. To improve computation efficiency, we developed a lightweight segmentation network, GCtx-UNet. This network can capture both global and local image features with accuracy comparable to the state-of-the-art approaches. GCtx-UNet utilizes vision Transformers that leverage global context self-attention modules combined with local self-attention to model long and short-range spatial dependencies. GCtx-UNet is more efficient than the state-of-the-art models in that it has less number of model parameters, lower computation workload, higher inference speed, and smaller model size. The efficiency and accuracy of GCtx-UNet make it a practical choice for image segmentation applications.

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Dr. Tian Zhao, for guiding me throughout my program and helping me overcome obstacles. I also want to acknowledge my friends, whose support and insights were invaluable to my study. Special thanks to my family for their unwavering support and encouragement.

Chapter 1

Introduction

Microscopy images have a wide range of applications in various fields, including biology, medicine, materials science, and nanotechnology. These images can be used to study the structure and function of biological tissues, cells, organs, and molecules, as well as to diagnose diseases and develop new therapies. In materials science and nanotechnology, microscopy images are used to study the properties and behavior of materials and devices at the nanoscale, quantifying defects in images of irradiated metal alloys, characterizing micro and nanoparticles, finding individual atoms in STEM images, and tracking detected objects from in situ EM video, as well as to develop new materials and technologies [11, 12].

Traditional image processing methods rely on hand-crafted features and mathematical models to extract information from images. These methods have been successful in many applications, but they can be limited by the complexity of the data and the difficulty of designing effective features. In recent years, Deep Learning (DL) has emerged as a powerful alternative to traditional image processing methods. DL models are trained on large datasets using end-to-end learning algorithms that automatically learn features from the data. DL can be used to improve image analysis and interpretation in microscopy by providing solutions for image processing and analysis, such as image reconstruction and super-resolution, classification and generation, denoising, segmentation, cell tracking, and feature selections. Additionally, DL can help increase the clinical and biological relevance of multimodal non-linear microscopy through DL-based image and video understanding. In microscopy, DL has the potential to revolutionize the field by enabling new types of analysis and discovery [13, 14, 12].

The most commonly used DL algorithms in microscopy include Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Generative Adversarial Network

(GAN). CNNs are particularly well-suited for image classification and segmentation tasks, while RNNs are often used for time-series analysis and cell tracking. GANs are used for image generation and can be used to generate synthetic microscopy images that can be used to augment existing datasets. Other DL algorithms that have been used in microscopy include autoencoder, Variational Autoencoder (VAE), and Deep Belief Network (DBN). The choice of algorithm depends on the specific task being performed and the characteristics of the microscopy images being analyzed. More recently, Transformer models have exhibited significant promise in the realm of microscopy analysis tasks, mirroring its success in natural language processing and its current growth in computer vision. DL has also been utilized in neuroimaging studies extensively due to its ability to detect abstract and complex patterns. However, the implementation of DL approaches is highly dependent on the quality and quantity of data used for training and, typically, on the availability of high-performance computers [15].

Semantic Segmentation Semantic segmentation is an important area of computer vision, serving as the foundation for many practical applications in pattern recognition and image analysis. It finds applications in various domains, such as microstructure and medical technology. Semantic segmentation aims to precisely outline the boundaries of objects in an image and assign each pixel a specific class label. Transformers and CNNs are DL architectures used in semantic segmentation tasks. A CNN uses convolution filters to extract features from local regions of the images (where the filters are typically small) and allows the model to capture local spatial relationships. In contrast, Transformers work by breaking an image down into patches and feeding them into a Transformer-based encoder. Transformer-based designs can capture long-range relations between the pixels across the images [16, 3].

Transfer learning Transfer learning is a foundational technique in DL, offers a range of advantages. It particularly addresses the challenge of working with limited data, a common practical obstacle due to the expense and time required for extensive data labeling. By

harnessing knowledge from a source domain rich in data and applying it to a target domain with limited data, transfer learning significantly enhances model performance. This approach not only expedites the training process but also bolsters the model’s ability to generalize to new, unseen data while mitigating the risk of overfitting. As a resource-efficient strategy, transfer learning empowers the development of robust models adaptable to varying data distributions between domains, rendering it a valuable asset in diverse DL applications [10, 17]. Transformer vision and CNN encoders pre-trained on natural images often lead to suboptimal results in microscopy analysis tasks. Transfer learning determined that the best performance was obtained when the source domain was visually comparable to the target domain (for example, using a different set of microstructural images instead of ImageNet.) This highlights the need to create large, publicly accessible annotated image datasets in the medical and materials domains [17].

Outline The remainder of this dissertation is organized as follows.

- Chapter 2 provides an overview of the challenges associated with applying DL to microscopy, including the complexity of the data, the need for large-scale datasets, and the importance of interpretability and explainability.
- In Chapter 3, we propose a computer vision pipeline for automating area fraction measurement in ductile iron microstructures. This research uses Machine Learning (ML) to automate the measurement of graphite particle area fractions in the microstructure of ductile iron.
- In Chapter 4, we investigate how effective it is to pre-train DL models using large microscopy datasets instead of relying on pre-trained models that pre-trained on large natural image dataset like ImageNet. Additionally, we examine the benefits of using both CNN and Transformer encoders together for semantic segmentation.
- In Chapter 5, we introduce a lightweight U-shaped method called GCtx-UNet, designed

to improve computation efficiency in microscopy segmentation tasks.

- Finally, in Chapter 6, we conclude our research by summarizing our findings and discussing future work. This includes proposing new U-shaped methods based on recent advancements in algorithms to further improve microscopy segmentation tasks.

Chapter 2

Literature Review

2.1 Deep Learning and CNNs

Deep learning (DL) methods have been applied in numerous fields, such as medical, health, biology, and material science, for feature learning and pattern classification [18, 17]. Past research showed that deep learning can outperform traditional machine learning approaches in several fields like image processing and computer vision [18]. This section briefly overviews deep learning concepts and discusses the most popular CNN (Convolutional Neural Network) techniques in material sciences.

Convolutional neural network (CNN) architecture is a powerful family of neural networks designed to work with images. It is among the most popular deep learning techniques and ubiquitous in the field of CV [19]. This is because of its success in solving challenging image classification problems previously solved by hand-crafted features [20]. In recent years, various enhanced CNN models have outperformed the state-of-art methods in a wide variety of visual recognition tasks [20, 21, 22, 23].

There are three distinctive characteristics of a CNN: local receptive fields, shared weights, and pooling (or sub-sampling). These characteristics help to identify similar local features and reduce the number of model parameters. The idea is not to connect the network fully but to replicate neurons at many places within a layer. A CNN uses various filters (or kernels) to apply convolution operations to the image pixels and generate feature maps that are layered on top of each other (Figure 5.1). Each feature map identifies a particular pattern relevant to a layer. Pooling is used to reduce the dimensions of feature maps and the number of network parameters. In the pooling layer, only a part of the previous layer is passed to the next layer, as shown in Figure 5.1 (a). This lowers the computational load and increases the

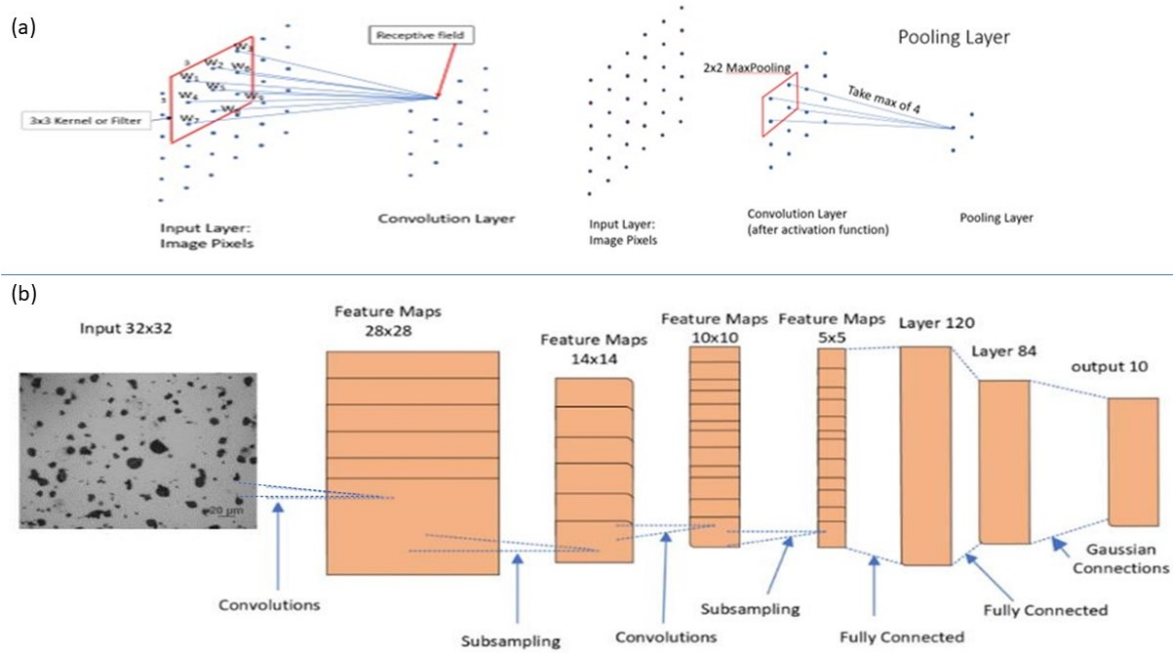


Figure 2.1: (a) Two typical layers in a CNN: convolution layer and max-pooling layer. (b) the architecture of a simple CNN for image classification. The input image is a grayscale image with 32 x 32-pixel size fed to the CNN. The first convolution layers applied different kernels (filters) to produce 28 x 28 feature maps. The pooling layer (subsampling) used a window of size 2 x 2 and take average or maximum value in that window to reduce the dimensionality of the feature maps from 28 x 28 to 14 x 14. This block is followed by another convolution layers to produce 10 x 10 feature maps, and 2 x 2 pooling layer to reduce the dimensionality of the feature maps to 5 x 5. Then the feature maps are flattened to get a 1-D vector which was fed to two fully connected layers with 120 and 84 neurons, respectively, finally producing the output value.

translational invariance by reducing the impact of the precise locations of visual features [24].

In the last decade, numerous CNN architectures have been developed for image analysis tasks such as image classification[20, 25, 26], image segmentation [27], object detection, and instance segmentation. For example, AlexNet [20] used five convolution layers for image classification tasks. VGG16 [25] increased the network depth to 16 weight layers using very small convolution layers of 3x3 filters. ResNet [24] was designed to address the vanishing gradient problem and proposed the residual connection. Following that, several variations of ResNet have been proposed [28, 2]. These architectures achieved high prediction accuracy on datasets such as ImageNet. For image segmentation, different types of CNN have been developed. Fully convolutional network (FCN) is an architecture mainly for semantic segmentation and can be efficiently trained to make dense predictions for per-pixel tasks [27].

The main idea of FCN is to replace the fully connected layers in a CNN with convolutional layers to enable a classification net that outputs a spatial map for pixel-wise learning. FCN consists of down-sampling (convolution), up-sampling (deconvolution), and skip connection layers. The skip connection layer is used to combine the coarse, semantic, and local appearance information to get accurate and fine predictions [27, 29]. To improve accuracy, variations of FCNs were introduced. For example, U-Net is a symmetric encoder and decoder network with skip connections. The U-Net architecture has many channels and skip connections to prevent information loss when reducing image resolution. SegNet [30] is a convolutional encoder-decoder network for image segmentation as well. The encoder network in the SegNet has the same topology as the convolutional layers of VGG16 without fully connected layers. The decoder part used the memorized max-pooling indices received from the corresponding encoder to perform non-linear up-sampling of their input feature maps. This step is used to produce sparse feature maps, which are convolved with a trainable filter bank to produce dense feature maps. The decoder outputs feature maps fed to a SoftMax classifier for pixel-wise classification [30]. For object detection, Region-based CNN (R-CCN) is used to detect the location of a set of objects. It has three steps: (1) generating region proposals using the selective search algorithm, (2) extracting a feature vector from each region proposal, and (3) using a pre-trained Support Vector Machine (SVM) algorithm to classify the region proposal to either the background or one of the object classes [21].

2.1.1 Generative Models

The generative machine learning methods were used in applications such as image super-resolution and microstructure reconstruction. The generative adversarial network (GAN) and variational autoencoder (VAE) are two particularly powerful methods.

2.1.2 Generative adversarial networks (GANs)

GAN contains a generator and a discriminator, which are trained against each other in an unsupervised learning process. As shown in Figure 2.2, the generator generates fake data by transforming the latent variables z with a prior distribution $p_z(z)$ to samples close to actual data. The discriminator learns to distinguish the real data from the fake ones produced by the generator. After the training completes, the discriminator should be unable to determine fake data from real data. The training is performed using mini-max objective learning as a loss function (or adversarial loss function) [1, 31]. The adversarial loss function is flexible and can be modified for other purposes. For example, Wasserstein GAN (WGAN) is designed to improve learning stability [32]. Super-resolution GAN (SRGAN) is developed for creating super-resolution images [33, 34]. CycleGAN is trained for image-to-image translation models [35]. Conditional GAN (cGAN) sets the conditional variables for both the generator and discriminator during the training [36].

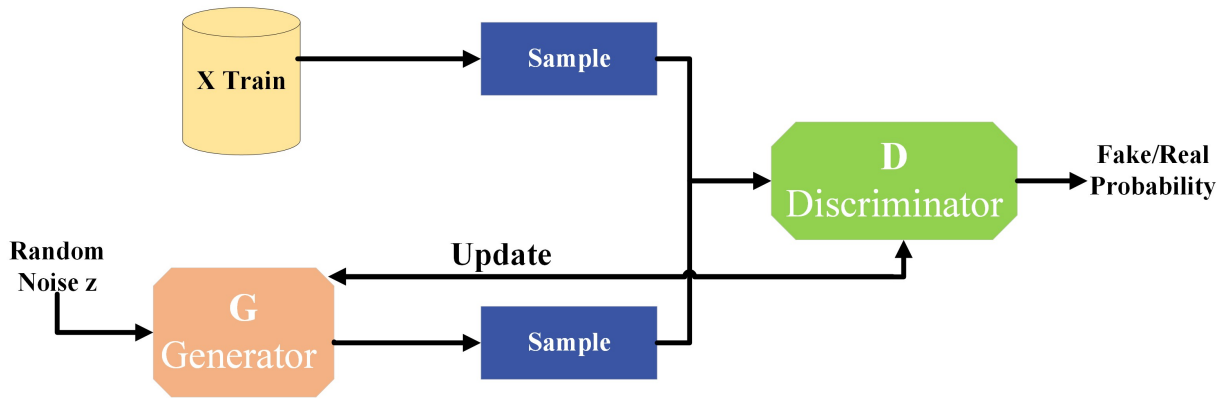


Figure 2.2: GAN architecture includes a generator G and a discriminator D , where the generator learns to output sample (fake) data similar to the training data. At the same time, the discriminator is trained to distinguish the generated data from the training (real) data [1]

2.1.3 Variational autoencoder (VAE)

Variational autoencoder (VAE) gives control over the data generated using latent variables. As shown in Figure 2.3, an autoencoder (AE) model includes an encoding and a decoding

network. In VAE, the encoder part maps its input into a probability distribution, where the encoder learns the latent variables from the input images and the decoder takes the samples' latent variables to generate new images. VAE assumes that the data is generated from a multivariate normal distribution $N \sim (\mu, \sigma)$, where the mean and standard deviation are two vectors [37, 38].

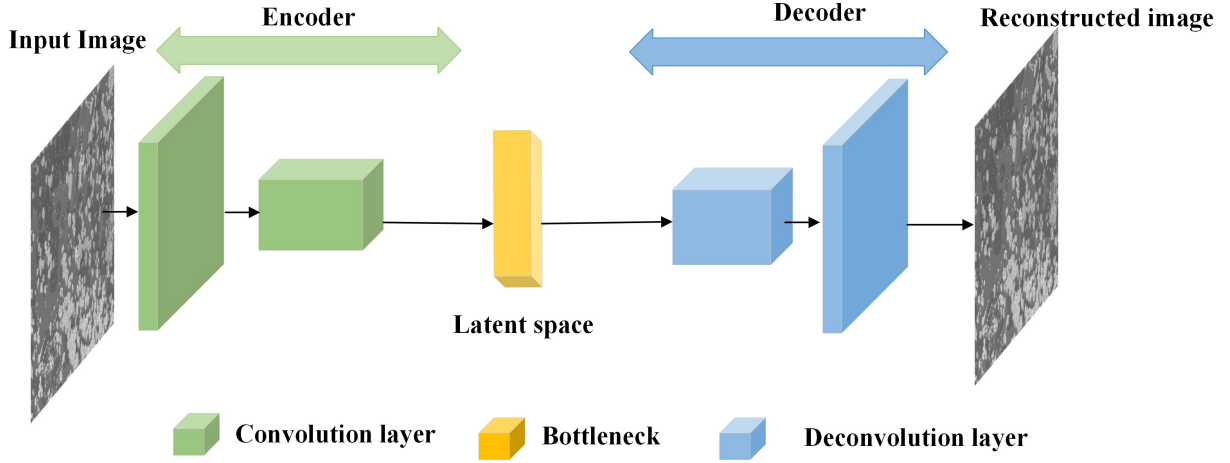


Figure 2.3: : The variational autoencoder (VAE) architecture. The input image map to a probability distribution over the latent space. Each point in the latent space is sampled from the probability distribution. The decoder takes the sample latent variable to generate a new image [2]

2.2 Transformers in CV

The core idea of CNN is its convolution layer, which operates locally and provides translational equivariance. However, the local receptive field in a convolution layer is limited in capturing the long-range relationship between pixels. The transformer-based models are solutions to this problem by capturing the long-range pixel relationships and the long-range dependencies within the input. Transformer is a type of deep neural network introduced by Vaswani *et al.* [39], which was successfully applied in Natural Language Processing (NLP) due to its ability to capture long-range dependencies in sequential data such as text. This approach led to significant improvements in NLP applications, such as language translation, text classification, and text generation. Compared to NLP tasks, using Transformer-based [3]

models on computer vision tasks is more challenging since images have size variations, noises, and redundant modalities. The critical component of a transformer-based model is self-attention. The idea behind the self-attention mechanism is to learn self-alignment that provides the ability to model long-range dependencies between image patches. The multi-head attention consists of multiple self-attention layers, where each self-attention is called head attention [1]. A transformer-based model’s training mechanism generally operates in two stages [3]. First, the pre-training stage is performed on a large-scale dataset in a supervised or self-supervised way. Second, the fine-tuning stage adapts the pre-trained weights to the downstream tasks (e.g., image classification, image segmentation, object detection) by using small or mid-scale datasets. This has led to much interest in the Transformer-based approach in CV domains such as image recognition, image segmentation, object detection, image super-resolution, and image generation.

Dosovitskiy *et al.* [3] introduced a Vision Transformer (ViT) that is built on a vanilla transformer network for Natural Language Processing (NLP) [39] with the fewest possible modifications to capture the global context of an input image. As shown in Figure 2.4, the ViT model architecture applied a pure transformer directly to sequences of image patches, flattened these patches, fed them to the linear projections, and provided the sequence of linear embeddings of these patches as an input to the transformer encoder model. The Multi-Layer Perceptron (MLP) head on the top of the model is used for the classification task. ViT was used for image classification tasks. The pre-training was performed using a large-scale dataset (14M-300M images) and was fine-tuned downstream for specific classification tasks using ImageNet dataset. ViT has better performance than traditional CNN and achieved 88.5% on ImageNet classification task. Despite ViT’s ability to capture the long-range dependencies between patches, it still suffers from poor performance in extracting local information. Additionally, the time complexity of the SoftMax function for each self-attention block is quadratic with respect to the length of the input sequence, which limits its applicability to high-resolution images.

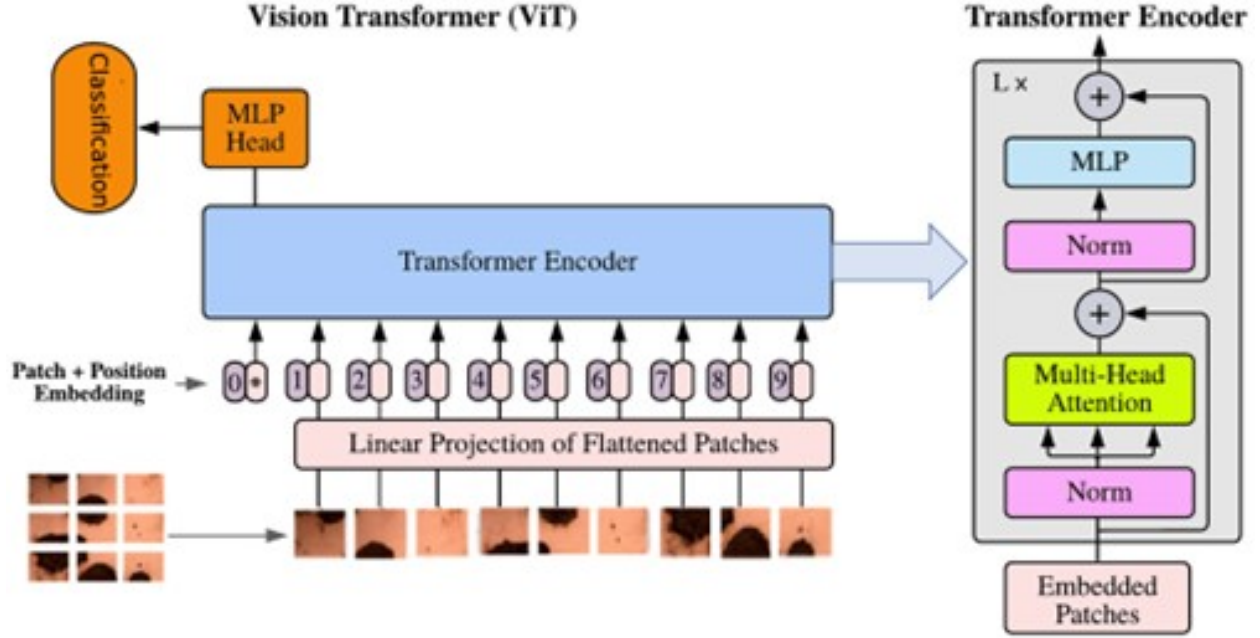


Figure 2.4: Vision Transformer (ViT) model architecture. ViT breaks the image into patches and treats each patch as a token. The patches are flattened and fed to the linear projection. The linear projection produces the sequence of linear embeddings that are fed to the Transformer encoder for the training. The MLP is used for the classification task [3].

To enhance the ability of a Transformer model to capture local information, Liu *et al.* [4] proposed a hierarchical Transformer architecture known as the Shifted Window Transformer (Swin Transformer). The key innovation of the Swin Transformer is its approach to processing image data in a hierarchical manner. Instead of treating the entire image as a flat sequence of patches, Swin Transformer computes the self-attention within the non-overlapping local windows to reduce the network complexity and processes them in a hierarchical manner. This method proposed a new general-purpose backbone for image classification and recognition tasks and achieved state-of-the-art performance. The model used a shifted-window scheme to capture large variations in the scale of visual entities and high-resolution pixels in an image with linear computation complexity to input image size. The overall architecture of the Swin Transformer is illustrated in Figure 2.5, the architecture includes four key components: patch partitioning, linear embedding, Swin Transformer blocks, and patch merging. When provided with a microscopy image of size $W \times H \times 3$ as input, the patch partition operation divides it into non-overlapping image patches of size $H/4 \times W/4 \times 48$, utilizing a shifting

window. Subsequently, the linear embedding layer projects the channel information of these image patches into a dimension denoted as C . Following this, the Swin Transformer block

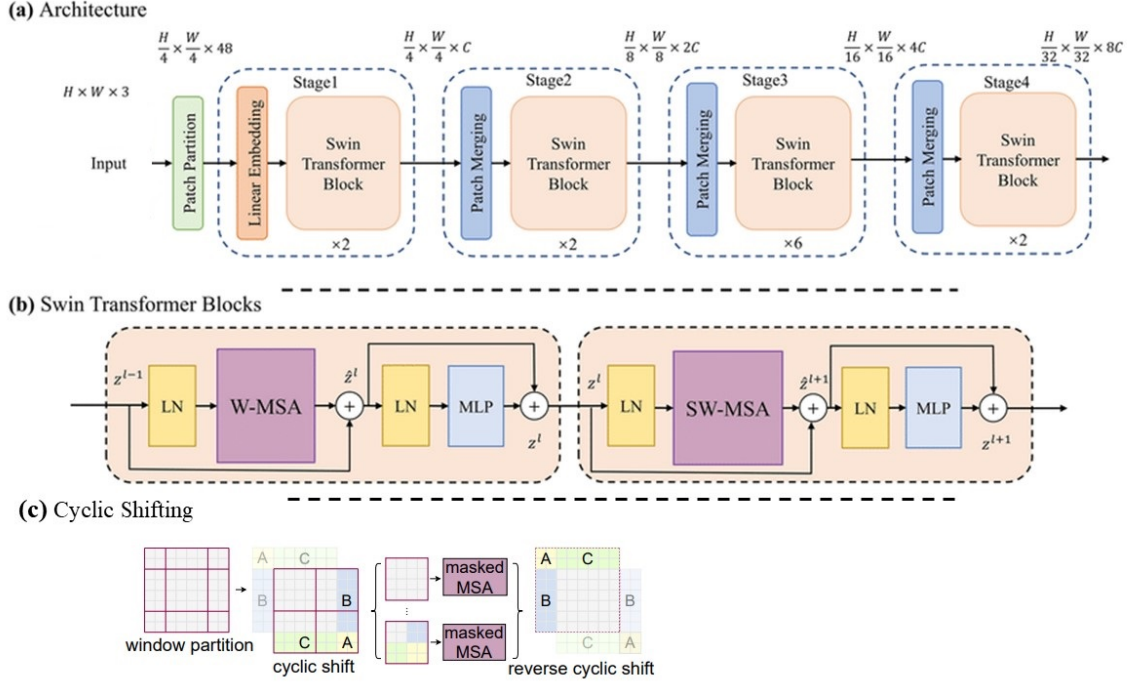


Figure 2.5: Swin Transformer architecture [4].

processes the image patches using shifted window-based self-attention in non-overlapping windows. By sequentially applying the patch merging layer and Swin Transformer block, a hierarchical architecture for the Swin Transformer is constructed. This results in output sizes of $H/4 \times W/4 \times C$ for Stage 1, $H/8 \times W/8 \times 2C$ for Stage 2, $H/16 \times W/16 \times 4C$ for Stage 3, and $H/32 \times W/32 \times 8C$ for Stage 4, respectively.

The core idea of the shifting window is shown in Figure 2.5 (c). The gray area is a patch, and the red outline indicates a window that groups neighboring patches in the first layer. Windows are of size 4 by 4, and each window has 16 patches. Once self-attention is performed locally within each window, in the next layer, windows are shifted towards the bottom right direction, resulting in a new group of windows on the right. The linear computational complexity is achieved because self-attention is computed locally within non-overlapping windows that partition an image. Swin Transformer achieves a good performance

of 87.3% on the ImageNet classification task, 58.7% box average precision score on COCO detection task, and 53.5% mIoU on the ADE20K dataset for segmentation task.

2.3 Microscopy images

Digital image processing technology for microstructural images is widely applied in materials science fields. Depending on the resolution of interest, images might come from different image characterization techniques (for both micro and macro images). Image characterization techniques such as Atomic Force Microscope (AFM), Scanning Electron Microscope (SEM), Transmission Electron Microscope (TEM), Optical Microscope (OM) are utilized for microscopic image acquisition, while computer tomography (CT) is used for macroscopic image acquisition. These techniques allow us to understand the unique characteristics of the materials at various scales. Examples of phenomenon captured by these techniques are defect evolution, second-phase identification, dislocation migration, phase transformation, and analysis of three-dimensional structure [40].

Microscopy images acquired using a microscope serve as the foundation of this field’s exploration. These images encompass various types, including brightfield, fluorescence, and electron microscopy, each offering distinct advantages and challenges. Unlike natural images found in datasets like ImageNet, microscopy images exhibit several unique characteristics. Firstly, they often boast significantly higher resolution, providing intricate details about the imaged objects. Secondly, some microscopy images, such as fluorescence images, have greater pixel depths and a wider dynamic range, further distinguishing them from their natural counterparts [41]. However, they also pose challenges in analysis due to factors such as low signal-to-noise ratios, complex backgrounds, and variations in sample preparation and imaging conditions. For instance, live imaging frequently demands low illumination intensities and rapid imaging, resulting in noisy acquisitions [42]. In disciplines like digital pathology and cell biology, high-dimensional microscopic data harbor intricate patterns and

relationships, making it challenging to predict how these images relate to diseases or other labels [13]. Lastly, microscopy images necessitate specialized image processing and analysis techniques tailored to the characteristics of the specific imaging modality in use.

2.4 CV for Microstructural Analysis

Recently, CV offered different techniques for extracting information from microstructural images, which can lead to the discovery of new materials with enhanced properties [43, 44]. CV analysis tasks such as image classification and semantic segmentation have been used for defect detection, morphological feature analysis, chemical composition, and material design. In this section, we explore the recently published research about the application of CV in material science.

2.4.1 Image classification

Classification models assign a label to each image in a dataset. The label is the best representation for the image. For example, wafer-defect detection is one of the key challenges facing semiconductor manufacturing companies. Chen et al. [5] proposed a novel CNN method called wafer defect detection network (WDD-Net) to detect wafer structural defects. The idea of WDD-Net is to reduce the number of parameters and calculations while preserving the depth of the network. As shown in Figure 2.6, the WDD-NET architecture consists of a 3×3 standard convolution and three depth-wise separable convolution (3×3 separation convolution and 1×1 standard convolution). The last layer was used for average pooling to ensure information integrity by taking global average pooling (GAP) while removing the fully connected layer. Two CNN architectures, VGG-16 and MobileNet-v2, were also trained for comparison purposes. The dataset has more than 10,000 grain images. The dataset was not balanced, where the qualified grains (positive data) were over 95% of a wafer and the defect grains (negative data) were only 5%. To enhance the generalization ability of the

model, data augmentation was applied to increase the number of defect images using affine transformation and GAN. The results show that the three methods are able to detect wafer defects with accuracy of more than 99%. They concluded that VGG-16 and MobileNetv2 [45] were better than WDD-Net for the limited data sets, but WDD-Net reduces the number of parameters, computation, and the model size.

Mulewicz et al. [46] developed Deep CNN (DCNN) by fine-tuning ResNet18 (that was trained on ImageNet) to classify different types of microstructures of low-alloy steels. DCNN was used to classify 8 microstructure classes of steel grades: C15 (bainite with carbide), C45 (ferrite with grain boundaries and small islands of perlite), C60 (martensite), C80 (perlite), V33(ferrite with grain boundaries), X70 (granular bainite), X70 (columnar bainite), and carbide free steel. The dataset has 33,283 photograph images of different types of microstructures. The proposed method achieved an accuracy of 99.8

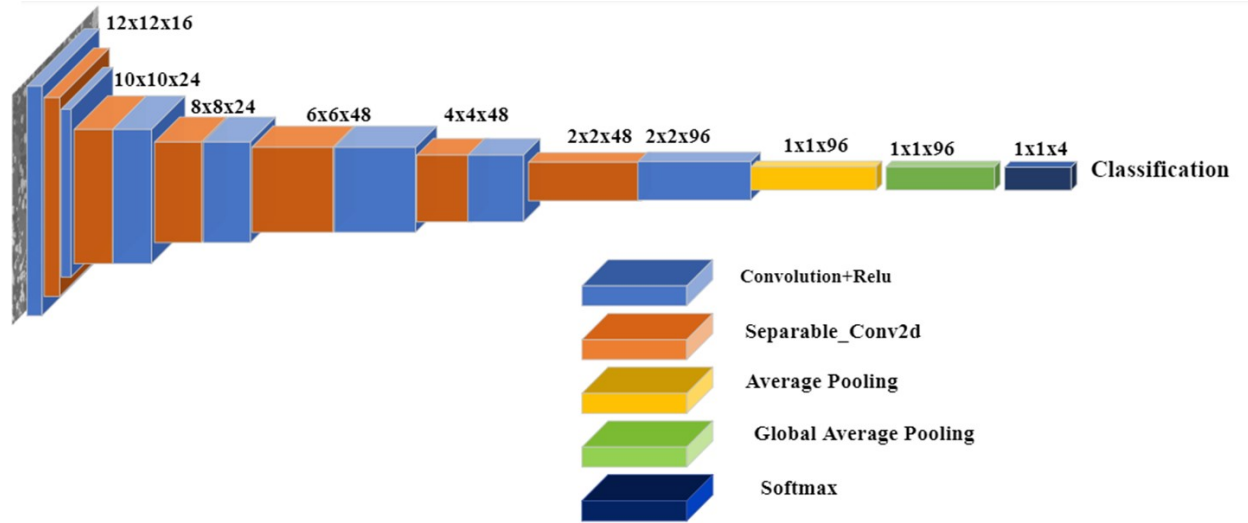


Figure 2.6: WDD-Net architecture consists of standard convolution, depth-wise separable and global average pooling to reduce parameters and computation [5]

Elbana et al. [47] used six CNN models (VGG-16, VGG-19, ResNet50, Mobile-Net [48], Inception-V3, and Xception [49]) for the microstructural constituents of ultrahigh carbon steel (UHCS) to classify Spheroidite microstructural images. The open-source UHCS dataset was used in this research, consisting of 961 micrographs including Pearlite, Spheroidite, Cementite network, Pearlite containing Spheroidite, Widmanstätten cementite, and Martensite.

There were 374 Spheroidite microstructural images in the dataset resulting from twenty-three distinct annealing schedules. The Spheroidite micrographs were divided into different classes based on different annealing conditions with the same magnification scale. Different datasets were constructed with distinct classes based on the magnification scales. Each image in each dataset was cropped to 224×224 sub-images. The six CNN models used transfer learning by either directly applying the pretrained model or fine-tuning the pretrained model. The results showed that CNNs incorporating pretrained models had the validation accuracies of 98.33% while the fine-tuned VGG16 model achieved the validation accuracy of 100%.

Banerjee and Sparks [50] implemented CNNs (VGG16, InceptionV3, and Xception), MLP, and Random Forest (RF) to classify microstructural images as dendritic or non-dendritic. The dataset consists of 133 dendritic and 444 non-dendritic structures from 21 alloy systems such as Al-Cu, Cu-Zn, Cast-Iron, and Carbon steel. They employed data augmentation via rotation and translation to increase the data points six-fold. The CNN methods used transfer learning by applying pre-trained networks of VGG16, InceptionV3, and Xception [49]. For MLP and RF, Binary Red Deer Algorithm (BRDA) optimization was employed for feature engineering and for improving the binary classification of microstructural images. The CNN methods with transfer learning achieved F1 scores of 80.1%–82.2% while MLP and RF achieved F1 scores in the range of 96%.

Baskaran et al [51] proposed a new paradigm for systematic segmentation of morphological features in titanium alloys. The proposed method used two-stage pipeline consisting of a classification step and a segmentation step. In the classification step, a CNN architecture consisting of three convolutional layers and one fully connected layer were developed to classify the microstructural images into three labels: lamellar, duplex, and acicular. The segmentation step used the watershed technique and the area fraction of the dominant-variant to process titanium microstructures. 1225 material microstructural images were used to train and test the CNN. The proposed method achieved an average accuracy of $93.00 \pm 1.17\%$.

2.4.2 Feature extraction

As we discussed in the background section, CNN architecture consists of convolutional layers to create and downsize feature maps and fully connected layers responsible for the classification of the image. CNN architecture can be employed as a feature extractor by removing the fully connected layers. In this format, each output vector represents a feature extracted from the microstructural image as shown in Figure 2.7. For example, after training VGG-16 using ImageNet and removing the last part of the model (fully connected layers), we can extract the features and feed them as inputs to the traditional machine learning models such as RF, Bayesian classifier, and I Bayes (NB). This framework is especially applicable to construct structure-properties relationships of materials.

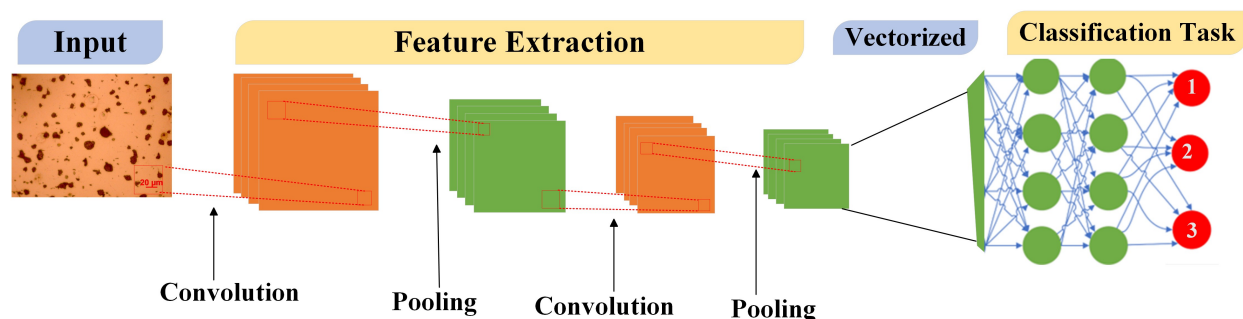


Figure 2.7: Backbone of CNN for feature extractor. The feature maps can be extracted from CNN by removing the last part which is the fully connected layer. For example, these feature maps can be used to fed to a traditional ML for classification analysis task.

Luo et al. [6] developed a CNN framework that automatically detects and classifies complex airborne carbon nanotubes/nanofibers (CNTs/CNFs) from TEM images. Airborne carbonaceous nanomaterials can have different physical-chemical properties and are very likely to form mixtures of individual nano-sized particles and micron-sized agglomerates with complex structures and irregular shapes, which makes the classification task extremely difficult. As shown in Figure 2.8, the framework was developed based on VGG16 that has five convolutional blocks with hypercolumn (Hcol)-based approach (VGG16+ 10% Hcol). Its result was compared with VGG16 with a linear SoftMax classifier (VGG16+ LS) that

also has five convolutional blocks and fully connected layer. The architecture of the framework consists of the hypercolumns that were used to extract information from each pixel at different length scales. The length of the hypercolumn vectors is the sum of these length scales. The hypercolumn vectors concatenated the information from various VGG16 convolutional layers (2, 4, 6, 9, and 12). Each pixel represents 1472 features, and each image can cover over 73 million features compared to VGG16+LS, which contains only 4096 features per image extracted from the fully connected layer. Hypercolumn feature vector was used to perform the classification of images, which includes three steps. The first step uses K-means to cluster the feature vectors in the high-dimensional space. The second step calculates the residuals for local features relative to the closest cluster centroids and concatenates the residuals to create Vector of Locally Aggregated Descriptors (VLAD) encoding. The third step tests three types of boosting algorithms (AdaBoost, random forest, and gradient boosting) for classification performance. The dataset has 5323 grey-scale TEM images of various nanostructured carbon samples. They defined five major classes representing the CNT/CNF structures: Cluster (Cl), Fiber (Fi), Matrix (Ma), Matrix-Surface (MS), and Non-CNT (NC). The sub-groups of NC classes are Graphene Sheets (GS), Soot Particles (SP), High-Density Particles (HDP), Polymer Residuals (PR), and Others (O). Since the data is not balanced, transfer learning and data augmentation have been applied to avoid overfitting. The proposed framework achieved 90.9% accuracy on the classification of a 4-class CNT dataset (including Cl, Fi, Ma, and MS) compared to VGG16+LS with 88% and 84.5% accuracy on a more complex 8-class dataset (including Cl, Fi, Ma, MS, GS, SP, HDP, and PR) compared to VGG16+LS with 80.2%. The framework can be also extended to the automated structural classification for other nanomaterials.

Ivo et al. [52] developed a method to classify photomicrographs of 1.26% Si non-grain-oriented (NGO) electrical steel. The electromagnetic efficiency of NGO electric steel is related to its grain size, which depends on the material's manufacturing process. However, it has been difficult to determine the efficiency of the electrical steel based on a particular

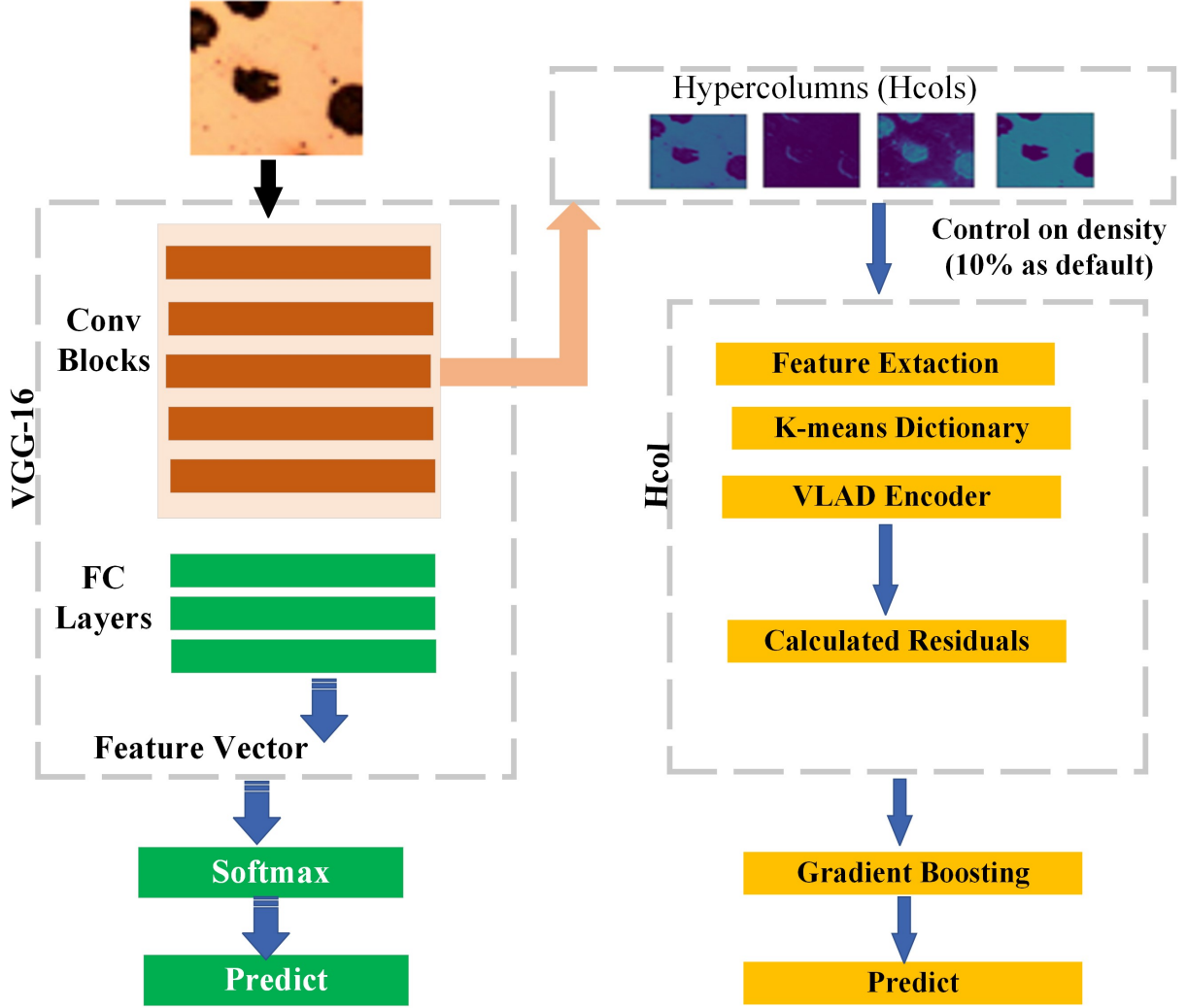


Figure 2.8: The left is a VGG-16 image classification pipeline with a linear-SoftMax classifier (VGG-16+ LS). The right is the VGG-16+ (10% Hcol) architecture, where the hypercolumn extracted the feature information from VGG-16 followed by VLAD classifiers [6].

manufacturing process. They proposed an intelligent, fast, and precise extractor-classifier combination method to classify NGO electrical steel using its photomicrographs by considering crystallographic texture and hysteresis curves data. Transfer learning using VGG, Neural Architecture Search Network (NASNet), Xception [49], MobileNet [48], ResNet, and Densely Connected Convolutional Network (DenseNet) [53] was applied as feature extractor and several traditional machine learning models (Bayesian classifier, Naive Bayes, MLP, kNN, Random Forest, and SVM) were employed as classifiers to classify data points using 127 images. The best performing extractor-classifier combination was the InceptionV3 architecture

with the kNN method, which achieved 100% for both accuracy and F1-Score.

CV has been successfully employed for heterogeneous dataset. Xiang et al. [54] combined the multi-layer perceptron (MLP) and CNN to predict the creep rupture time of Fe–Cr–Ni heat-resistant from various centrifugally cast HP40Nb alloys ($\approx Fe-25Cr-35Ni-1.5Nb-0.4Cwt\%$) tubes.

The dataset consists of textual data (chemical composition and creep testing conditions) and visual (as-cast microstructure) images. MLP was trained to analyze the textual data and CNN was trained to analyze the visual data. The proposed method used fusion deep learning model by concatenating the textual and visual features extracted from MLP and CNN layers, respectively. It then fed the concatenated features to a densely connected multilayer to estimate the creep rupture time as its single output. The proposed method was trained in multi-source heterogeneous data, and it achieved a significant enhancement in prediction accuracy.

2.4.3 Semantic Segmentation

Semantic segmentation has been applied to various types of microstructural images. In semantic segmentation, a label is assigned to each pixel in an image, which provides information about the size, shape, and position of features of interest in an image. For example, Sadre et al. [55] compared two methods to perform the segmentation of complex features in phase-contrast high-resolution transmission electron microscopy (HRTEM) images of monolayer graphene. A conventional Bragg filtering algorithm and UNet architecture were used. UNet outperformed the Bragg filtering algorithm in all performance metrics including F1, Jaccard, precision, and recall scores. UNet was robust in preventing incorrect classification of structurally important regions compared with Bragg filtering. UNet achieved 97.9% accuracy compared with Bragg filtering with 95.6% accuracy.

Alrfou et al. [56] proposed a ML pipeline to create an automatic model to segment the graphite nodule in the microstructure of ductile iron. The ML pipeline consists of two steps.

The first step was unsupervised learning using K-means clustering with manual thresholds to generate segmentation masks for the training dataset. This step significantly reduces the amount of time for preparing the training dataset. The step two was supervised learning by applying UNet to predict the percentage of graphite in the microstructure of ductile iron. To reduce memory usage and augment the training dataset, each 1280×964 input image is divided into 256×256 tiles. The algorithm achieved an Intersection Over Union (IoU) value of 96.3 Decost et al. [57] applied PixelNet CNN with transfer learning (using a VGG16 network pretrained on ImageNet) for the Ultra High Carbon Steel Database. The database includes 24 micrograph images of the 645×484 resolution. They performed two segmentation tasks: semantic microconstituent segmentation of steel micrographs into four regions (grain boundary carbide, spheroidized particle matrix, particle-free grain boundary denuded zone, and Widmanstätten cementite) and segmenting cementite particles within the spheroidized particle matrix. The models were trained on 20 hand-annotated images using sixfold cross-validation with batch normalization, dropout, weight decay regularization, and data augmentation to avoid overfitting. Two loss functions were used: the standard cross-entropy classification loss and focal loss. For semantic microconstituent segmentation task, the predictions of model1, which used the focal loss function, achieved the IoU of 62.6 ± 2.5 , while model2, which used cross-entropy loss function, achieved the IoU of 75.4 ± 3.7 . For the spheroidite particle segmentation task, the model used focal loss function and achieved the IoU of 72.4 ± 3.1 .

Azimi et al. [7] used pixel-wise segmentation of steel microstructure dataset. They applied an ensemble of fully convolutional neural network called Max-Voted FCNN (MVFCNN) to segment martensite, tempered martensite, bainite, and pearlite in carbon steels. As shown in Figure 2.9, the FCNN architecture is encoder-decoder architecture similar to VGG16 except that they replaced the fully connected layers with convolutional, up-sampling, and skip layers. The output of FCNN is a 3D matrix with the number of channels equal to the number of classes. A SoftMax layer is used to classify each pixel to the corresponding

microstructure class C_i . In order to classify objects rather than pixel, max-voting policy was applied on each object, assigning it to the class of the majority of the pixels. The steel image dataset consists of 21 LOM and SEM images with $7000 \times 8000 \text{ pixels}$. To reduce memory use, each image is cropped into $1000 \times 1000 \text{ patches}$ using sliding windows. Data augmentation was applied to the unbalanced dataset by rotating each patch by 90° , 180° and 270° to reduce overfitting. This method showed 93.94% accuracy.

Ajioka et al. [58] developed two DL methods using SegNet and UNet to perform segmentation for the microstructural images of ferrite-martensite dual phase steel. These methods were used to segment the ferrite phase, martensite phase, and ferrite grain boundary. The DL methods were compared to the existing methods such as segmentation by thresholds. The training and testing datasets consist of 40 and 10 Optical Microscopy (OM) images, respectively. It shows that DL methods are more accurate than the existing method. UNet achieved an accuracy of 95.10% for all phases compared to SegNet with 91.42% accuracy.

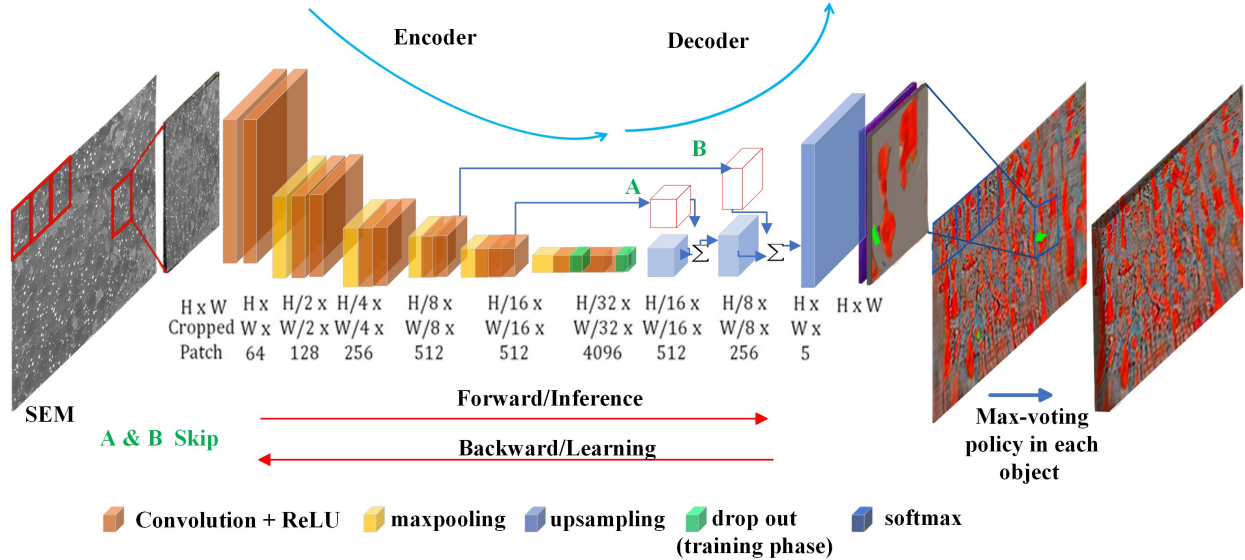


Figure 2.9: Max-voted segmentation-based microstructural classification approach using FCNNs (MVFCNN) [7]

Lu et al. [4] proposed a new segmentation model called Swin-UNet++ to perform the automatic representation of dimples micro-features of an alloy fracture surface with complex

morphology of 2.25Cr1Mo0.25V steel. The proposed method is based on Swin-UNet [59] and UNet++ [60]. Swin-UNet is a U-shaped architecture based on Swin transformer model. The Swin-UNet consists of encoders, a bottleneck, skip connections, and a decoder. Figure 2.10 shows the Swin-UNet++ architecture, where the decoder was redesigned to fuse additional feature representation by introducing a vice path to up-sample the feature representations from the last Swin-T block encoder. The proposed method can efficiently predict the boundaries and interior regions since it is based on Swin-UNet, which can capture the global and long-range dependency from the input image. Swin-UNet++ can accurately identify dimples, has a much higher prediction accuracy, and is more robust than Swin-uNet and UNet. It has 94.65% dice similarity coefficient (DSC) compared to Swin-UNet, which has 86.95% DSC.

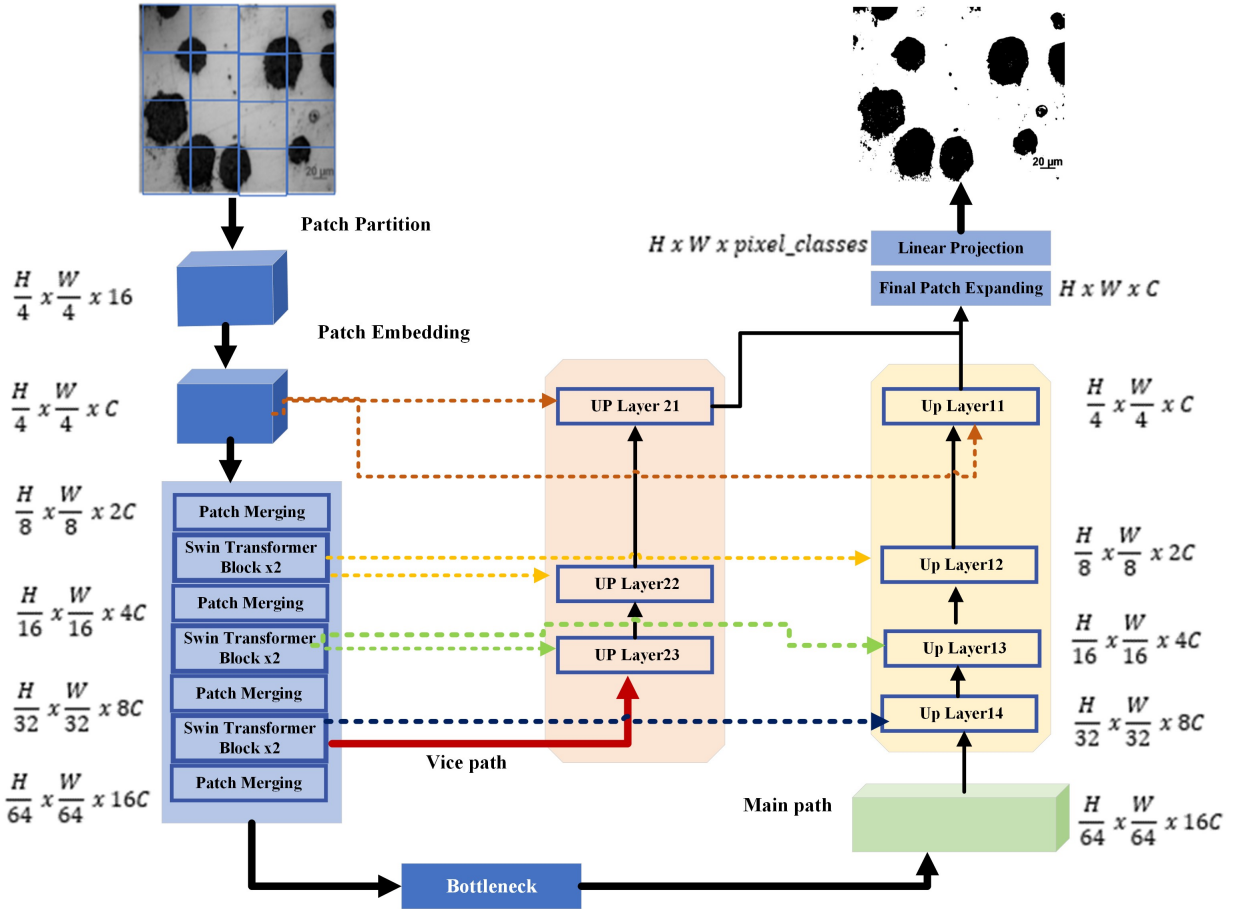


Figure 2.10: Swin-UNet++ architecture consists of an encoder, bottleneck, skip connections, vice path and decoder [4].

2.5 CV for medical Analysis

2.5.1 Classification

Healthcare professionals make diagnoses by examining medical images, enabling them to identify the presence and specific type of diseases. Traditional diagnostic methods can benefit from the integration of deep learning techniques. These technologies can classify images into different categories, such as identifying various types of tumors, fractures, organ anomalies, or even assessing the progression of a disease. Therefore, the results achieved by CNN and Transformer are not only time efficient but also can assist healthcare professionals.

Deep learning can be used in medical image classification for example: analyzing dermatological images to classify skin lesions as benign or malignant, classifying chest X-rays into categories such as pneumonia, tuberculosis, or other lung diseases, or analyzing mammograms to detect and classify breast tumors. Chen *et al* [61], combined texture feature and morphology detection methods [62, 63] and employed the AlexNet model to identify nodules within images while also predicting their benign or malignant nature. This approach resulted in an impressive AUC value of 0.9325. Shalev-Shwartz *et al.* [64] applied deep learning techniques to enhance the segmentation of left ventricular ultrasound images of the heart, achieving notably improved segmentation results. Russo *et al.* [65] introduced a CNN-based approach to detect lung cancer in histopathology images. Their study involved a comparison between the VGG16 and ResNet50 architectures using the ACDC@LUNGHP dataset, designed for automatic cancer detection and classification in whole-slide lung histopathology images. It's worth noting that the authors specifically focused on a region of interest (ROI) within the images. Their findings revealed that VGG16 achieved an impressive accuracy of 97.9% for patch classification.

Wu *et al.* [66] conducted a comprehensive assessment and extensive evaluation of the ViT model's performance in emphysema classification. They began by cropping large image

patches (16x16) from CT scans. After resizing, these patches were flattened and linearly embedded to create a sequence of patch embeddings. To retain positional information, the class embeddings were concatenated with the patch embeddings. The final embedding sequence was then processed through the transformer encoder module, and a learnable class embedding was used for emphysema classification via a softmax layer, ViT model achieves 95.95%. Than *et al.* [67] conducted a preliminary study to investigate the effectiveness of using ViT with differently sized patches on CT scans of diseased lungs, COVID-19 infected lungs, and normal lungs. They used a default positional embedding and passed the combination to a transformer encoder composed of alternating layers of multi-headed self-attention and multi-layer perceptron (MLP) units. The transformer encoder’s output was fed into an MLP head, which produced the predicted class. Their methodology was simpler and less computationally demanding compared to CNNs, but they suggested that pretraining the transformer and adding a convolutional layer in front of it (i.e., Convolutional Vision Transformers) might enhance performance. ViT achieved an accuracy of 95.36%. Tuagi *et al.* [68] proposed a vision transformer (ViT) model for pneumonia classification. To validate the model’s performance, it was compared with a CNN model and VGG16. The results revealed that ViT outperformed other techniques, achieving the highest accuracy of 0.96. Furthermore, it exhibited superior computational efficiency.

2.5.2 Segmentation

Chen *et al.* [69] proposed TransUNet, a U-shaped architecture that employs a hybrid CNN-Transformer encoder followed by multiple up-sampling layers in the CNN-decoder. This method leverages both detailed high-resolution spatial information from CNN features and the global context encoded by Transformers. The TransUNet architecture includes 12 Vision Transformer (ViT [3]) layers in the encoder, which encode tokenized image patches obtained from CNN layers. These encoded features are then upsampled using up-sampling layers in the decoder to generate the final segmentation map, with skip-connections incorporated.

TransUNet achieved high performance compared with CNN-based models and achieved accuracy on the Synapse multi-organ dataset with 77.3% in mean DSC and a mean Hausdorff distance (HD) of 31.9 mm. Additionally, it achieves an accuracy of 89.71% in DSC on the Automated Cardiac Diagnosis Challenge (ACDC) dataset. TransUNet++ [70] present an enhanced design that demonstrates promising outcomes in CT segmentation tasks. In the context of volumetric medical segmentation, TransBTS [71] and TransBTSV2 [72] introduce the Transformer for modeling spatial patch embedding within the bottleneck features.

Cao *et al.* [59] proposed a UNet-like pure transformer model called Swin-Unet for medical image segmentation. Swin-Unet has a U-shaped encoder-decoder architecture with skip-connections for local-global semantic feature learning by feeding the tokenized image patches into the Swin-Unet model. Both encoder and decoder structures are inspired by a hierarchical Swin-Transformer [4] with shifted windows. MISSFormer [73] is a Transformer network that focuses on utilizing long-range dependencies by incorporating a feed-forward enhanced Transformer block with a context bridge. It effectively models local features at various scales.

Hatamizadeh *et al.* [74] proposed UNet TRansformer (UNETR) model, a U-shaped encoder-decoder architecture that uses a ViT [3] as an encoder to capture global multi-scale information. The Transformer encoder is connected with the CNN-decoder using skip connections to compute the final semantic segmentation output. The proposed method has excellent accuracy and efficiency in various medical datasets for image segmentation tasks. Hatamizadeh *et al.* [75] also proposed a novel model for medical image segmentation of brain tumors using multi-modal MRI images. This model is called Swin UNETR. It uses the Swin Transformer [4] as encoder and connects to a CNN-based decoder via skip connections at different resolutions. In BraTS 2021 segmentation challenge, the team showed that the proposed model for multi-modal 3D brain tumor segmentation ranked among the top-performing approaches in the validation phase.

Heidari *et al.* [76] proposed HiFormer that bridges a CNN and a Transformer for medical

image segmentation. HiFormer efficiently uses two multi-scale feature representations and a Double-Level Fusion (DLF) module to fuse global and local features. Extensive experiments show that HiFormer outperforms other CNN-based, Transformer-based, and hybrid methods in terms of computational complexity and quality of results. HiFormer provides an end-to-end training strategy that integrates global contextual representations from Swin Transformer and local representative features from the CNN module in the encoder, followed by a decoder that outputs the final segmentation map.

Azad *et al.* [77] proposed a novel approach, TransDeepLab, which is a DeepLabV3+ architecture based on pure Transformer for medical image segmentation. The proposed model uses a hierarchical Swin-Transformer with shifted windows to model the Atrous Spatial Pyramid Pooling (ASPP) module. The encoder module splits the input image into patches and applies Swin-Transformer blocks to encode local semantic and long-range contextual representation. A pyramid of Swin-Transformer blocks with varying window sizes is designed for ASPP to capture multi-scale information. The extracted multi-scale features are then fused into the decoder module using a Cross-Contextual attention mechanism. Finally, in the decoding path, the extracted multi-scale features are up-sampled and concatenated with the low-level features from the encoder to refine the feature representation.

2.6 Challenges and Future Directions of CV for Microstructural Analysis

There are several challenges in using CV for microscopy image analysis, which includes data preparation and deciding the specific image analysis task needed for a given problem. This section lists the challenges and several ways to overcome these challenges.

2.6.1 Preparing Dataset for Training

a) Noisy raw data: As shown in figure 2.11 (a), the microscopy images often contain noises such as salt-and-pepper noise (sparsely occurring white and black pixels), Gaussian noise, blurring problems, and low contrast. These issues can be considered one of the major complicating factors for CV algorithms. Different image-processing techniques have been used to resolve these issues. For examples, anisotropic diffusion was used to reduce image noise without removing significant parts of the image content compared to traditional filtering algorithms that might cause blurring of relevant details [40, 78, 79, 80].

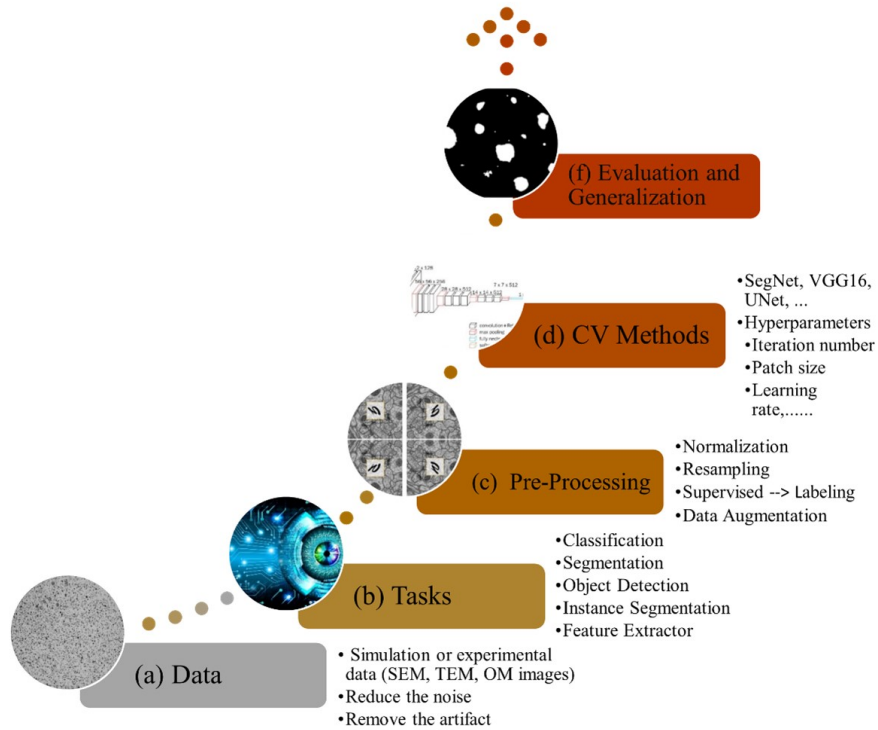


Figure 2.11: The general workflow of computer vision tasks.

b) Unlabeled dataset: Extracting different kinds of information from microscopy images depends on image analysis tasks as shown in figure 2.11(b) [15]. In supervised learning, the training dataset contains images and labels to train the model by examples. For instance, in semantic segmentation, labels provide the ground-truth of the semantic classes of the im-

age pixels. The labels can be generated manually by material experts who annotate images using software such as ImageJ, GIMP (GNU Image Manipulation Program), label Studio, and QuPath. The labels can also be generated using ML techniques such as K-means [56]. Potential problems with image labelling include boundary distortion and interpolated edges. These problems may be mitigated by coding or postprocessing to reduce interpolating between regions or classes.

c) Small dataset: Another challenge in CV is that it usually needs large dataset to achieve good performance. However, the datasets in materials or medical research are usually small and sometimes more diverse than data in other fields. To address this problem, there are three possible solutions:

1. Data augmentation: It provides more samples in the training set, which reduces the risk of overfitting and improves the generalization of CV methods [43, 81]. Data augmentation techniques include image transformations such as random cropping, flipping the image horizontally or vertically, resizing, and translation. Albumentations library [82], Tensorflow-Keras, and CV methods such as Generative Models [83] can be used to provide more samples to the training set as shown in Figure 2.11 (c). For example, Alrfou et al. [56] used Albumentations library to generate more than 500 samples for the training step while the original dataset contains only 10 images for training and 3 images for testing. Stan et al. [84] also used image augmentation to train SegNet to predict the solidification dendrite.
2. Transfer-learning and fine-tuning: Transfer learning includes training the DL model (the source domain) with a large amount of data and taking the knowledge gained from source domain by modifying the last part of the network for a new task while freezing the feature extraction part. Fine-tuning is similar to transfer learning except that the fine-tuning updates the parameters on the feature extraction part as well. Both techniques can be used to train the model (target domain) with a relatively small

dataset [85]. CNN models such as AlexNet, GoogleNet, ResNet, and VGG have been trained on large datasets such as ImageNet [20] and MS-COCO [86] for image classification tasks. The weights and parameters can be transferred or fine-tuned to a new model, which reduces the amount of time for training the new model from scratch with smaller datasets. Transfer learning has also been widely used for analyzing materials microstructure images. For example, Modarres et al. [87] applied CNN with transfer learning (using Inception-V3 pre-trained on the ImageNet 2012) for SEM images of nanostructures. They used around 20,000 SEM images that were manually classified into 10 categories (Porous_Sponge, Patterned_surface, Particles, Films_Coated_Surface, Powder, Tips, Nanowires, Biological, MEMS_devices_and_electrodes, and Fibers). CNN-based model was then used for automatic categorization and labelling of images. Li et al. [88] proposed a transfer learning-based approach using pre-trained VGG19 for reconstructing statistically equivalent microstructures from a wide range of material systems. The transferred deep convolutional network is pruned to reduce the computational cost by dropping the higher-level convolutional layers. The knowledge learned in model pruning can be transferred to developing structure-property predictions.

3. Simulation: Collecting experimental data is time consuming and requires human expertise. Simulation has lower cost and can be used to provide more data. Table 2 shows a list of available datasets generated by experiments or simulations that can be applied to CV tasks. For example, CrystalKit is a program used to build crystalline defects of various kinds from point defects to grain boundaries and precipitates. Landyne 2 published in 2016 is used as a practical tool in research and as teaching aid on microscopy and crystallography. It can be used for generating electron diffraction patterns. However, in some cases, the simulated data may carry less information than the experimental data. To address this issue, Ma et al. [89] used Monte Carlo Potts model to represent the polycrystalline microstructure of materials images and labels.

Table 2.1: List of some experimental/simulation Datasets

Experimental/ Simulation Dataset	Description
Atomagined [90], [91]	A simulated atomic-resolution high angle annular dark field (HAADF) imaging dataset containing unique ICSD structure prototypes.
DeepDamage [92]	Deep learning to quantify and classify damage in dual phase steel
UHCSDB [93]	Ultra-High Carbon Steel Micrograph Database
UHCS [57]	Ultrahigh carbon steel (UHCS) micrographs
Warwick electron microscopy dataset [94]	TEM and STEM Images/Crops datasets
Powder bed anomaly [95]	The dataset was collected and annotated using the internally developed Peregrine software tool. The dataset consists of three different types of layer-wise powder bed images including: laser powder bed fusion, electron beam powder bed fusion, and binder jetting.

The method extracts useful features from the real images and modify the simulated images by applying GAN to generate synthetic images and make them more relevant to the segmentation task.

d) Imbalanced dataset: Another challenge is the imbalanced dataset in materials microstructures and medical imaging, where there are far more negative samples than positive ones. Alrfou et al. [56] used an imbalanced dataset, where most of the pixels belonged to the background class (Iron) as opposed to the particle (graphite) pixels. They used focal-loss function [96], which generalizes the binary cross-entropy loss, to improve the imbalance dataset by assigning more weights to hard misclassified examples. Zhou et al. [97] proposed novel defect detection model of material microscope image called Material Defect Detection Network (MDD-Net), where focal-loss function is also applied to address the imbalance between foreground and background.

2.6.2 Explainability and Generalizability

Deep learning models are often considered as black boxes, where the processing mechanism of the internal structure does not explain why a given input produces a specific output. In CNN models, each convolutional block uses convolution function with multiple kernels or filters. These filters have relatively small number of columns and rows and they convolve over the

input to extract high-level local features in the form of feature maps [98]. Visualizing feature maps in CNN can provide some insights about the prediction process. Figure 14 shows the features maps for the second and fifteenth layers in a UNet model for the segmented graphite nodules in ductile iron microstructure [56]. In the second layer, different filters were applied to extract the feature maps representing edges, textures, and blurring, which is human readable. However, going deeper in the network, the filters concentrate more and learn more abstract features. As shown in Figure 2.12, in the last down-sampling layer (layer 15), feature maps are very abstract and cannot be interpreted by human.

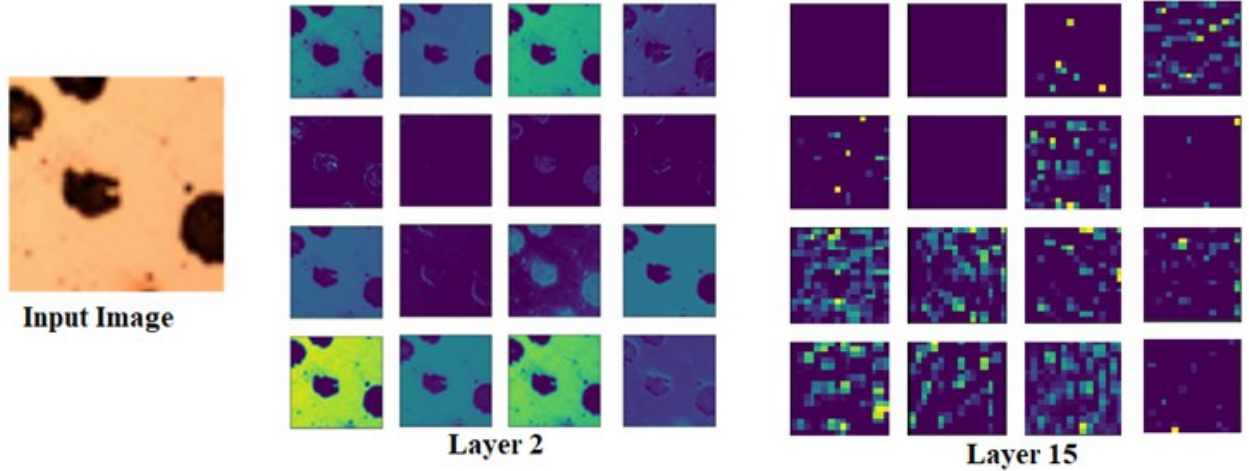


Figure 2.12: Visualization of a 2D image of graphite in ductile iron, and the resulting 16 feature maps in the second convolutional layer, and 16 feature maps in the last convolutional layer in the encoder of the UNet architecture.

To address the black-box nature of the CNN models, a number of studies have been conducted to develop interpretation techniques and tools in order to explain and visualize the decision-making process [99, 100]. Computing the heatmap is the most popular approach for visualizing and understanding CV tasks. The heat map is the class activation map, highlighting the importance of the image region to provide more details of why the model predicts a specific output. Zhou et al. [101] proposed a Class Activation Mapping (CAM) method for explaining CNN image prediction classifiers. This technique generates a CAM using the global average pooling (GAP) and adds additional layers on the top of GAP to learn classes specific in CNN. CAM obtains the weighted average of the last convolutional features using

the fully connected layer weights for each class of the network. For each class, the CAM indicates which parts of the image are effective in classification. The drawback of CAM is that it requires GAP, which cannot apply to every CNNs. Selvaraju et al. [102] proposed gradient-weighted CAM (Grad-CAM) to solve the limitation of CAM on CNN architecture and introduced the gradient information to CAM. The proposed method generalizes well for most CNNs. Zeiler and Fergus [22] proposed deconvolutional network (deconvnet) that performs an inverse convolution model. The method is used to highlight the part of the image that is responsible for the activation of each neuron and show what input pattern originally causes a given activation in the feature maps. In another research, an integrated gradients optimization saliency (I-GOS) and I-GOS++ were developed to avoid local optima in heatmap generation, which advanced single image explanations by improving the performance across all resolutions [103, 104].

Visualization methods have been used to interpret the CV methods applied to classify material microstructure images. Xing et al. [105] used vanilla CNNs and four state-of-the-art CNN models AlexNet, ResNet, SqueezeNet, and InceptionV3 to classify solidified melt pool images captured in a pulsed selective laser melting (P-SLM) process. They employed the images captured at five input laser energy levels to train and test CNN models. To avoid overfitting, data augmentation and dropout layers were integrated into the model. The experiments were performed using Q235A steel powder with an average diameter of $27\mu m$. The results show that ResNet has improved the classification accuracy of melt pools to 96.6%, which is resulted from the ability to utilize complex image features from the melt pool region. To ensure the classification results are reasonable, three visualization methods were used to add interpretation of the AlexNet model: CAM, deconvnet, and Grad-CAM. The visualization methods show that the melt pool area has higher weights than other areas such as the background and noise.

Deep learning models are often trained on specific datasets, which can limit their ability to generalize to new data or different experimental conditions. This can be particularly

challenging in microscopy, where images can vary widely depending on factors such as sample preparation, imaging conditions, and instrument settings [106].

2.6.3 Transformers for microstructural analysis

Transformer based models can be applied to various computer vision tasks such as image classification, image segmentation, object detection, and image synthesis domains. The existing transformer-based models are mainly trained with 2D natural images such as ImageNet and other large-scale datasets for pre-training and fine-tuning. It is shown that transformer models have achieved impressive results over benchmark datasets including medical datasets compared to CNN-based models. This is due to the fact that the original transformer model ViT [3] was able to capture long-range dependencies between pixels and global information. However, ViT still has poor performance in the extraction of local information. Several transformer models were introduced to help capturing local information. In medical segmentation, the transformer-based models are pre-trained on ImageNet for the fine-tuning task [59]. These approaches have been sub-optimal due to the large domain gap between natural and medical images. Recent GAN transformer-based approaches can effectively generate more realistic images than GAN CNN-based models and avoid the limitation of GAN CNN models such as instability.

We believe that the transformer models in both pure transformer and hybrid architecture can be adopted for material and medical imaging and they may demonstrate better performance compared to CNN-based models. This is due to their ability to capture long-range dependencies and global information between pixels, which might capture complex structures and irregular shapes and improve the prediction results. We believe that the training datasets will play a critical role in exploring the limits of transformers for CV in material science. Since transformer-based models are pre-trained on natural images, the features detected do not translate well to material microstructural analysis due to the significant semantic gap between natural images and microstructural images. Creating a large, labelled

dataset for microstructural or medical images and pre-training the transformer models using the dataset will improve the model performance.

2.7 Conclusion

This section highlighted the latest computer vision (CV) methods for the microstructural analysis of material and medical imaging. We briefly reviewed the concepts behind the success of convolutional neural network models and discussed the tradeoffs of various models and their impact on material sciences and medical fields. We characterized the main image analysis tasks i.e., classification, semantic segmentation, and feature extraction that can be performed by variations of CNN algorithms. We underscored the key strengths and weaknesses of the existing methods. Moreover, some recent applications in designing new materials were highlighted, including predicting complex stress and strain fields in composite materials, identifying, and classifying particles with different sizes, shapes, and thicknesses, and inverse designs.

Despite the success of CNN for materials characterization, the approach is still in its early stage and has great potential for the discovery, design, and development of new materials. The availability of large training datasets is critical to the success of applying CV methods to material science and medical imaging. Other challenges pertain to the presence of noisy raw data, unlabeled data, and imbalanced dataset. Some solutions to resolve these issues have been suggested.

Finally, we identified two main venues for future research about the application of CV methods applied to materials science and medical fields. First, to develop new techniques to enhance explainability of the learning process of CNNs to ensure the output of the algorithms are interpretable by the underlying physics. Second, we covered the concepts of transformer models and how to adapt these techniques to material science and medical fields. In several fields, including health science, it has been observed that there is a growing interest in

transformer-based models. This is due to the fact that the transformer-based model identifies long-range pixel relationships, leading to a global understanding of images, while CNN models operate locally. We discussed that through the synergy of the self-attention mechanism with CNN models, local and global features can be captured, which can outperform the traditional CNN models.

Chapter 3

Semi-supervised ML for image segmentation: measurement of graphite in ductile iron microstructure

3.1 Abstract

In this research, a machine learning (ML) approach was used for automating the measurement of the area fraction of the graphite particles in the microstructure of ductile iron. The machine learning pipeline includes an unsupervised learning step followed by a supervised learning step. The unsupervised learning step uses K-means clustering to produce the segmentation masks, which are used as the training data for the supervised learning step. For the supervised learning step, the U-Net architecture, which is a fully convolutional neural network, is used to build the model and to predict the area fraction of graphite in the microstructure of ductile iron. To prevent overfitting, data augmentation is used to generate more samples for the dataset. The focal loss function is applied to the training process to reduce the effect of the unbalance data in the training dataset and to enhance the classification accuracy of the model. The developed ML methods show promising results in terms of the accuracy of the graphite segmentation and the matching with the measured values by human experts. This outcome can be readily tailored for the microstructural analysis of other composite materials, which can lead to better design of materials with enhanced properties.

3.2 Introduction

From the early microstructural studies of acid-etched steel by Sorby in 1863 and his subsequent research which led to the first establishment of the relationship between materials structure, composition, processing, and properties, until recently, the microstructural investigation mostly relies on qualitative, instead of quantitative, descriptions [107, 108, 15]. However, by the advent of computer vision (CV) and machine learning (ML) techniques, the way we analyze materials microstructure can substantially change by quantitatively relating microstructure to composition, processing, and properties. The field of computer vision is concerned with quantifying the visual content of digital images. A digital image is constructed by pixels in which each pixel is either an integer value in black/white images or a short vector value (i.e. RGB) in color images [109]. Machine learning techniques can be applied to map the high-dimensional digital image representation to the content of the images. CV/ML can perform a variety of tasks in microstructural analysis such as classification, semantic segmentation, object detection, instance segmentation, microstructure reconstruction, and resolution enhancement [33].

The content of a microstructure or data generated by various microstructural analysis can be classified as belonging to a given class using image classification. For examples, CV/ML have been used to classify the diffraction pattern generated by four-dimensional scanning transmission electron microscopy (4D-STEM) [110], powder feedstock materials for metal additive manufacturing investigated by scanning electron microscope (SEM) [111], carbon nanotube structures studied by TEM images [112, 6], crystal structures using powder X-ray diffraction (XRD) data [113, 114], and crystal structures studied by electron images and electron diffraction patterns [115]. Semantic segmentation classifies pixels as one of the classes present in the microstructure. This method have been used for the segmentation of 3D transmission x-ray microscopy (TXM) nanotomography image data [116, 117], high-resolution transmission electron microscopy (HR-TEM) images [118], and electron backscat-

ter diffraction (EBSD) data [119]. Object detection finds each individual class and instance segmentation assigns different pixels to each instance of a class in the microstructure. For instance, this method can detect defects in micro-CT scan results of additive manufactured fiber reinforced composite material [120] and 2D materials such as graphene, hBN, MoS₂, and WTe₂ in optical microscope images [121]. Generative adversarial network (GAN) [122] has been applied in microstructural studies to synthesize new data from a collection of training data in such a way that the synthesized data closely matches the actual data. For example, Kench & Cooper [123] introduced a GAN architecture that is able to synthesize 3D datasets using representative 2D images. Lee et al. [124] applied GAN to generate two-phase morphologies for developing an inverse models optimize photovoltaic properties.

This research is a continuation of the research conducted by Kordijazi et al. [125], in which an artificial neural network (ANN) was developed for establishing a forward model of the processing-structure-properties (PSP) relationship in ductile iron materials. One of the independent factors they used in their study was the graphite percentage in the microstructure, which was measured manually using ImageJ software. This research aims to automate this process by developing a deep learning platform to segment graphite particles and consequently measure its percentage in the microstructure. This allows to establish more accurate relationship of the microstructural features and the surface properties such as wettability, corrosion, and wear resistance of the materials. The significance of this work, in addition to automating the microstructural analysis using ML methods, is related to the importance of the cast and ductile iron materials in water industry. Cast iron pipes have been used for more than 500 years for transporting drinking water [126]. Nowadays almost 60% of pipes used for distribution system are made of cast iron and ductile irons [127]. Ductile Iron is a type of cast iron known for its wear, fatigue, and impact resistance due to the spherical graphite structures in the metal. The difference between ductile and cast iron is the form of graphite present in their microstructure. In a regular cast iron, the graphite is in the form of flakes. However, in the ductile iron, the graphite particles look like little

spheres [128, 129]. One of the most common problem regarding the usage of ductile iron in water distribution system is its susceptibility to corrosion, which is known to correspond to the presence of graphite particles in the microstructure [130]. Thus, developing a robust methodology to accurately measure the graphite percentage and to relate it to the surface properties can lead to better design of this important infrastructure material.

Convolutional neural networks (CNNs) are types of deep neural networks that have been used in different computer vision tasks such as image classification, object detection, and image segmentation and they have been applied in various domains such as self-driving cars, manufacturing, and medical imaging [131, 132]. The key distinction between a densely connected layer and a convolutional layer is that the dense layers learn global patterns in their input feature space, while the local patterns are learnt in the convolution layers. This leads to two important properties for CNNs. First, the learnt patterns are translation invariant meaning that after learning a pattern, a CNN can recognize it anywhere in the images no matter to its location. This makes CNNs suitable choices for image processing as the visual world is basically translation invariant. Second, they have the ability to learn spatial hierarchies of patterns. A first convolution layer will learn small local patterns like edges, whereas a second convolution layer will learn larger patterns based on the first layer's features, and so on. This enables CNNs to learn increasingly complicated and abstract visual concepts more quickly [133]. In this research, we performed semantic image segmentation using a particular class of CNN algorithms, U-Net [28]. It is one of the robust segmentation algorithms with a fully connected network which labels each pixel of an image with a class or category. Semantic image segmentation is traditionally performed manually by materials scientists using various software such as ImageJ, which can introduce inconsistencies in the results due to varying user judgement and software settings. Moreover, this process is labor-intensive and is not scalable to large sets of images. The goal of this work is to create an automatic segmentation model using deep neural network designed for supervised machine learning. In supervised machine learning, a set of training data including the input (raw

images) and the output (segmentation masks) is used to train the ML model so it can produce segmentation mask for a new input image. To generate the segmentation masks for the training dataset, we used K-means clustering (an unsupervised learning algorithm) and manual thresholding to classify the pixels in the input images. This significantly reduces the amount of time for preparing the training dataset.

K-means is an unsupervised learning algorithm with low computational complexity that can be used to segment the areas of interests from the background. It classifies a collection of data into a K number of groups. K-means algorithm consists of two steps [16, 134]. In the first step, it calculates the centroid for each cluster. In the second step, it assigns each data point to a cluster whose centroid is closest to the data point based on the Euclidean distance. After the step two completes, the algorithm goes back to the step one and recalculates the centroid of each cluster and then based on the new centroids, it reassigns each point to a cluster. The two steps are repeated until the centroids are no longer updated. K-means clustering is efficient for image semantic segmentation if the number of clusters is known. We need to find the optimal number of clusters since too many clusters increases runtime cost while too few clusters can cause over segmentation [135]. To this end, we use the elbow method to find the optimal cluster number K. The elbow method runs the K-means cluster method on the dataset for a varying number of K clusters (e.g., from 2 to 6 clusters). It starts with 2 clusters and increases the number of clusters 1 at a time. For each K value, the elbow method calculates a cost function such as the Within-Cluster Sum of Square (WCSS) function, which is the sum-of-squared distance between each point and the centroid in a cluster. As the number of K is increased, the cost function will start to decrease. The graph of the cost will look like an elbow, where at some point of the graph, the cost function will drop dramatically giving an angle in the graph. The optimal K number of clusters is chosen at this elbow point [136].

After generating the training data using K-means clustering, we used the U-Net architecture to create an automatic segmentation model. After that step, we used the morphological

opening operation, calculated the percent of the graphite nodules for each micrograph sample, and compared the predicted values with the experimental results.

3.3 Experimental methods and procedures

In this study, four types of ductile iron samples with varying weight percentages and area fractions of the graphite nodules were used. The details of the samples and their manufacturing process can be found in this reference [137]. Figure 3.1 is a diagram indicating each step of the proposed methodology. The red box represents the preprocessing and training steps and the blue box represents the final automated model.

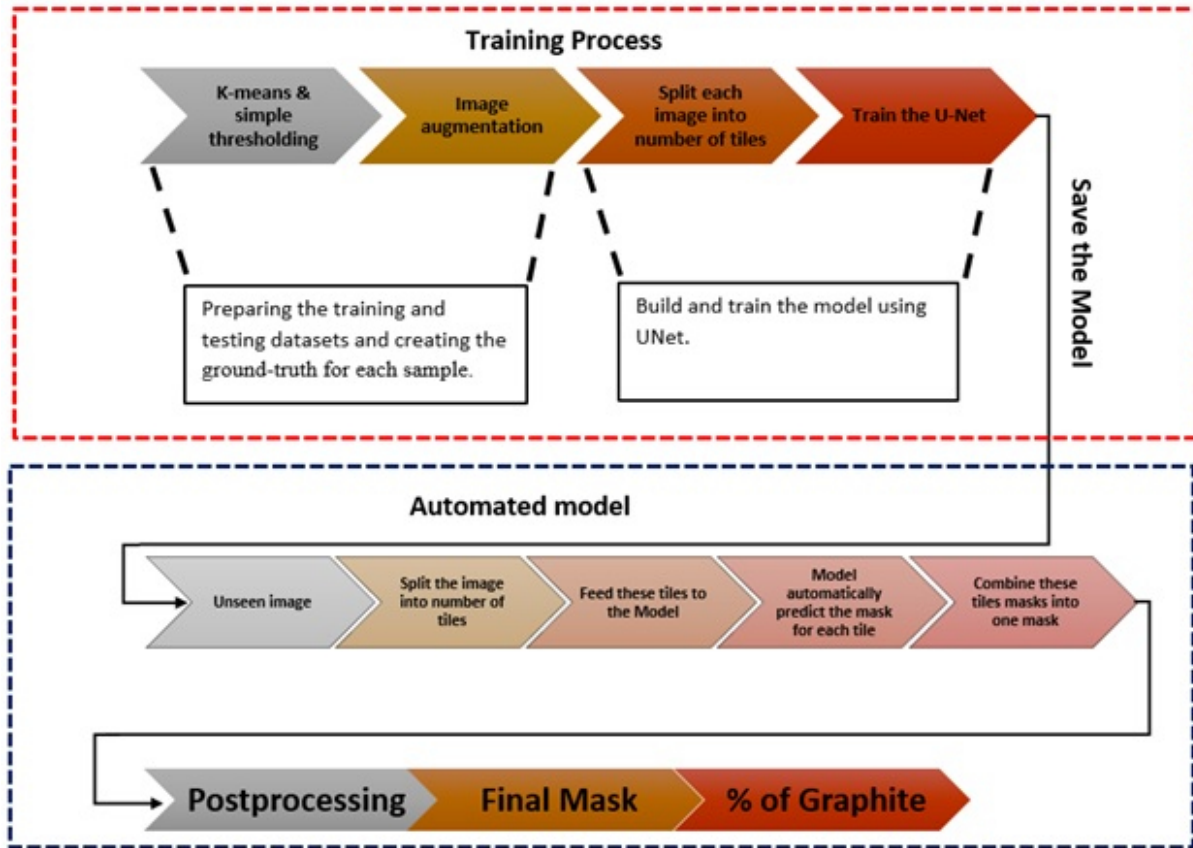


Figure 3.1: Research framework.

3.3.1 Image Preprocessing and Finding the Segmentation Mask

The segmentation masks (ground-truth) were obtained using K-means clustering and manual thresholding techniques. In the first step, we partition the images into (graphite) particle region and background region using K-means clustering algorithm. The images were transformed into gray scale and then K-means clustering was applied with different number of clusters where the elbow method was used to find the optimal number of clusters. For our data, the optimal cluster number is between 2 and 3 depending on the samples and the optimal threshold values that separate the graphite nodules from the background. In the second step, we applied a simple thresholding technique to create a pixel-wise mask for each image, which sets each pixel to either 0 or 255 based on whether the image pixel value is greater than a manually chosen threshold. This mask is a binary image where each pixel is either 0 or 255 (i.e., a black and white image). Figure 3.2 shows a sample of our dataset. The first row contains micrographs of samples showing the distribution of graphite nodules. The second row consists of segmentation masks generated by K-means and the simple thresholding technique.

3.3.2 Data Augmentation

Image augmentation is a technique that can be used to provide more training samples. Training a CNN with augmented data will reduce the risk of overfitting and improve the generalization ability of the CNN [138]. Since we have a small dataset with 10 images for training and 3 images for testing, we used data augmentation to increase the sizes of the training and testing dataset. We augmented the datasets by flipping (horizontal and vertical), transpose, random rotate, and grid distortion using a Python library `albumentations`. Table 3.1 shows the parameters of the data augmentation method [82]. After data augmentation, the training set contains 500 images, and the testing dataset contains 30 images.

The training and testing datasets contain images of different sizes such as (1280 by 964)

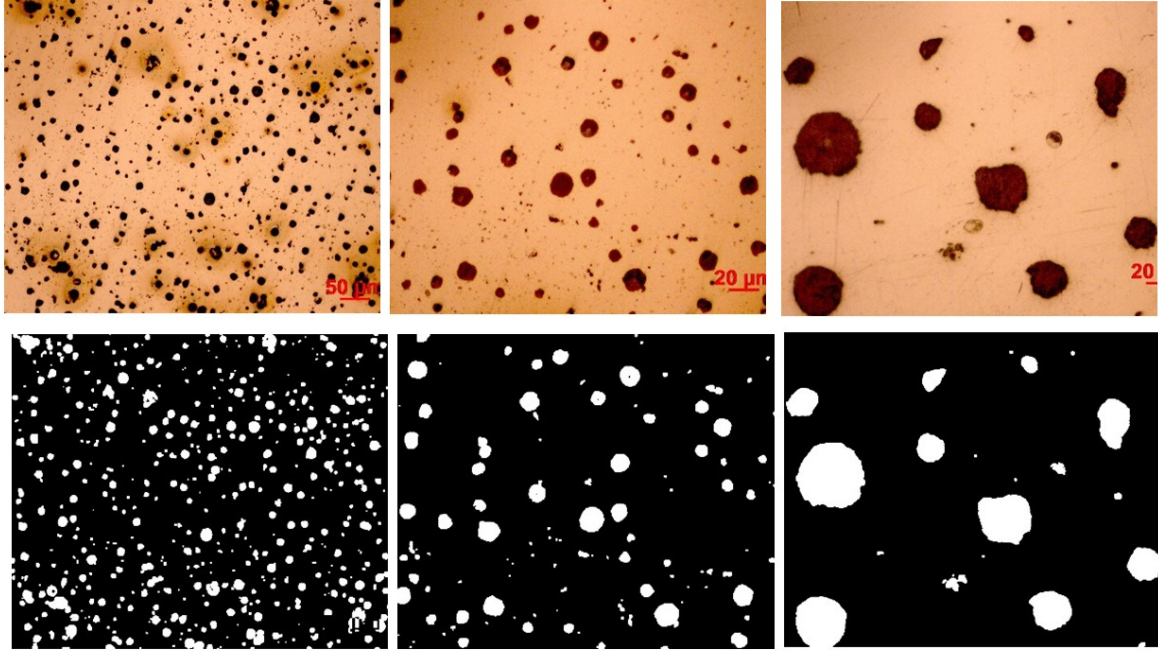


Figure 3.2: : samples of the the ductile iron microstructure. Row 2: the related mask for each sample.

Table 3.1: Data augmentation methods

Method	Probability
Flip horizontal	1
Flip vertical	0.5
Grid distortion	1
Transpose	1
RandomRotate90	0.5

and (964 by 1280), which creates two issues that must be addressed before training the neural network. Firstly, deep neural networks typically require the input images to have the same dimensions. Secondly, the size of the images is too big for our training model. To solve the issues, we split each image into 20 sub-images or tiles, each of which has the size of 256 by 256 pixels. Figure 3.3 shows an example of how we split an image into 20 tiles. After this step, the training dataset contains 10,000 tiles representing 500 images and the testing dataset contains 600 tiles representing 30 images.

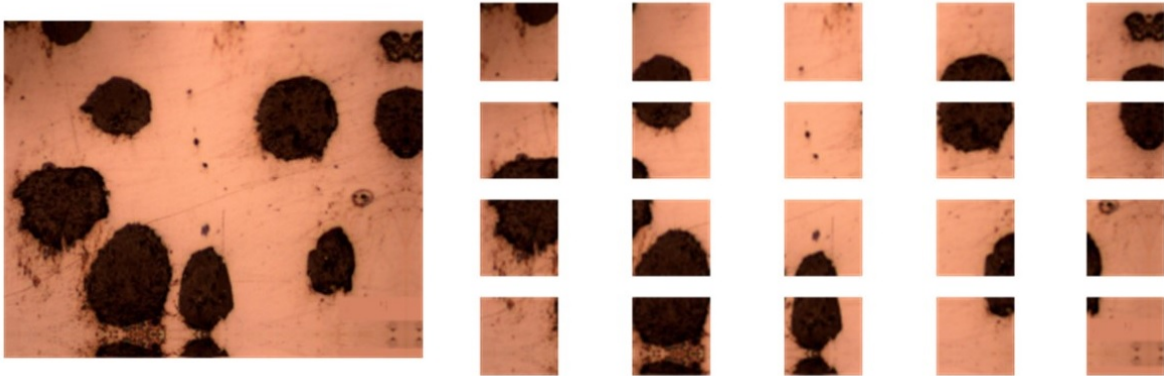


Figure 3.3: Original microstructure (left), and 20 patches with size of 256 by 256 pixels (right).

3.3.3 U-Net

U-Net is a symmetric, U-shaped, encoder-decoder neural network for semantic image segmentation. Originally it was developed for medical image processing [116]. Our U-Net architecture was adapted from the U-Net design of Ronneberger et al [29]. The U-Net architecture consists of a contracting (encoder) path and an expansion (decoder) path. The details of the U-Net architecture are shown in Figure 3.4. In the contracting (encoder) path,

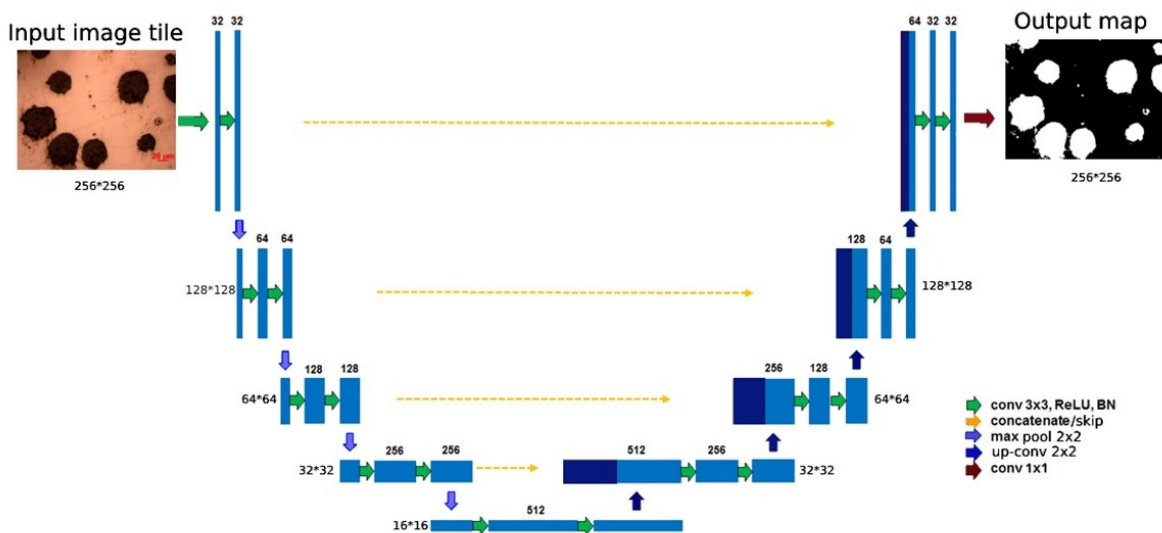


Figure 3.4: U-Net Architecture.

each block has two convolutional layers with 3×3 filter and 1×1 stride, followed by a batch

normalization layer, a Rectified Linear Unit (ReLU) activation function, and a max pooling layer with 2×2 stride. At each of the down-sampling block, the dimensions of the input image are reduced by half and the number of feature channels is doubled. The base or the bridge includes two 3×3 convolutional layers without a max-pooling layer.

In the expansion (decoder) path, each block has an up-sampling layer for the feature map, which is followed by a 2×2 convolution layer, a concatenation layer that combines the corresponding feature map from the contracting path, and two 3×3 convolutional layers, each of which is followed by a batch normalization, and a ReLU activation layer. The final layer is a 1×1 convolution layer to map the feature vector to the binary prediction. This model was trained with 15 epochs, the batch size of 32, and Adam optimizer.

3.4 Results and discussions

3.4.1 Training, Testing, and Evaluation of U-Net Architecture

Both of our training and testing datasets are unbalanced, where there are far more pixels in the background class than the pixels in the graphite class. To reduce the impact of the class imbalance during training, we used the focal loss function, which generalizes the binary cross-entropy loss function. The focal loss function assigns more weights to the hard misclassified samples [96], which are graphite pixels classified as background. In our dataset, we have 10000 tiles for training and 600 tiles for testing. During the training step, the masks were used as target output to reduce the focal loss error, where 10% of our training dataset is randomly chosen for validation.

To evaluate the model for the semantic segmentation, we used the Intersection over Union (IoU) score as the evaluation metric. Eq 1 shows the formula of the IoU score, which measures the ratio of the correct prediction over the sum of the correct prediction and the prediction error.

We tested the model using 600 tiles from 30 images, where we merged the 20 tiles belong-

ing to each predicted mask into one image. The IoU score for testing is 96.3% indicating high accuracy in the segmentation of the graphite nodules. Figure 3.5 shows the IoU confusion matrix for U-Net model where 0 represents the background and 1 represents the graphite nodules. Figure 3.6 shows the prediction for the original testing dataset, and the comparison between the true and predicted masks. It demonstrates that in most cases the CNN model was able to accurately segment the graphite nodules in the ductile iron microstructure.

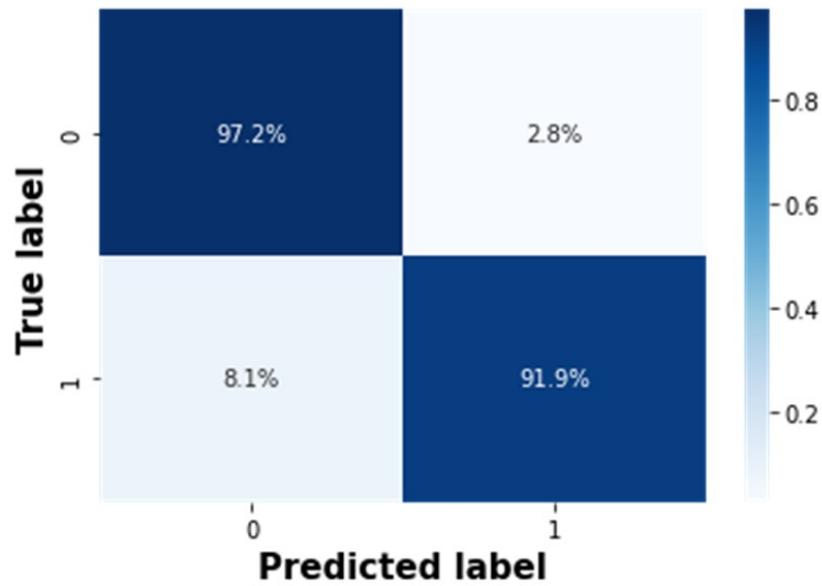


Figure 3.5: Normalized IoU confusion matrix based on the testing data.

Few incorrect classifications were observed for some graphite nodules. Figure 3.7 shows an example of misclassification of the output for a micrograph sample using the U-Net model. It was found that some pixels belong to graphite, but the model classifies them as background.

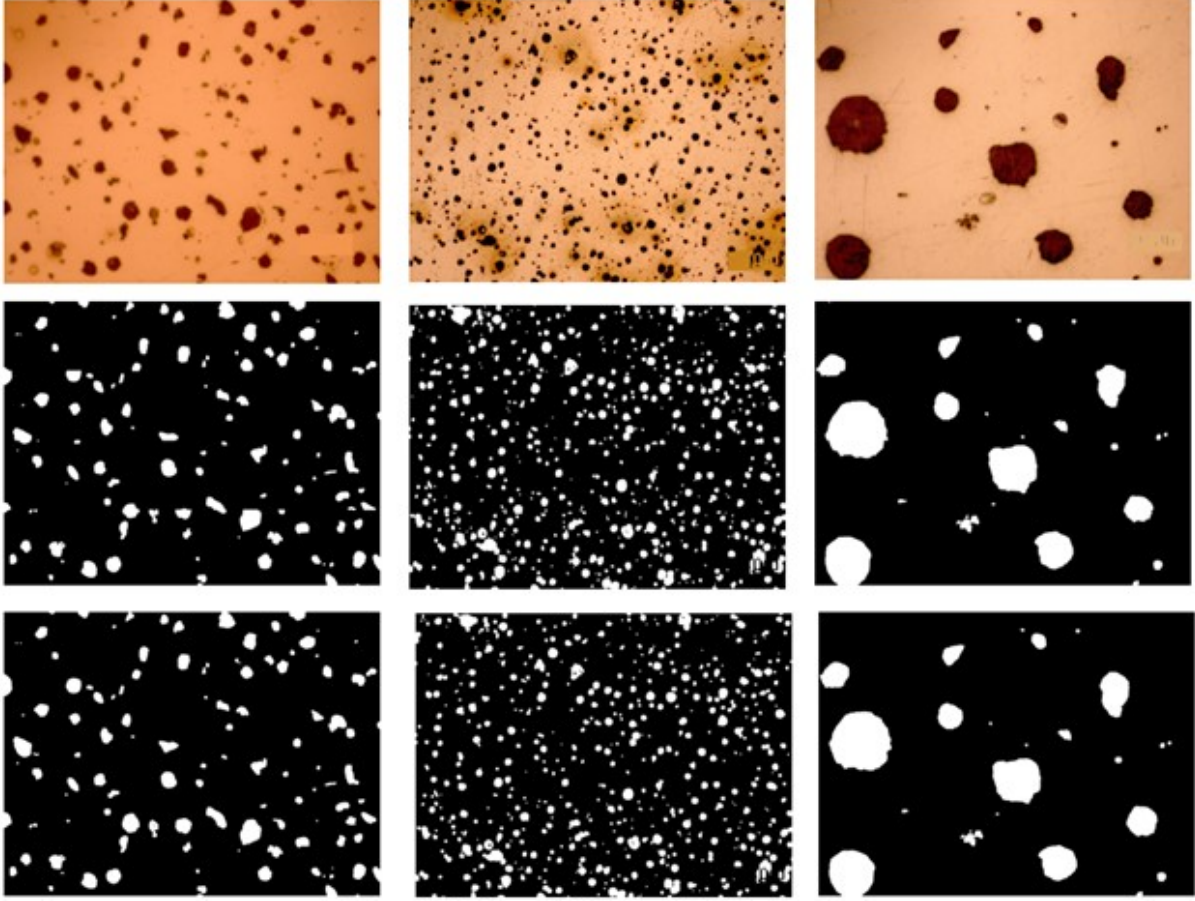


Figure 3.6: : The first row is the original images, the second row is the predicted masks, and the third row is the true masks.

3.4.2 Postprocessing

After the semantic segmentation, we need to calculate the percentage of the graphite for each micrograph sample. We used the morphological opening operation in our prediction output to isolate graphite regions. The morphological opening operation is the compound operation of erosion followed by dilation to improve the binary images [139]. Erosion will shrink or remove the connected set of one's in the output prediction and dilation will fill the holes and gaps. This is defined by the kernel which is set to 2 by 2.

The U-Net model was used to predict the percentage of the graphite for four types of ductile iron samples used by Behera et al [35]. These four types have varying area fraction of the graphite nodules as listed in Table 3.2.

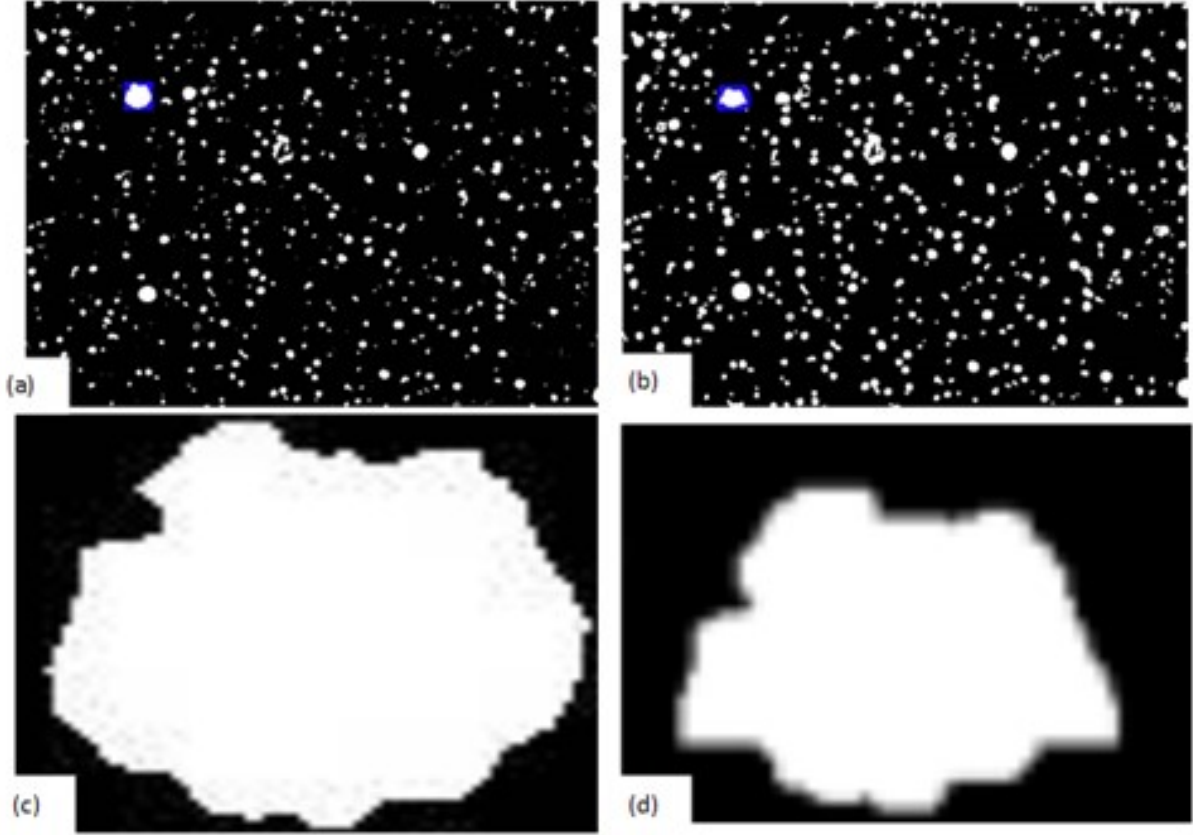


Figure 3.7: Examples of misclassified of graphite nodules where (a) is the true output and (b) is the predicted output. The figures (c) and (d) show where the U-Net misclassified the graphite nodule where (c) is the true output and (d) is the predicted output.

Table 3.2: Comparison of the area fraction of the graphite nodules in ductile iron samples obtained by ImageJ and CNN

Image label	Area fraction of graphite reported in [35]	Predicted values of area fraction of graphite
a	10.13	10.14
b	12.68	12.33
c	15.60	15.67
d	18.46	18.33

Behera et al [35] measured the percentage of the graphite for each sample using ImageJ. Figures 3.8, 3.9 and 3.10 represent, respectively, the raw micrograph of each ductile iron sample, the prediction output after applying morphological opening operation, and the raw images labelled with the prediction output. Table 2 compared the predicted results by the CNN models and the measured outcomes. As we can see, the predicted values for all four samples are very close (± 0.4 % of area fraction) to the measured values by ImageJ, which was operated by human expert. This shows that the machine learning methods intro-

duced in this paper can replace the human-operated procedure for the quantification of the area fraction of the graphite nodules in the ductile iron microstructure. This leads to the process automation with more consistency and accuracy with lower operation time. More importantly, as we only use a few microstructures (total of 13) for training and testing, the outcome of this research can be easily translated for the microstructural analysis of other composite materials leading to establish more accurate and robust PSP relationship, which is essential for better design of materials with enhanced properties.

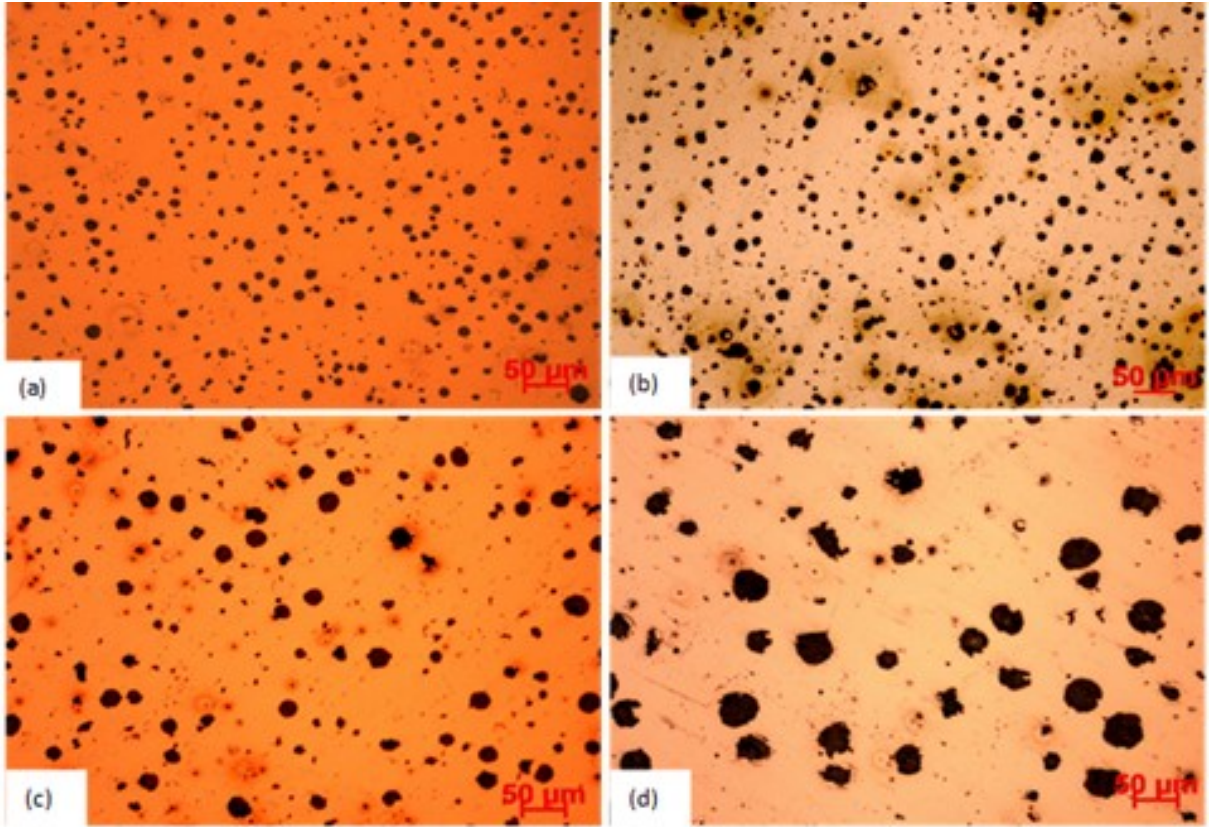


Figure 3.8: The distribution of graphite nodules in the original ductile iron microstructures

3.5 Conclusions

A combination of unsupervised and supervised machine learning methods was developed for the segmentation of the graphite nodule in the microstructure of ductile iron. K-means clustering, as the unsupervised method, was applied to generate the training dataset. This is

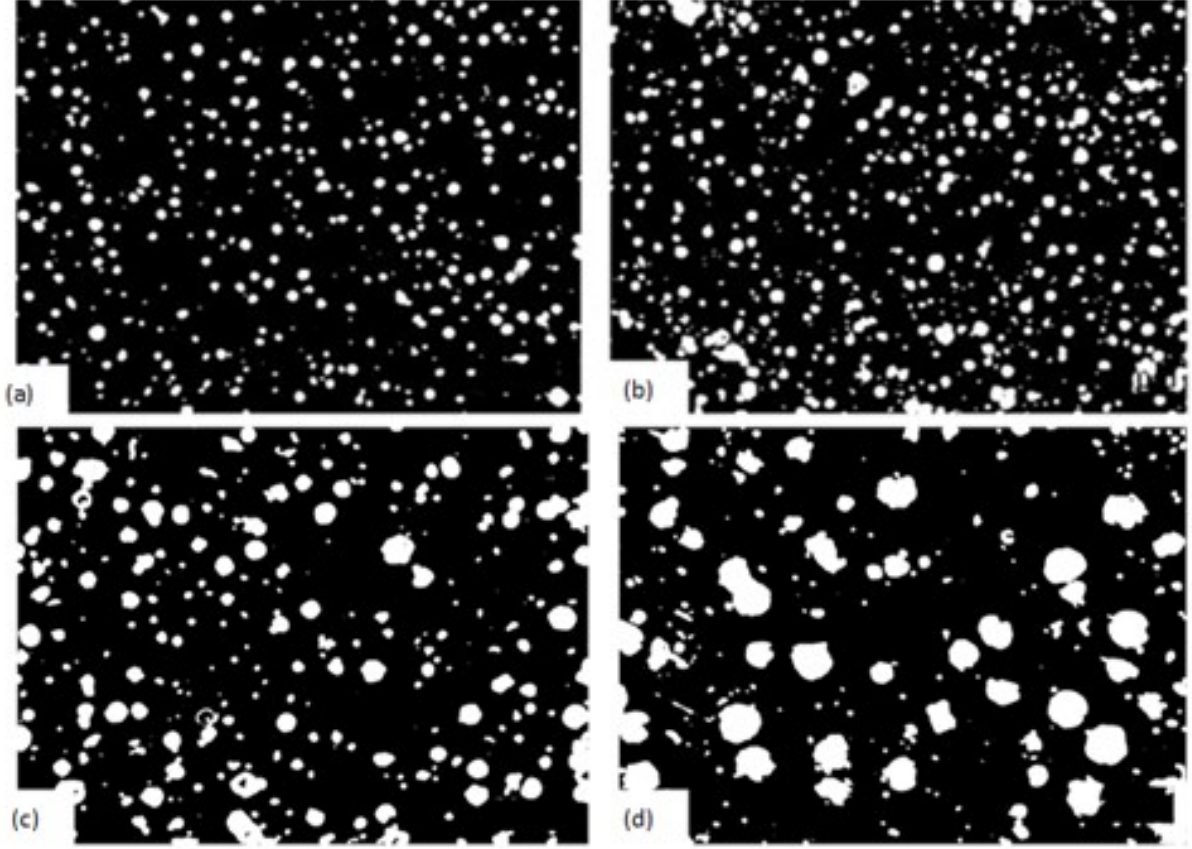


Figure 3.9: Prediction output that shows the distribution of graphite nodules.

done by finding the optimal K clusters and then applying a thresholding technique to obtain the segmentation mask for each image sample of the microstructure dataset. This dataset was then used for training a convolutional neural network based on the U-Net architecture as a supervised machine learning method. In addition, data augmentation was applied to reduce the likelihood of overfitting. Focal-loss instead of cross-entropy loss was used as the loss function to enhance the accuracy of the prediction since we work with unbalanced datasets. IoU score was used for evaluation metric. The developed U-Net architecture shows promising results as the IoU score for the testing dataset is 96.3% indicating high accuracy in the segmentation of the graphite nodules. Moreover, the predicted percentages of the area fraction of graphite are within ± 0.4 of the measurements made by a human expert using ImageJ. This demonstrates that the automated ML method of this study can replace the human-operated procedures for the quantification of the area fraction of the graphite

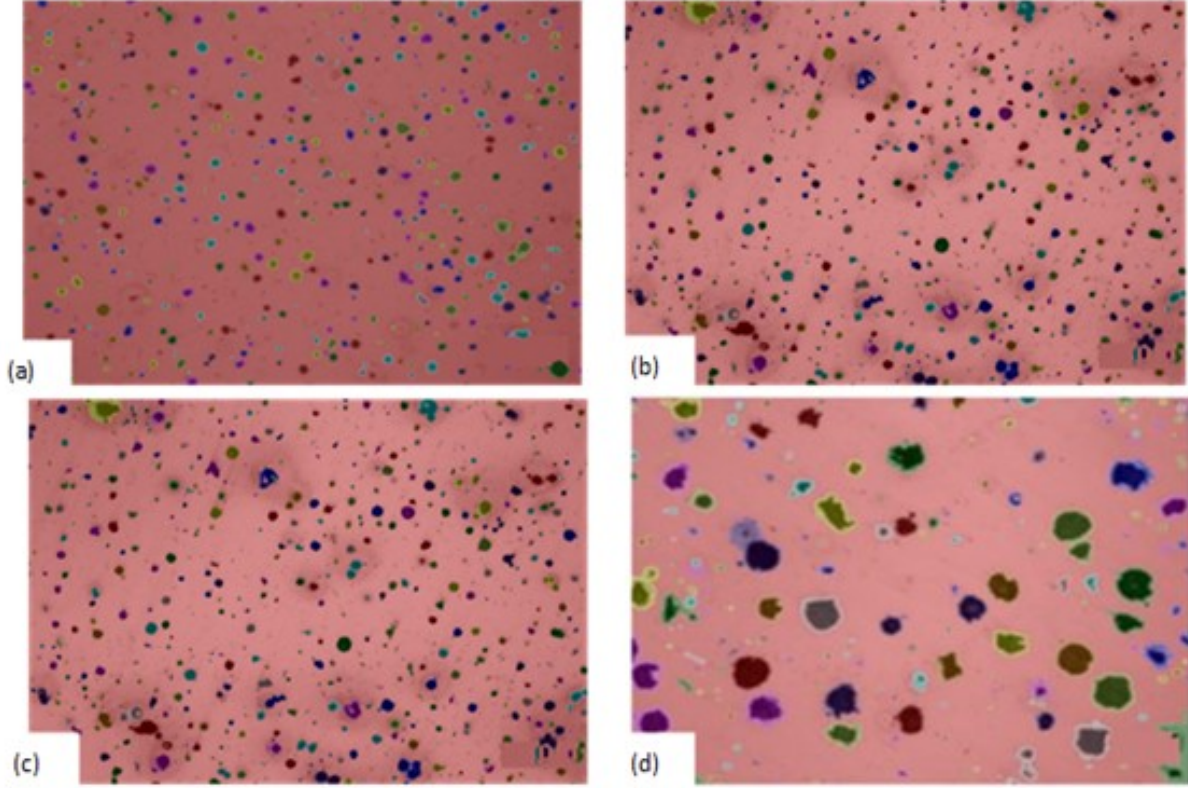


Figure 3.10: The original micrographs labelled with the prediction output, which shows the distribution of graphite nodules.

nodules in the ductile iron microstructure. In addition, since we only used a few microstructures (13 in total) for training and testing, the ML methods can be easily translated to the microstructural analysis of other composite materials leading to more accurate and robust PSP relationship essential for better design of materials with enhanced properties. As future work, we plan to experiment with different convolutional neural networks such as U-Net++, SegNet, and PSPNet and compare their performance with this U-Net model.

Chapter 4

CS-UNet: A Generalizable and Flexible Segmentation Algorithm

4.1 Abstract

This study introduces a novel U-shaped image-segmentation algorithm, CS-UNet, which contains parallel CNN and Transformer encoders. This algorithm leverages the relative strength of CNN and Transformers, and enables flexible combination of encoders pre-trained on different datasets to extract the maximum benefit of transfer-learning. CS-UNet is evaluated for its segmentation accuracy on microscopy images of materials science. The performance of CS-UNet is comparable or better than state-of-the-art algorithms based on CNN or Transformer encoders. As expected, the performance of CS-UNet is better when its encoders are pre-trained on microscopy images than when its encoders are pre-trained on natural images. However, the strength of in-domain pre-training is more significant in use cases such as out-of-distribution learning and one-shot learning. In particular, the Intersection over Union (IoU) accuracy of nickel-based super-alloy images is improved from 77.89% to 82.13% for out-of-distribution learning and IoU accuracy of environmental-barrier-coating images is improved from 65.9% to 70.45% for one-shot learning.

4.2 Introduction

CS-UNet also has surprisingly good performance on medical images. For Synapse multi-organ dataset, CS-UNet pre-trained on materials microscopy images has an average accuracy of 84.2% in Dice Similarity Coefficient (DSC), and 8.89 mm in 95% Hausdorff Distance (HD).

In comparison, state-of-the-art segmentation algorithms pre-trained on ImageNet have an average DSC ranging from 76.5% to 80.39% and average HD ranging from 14.7 to 39.7 mm. Even when pre-trained on ImageNet, CS-UNet still has DSC of 83.27% and HD of 15.26 mm. This suggests that Transformer and CNN complement each other and pre-training on images with similar attributes is beneficial to the downstream tasks.

Deep Learning (DL) has been widely applied to complex systems because of its ability to extract important information automatically. Researchers have applied DL algorithms to image analysis to identify structures and to determine the relationship between microstructure and performance [140]. DL has been demonstrated to complement physics-based methods for materials design. However, DL requires large amount of training data while the limited number of microscopy images tends to reduce its effectiveness. Learning techniques, such as transfer-learning, multi-fidelity modeling, and active learning, were developed to make DL applicable to smaller datasets [140, 141]. Transfer-learning uses the parameters of a model pre-trained on a larger dataset to initialize a model trained on a smaller dataset for a downstream task. For example, a Convolutional Neural Network (CNN) pre-trained on natural images can be used to initialize a neural network for image segmentation such as UNet to improve its precision and reduce the training time.

In recent years, attention-based neural networks called Transformers are widely adopted in computer vision. While CNN extracts features from local regions of images using convolution filters to capture the spatial relation between the pixels, Transformer divides an image into patches and feeds them into a Transformer-based encoder to capture the long-range relation between pixels across the images [3, 11]. Thus, it is possible that a combination of CNN and Transformer may be more effective in transfer-learning than either of the models alone.

In this paper, we present a novel segmentation algorithm called CS-UNet that includes a parallel composition of CNN and Transformer encoders. The high-level architecture of CS-UNet is illustrated in Figure 4.1, which shows a U-shaped encoder-decoder architecture. The

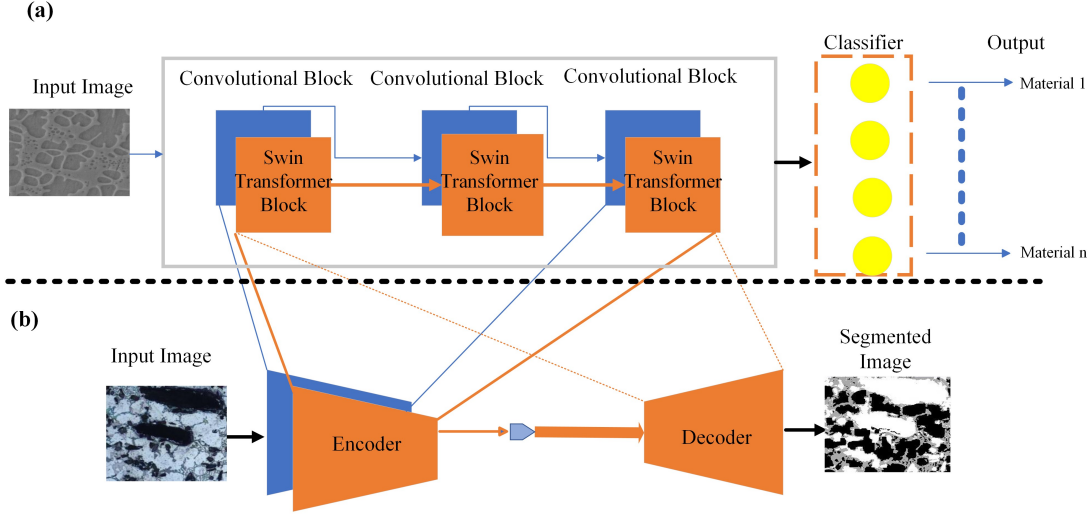


Figure 4.1: The high-level architecture of CS-UNet, where the weights of the CNN and Swin-T encoders (below the line) can be initialized from CNN and Swin-T blocks (above the line) pre-trained on natural and/or microscopy images through classification tasks.

parameters of the encoders are initialized from models pre-trained on natural or microscopy images. Each encoder transforms the input image into a latent representation vector to extract semantic information. Each decoder maps the extracted information back to each pixel in the input image to generate a pixel-wise classification of the image [140, 142]. The output of the CNN and Transformer encoders are fused before connecting to the decoder. CS-UNet allows great flexibility in combining different types of CNN and Transformer encoders pre-trained on different types of data to allow optimal choices of encoders for the segmentation tasks.

Encoder-decoder architecture allows pre-training to improve segmentation accuracy. It was speculated that pre-training with microscopy images is better for microscopy image segmentation since natural images has high-level features that do not exist in microscopy images. Recent work by Stuckner *et al.* [10] confirmed the benefit of pre-training CNN encoders on a microscopy dataset called MicroNet with over 100,000 images. They evaluated the CNN encoders with the segmentation of nickel-based super alloy (Super) and environmental barrier coating (EBC) images. Their tests showed higher accuracy measured in Intersection over Union (IoU) for one-shot and few-shot learning and for out-of-distribution images that have

different compositions, etching, and imaging conditions than the training images.

To evaluate the performance of CS-UNet, we pre-trained CNN and Transformer encoders on different types of datasets and performed segmentation on the same test sets used by Stuckner *et al.* [10]. We chose the tiny version of Swin-Transformer – Swin-T [4] as our Transformer encoder. While we can initialize the CNN encoders using the CNN models of Stuckner *et al.* [10], we are unable to obtain their dataset MicroNet to train our Swin-T encoder. To this end, we created a similar pre-training dataset with about 50,000 microscopy images in 74 classes, which we will refer to as MicroLite.

Our experiments showed that CS-UNet has similar or better accuracy than the state-of-the-art algorithms based on CNN or Transformer encoders including the ones included in Stuckner *et al.* [10]. We also compared the performance of CS-UNet using encoders pre-trained on microscopy images with the encoders pre-trained on natural images. The results showed improvement in segmentation accuracy for one-shot learning and out-of-distribution learning, which is largely in agreement with the result of Stuckner *et al.*. Due to visual similarity between microscopy and the computed tomography (CT) images, we speculate that CS-UNet pre-trained on microscopy image can improve segmentation accuracy of CT images. To this end, we compared the performance of CS-UNet with 6 state-of-the-art algorithms on the Synapse multi-organ segmentation dataset. Our results showed that CS-UNet pre-trained on microscopy images outperforms the previous algorithms in both Dice Similarity Coefficient (DSC) and 95% Hausdorff Distance (HD) by significant margins. This offers further evidence that the parallel combination of CNN and Transformer encoders is a good choice for U-shaped segmentation networks and pre-training on visually similar images help improving downstream tasks. The implementation is freely available¹.

In the rest of the paper, we first survey Transformer-based algorithms on image analysis and segmentation in Section 4.3. We then describe the CS-UNet architecture, the pre-training dataset MicroLite, and the pre-training process of the Transformer encoders of

¹<https://github.com/Kalrfou/SwinT-pretrained-microscopy-models>

CS-UNet in Section 4.4. We discuss the evaluation result of CS-UNet on test datasets of microscopy images in Section 4.5 and of medical images in Section 4.6.

4.3 Related Work

In this section, we review the recent Transformer networks for image analysis and the Transformer-based segmentation algorithms. We focus on the algorithms that are most relevant to this study.

4.3.1 Transformers for Image Analysis

CNN uses the convolution operators to provide translational equivariance but its local receptive field has limitation in capturing long-range relation between pixels [11]. In recent years, Transformer has been used in place of CNN for computer vision (CV) tasks to overcome this limitation. Transformer is a type of deep neural network introduced by Vaswani *et al.* [39], which was successfully applied in Natural Language Processing (NLP) due to their ability to capture long-range dependencies in sequential data such as text. This approach led to significant improvements in NLP applications, such as language translation, text classification, and text generation. Compared to NLP tasks, using Transformer-based models on CV tasks is more challenging since images have size variations, noises, and redundant modalities. Self-attention process is the fundamental building block of Transformer aiming to learn self-alignment that provides the ability to capture the long-range relation between image patches [11]. This has led to much interest in the Transformer-based approach in CV domains [3] such as image recognition, image segmentation [143], object detection [144, 145], image super-resolution, and image generation [146]. Dosovitskiy *et al.* [3] proposed a Vision Transformer (ViT) based on a vanilla Transformer network for NLP [39] with as few modifications as possible to capture the global context of an input image. ViT splits each image into patches and provides the Transformer with the linear embedding of each patch in order.

Image patches are handled in the same manner as tokens in an NLP application. Supervised learning is used to train the model for image classification. The model is fine-tuned using downstream recognition benchmarks such as ImageNet classification after pre-training on a JFT dataset with 300 million images [147]. ViT has better performance than traditional CNN and achieved 88.5% on ImageNet classification task. However, ViT required more computational resources to train. In addition, the complexity of computing SoftMax for each self-attention block is quadratic with respect to the length of input sequence, limiting its applicability to high-resolution images [3, 11].

To improve a Transformer model to capture local information, Liu *et al.* [4] proposed a new vision Transformer called Shifted Window Transformer (Swin Transformer). This method proposed a new general-purpose backbone for image classification and recognition tasks and achieved state-of-the-art performance. The model used a shifted-window scheme to capture large variations in the scale of visual entities and high resolutions pixels in an image with linear computation complexity to input image size. In contrast, ViT [3] model produces feature maps of a single low resolution and have quadratic computation complexity to input image size because self-attention is applied globally to all the patches. Swin Transformer achieves a good performance of 87.3% on the ImageNet classification task, 58.7% box average precision score on COCO detection task, and 53.5% mIoU on the ADE20K dataset for segmentation task. Swin Transformer V2 [148] can scale up to 3 billion parameters and train with images as high quality as 1536×1536 pixels. Swin Transformer V2 modified the Swin attention module for better window resolution and scale model capacity. This is done by replacing the pre-norm with post-norm configuration, using scaled cosine attention instead of dot product attention, and replacing the previous parameterized approach with a log-spaced continuous relative position bias approach.

The concept of unboxing the decision-making strategy in image segmentation aims to move beyond just achieving results but to understand how those results are obtained. This transparency is vital in medical applications where trust and interpretability are paramount.

Deep Nearest Centroid (DNC) [149] is an approach that utilizes the Sinkhorn-Knopp algorithm [150] to expedite the clustering of features into sub-centroids corresponding to different categories. This approach replaced the traditional softmax classification layer, leading to improved optimization of network parameters and enhancements in tasks such as image classification and semantic segmentation. DNC draws inspiration from intuitive case-based reasoning. It summarizes each class into sub-centroids by clustering training data. Classifications are made based on the proximity of test data to these sub-centroids, allowing for the generation of IF-THEN rules and visualization of representative images, making the decision process clear and understandable. ClusterFormer [151] and CLUSTSEG [152] leveraged clustering in the Transformer architecture. ClusterFormer utilizes recurrent cross-attention clustering and feature dispatching to handle various tasks with varying granularity. CLUSTSEG employs task-aware initialization and recursive clustering to tackle diverse segmentation tasks like superpixel, semantic, instance, and panoptic segmentation, all within a single framework. Both models achieve state-of-the-art results while offering transparency through their clustering-based approaches. CS-UNet may be adapted to incorporate these clustering-based methods such as DNC [149] to introduce interpretability to the model and to improve performance.

4.3.2 Image Segmentation Algorithms

In recent years, many variations of U-shaped networks have been used in image segmentation. U-Net is a Fully Convolutional Network (FCN) proposed by Ronneberger et al. [29, 56], which is a symmetric, U-shaped, encoder-decoder neural network for semantic image segmentation. U-Net typically consists of a down-sampling encoder and an up-sampling decoder structure and a “skip connection” between them. These connections copy feature maps from the encoder and concatenate them with the feature maps in the decoder.

Transformer encoder was used in SegFormer [153], which is a semantic segmentation framework that combines Transformer encoders with lightweight MultiLayer Perceptron

(MLP) decoders. SegFormer is also based on encoder-decoder architecture, where the encoder is a hierarchically structured Transformer that outputs multi-scale features without the need for positional encoding and the lightweight All-MLP decoder aggregates the information from different layers and combines local and global attention to produce the final semantic segmentation mask. SegFormer uses a patch size of 4×4 pixels to output a segmentation map. This approach helps improve dense prediction tasks and has resulted in impressive mIoU scores of 50.3% on the ADE20K dataset and 84% on the Cityscapes dataset.

CNN and Transformer were combined in TransUNet [69], which is a U-shaped architecture that employs a hybrid CNN-Transformer encoder followed by multiple up-sampling layers in the CNN decoder. This method leverages both detailed high-resolution spatial information from CNN features and the global context encoded by Transformers. The TransUNet architecture includes 12 ViT [3] layers in the encoder, which encodes tokenized image patches obtained from the CNN layers. These encoded features are then up-sampled in the decoder to generate the final segmentation map, with skip-connections incorporated. TransUNet achieved high performance compared with the CNN-based models.

Swin-Unet [59] uses only Transformer encoders in its U-shaped encoder-decoder architecture for medical image segmentation. Swin-Unet includes skip-connections for local-global semantic-feature learning by feeding the tokenized image patches into the model. Both the encoder and decoder structures of Swin-Unet were inspired by the hierarchical Swin-Transformer [4] with shifted windows.

UNET Transformer (UNETR) model [74] is a U-shaped encoder-decoder architecture for 3D medical image segmentation, which uses ViT [3] as encoders to capture global multi-scale information. The Transformer encoder is connected with the CNN decoder using skip connections to compute the final semantic segmentation output. UNETR has excellent accuracy and efficiency in various medical datasets for image segmentation tasks. Swin UNETR [75] is a similar model for segmenting brain tumors in multi-modal MRI images, which uses Swin Transformer [4] as encoders and connects to CNN decoders via skip connections at differ-

ent resolutions. This model had one of the top performance in BraTS 2021 segmentation challenge for multi-modal 3D brain tumor segmentation.

HiFormer [76] bridges CNN and Transformer encoders for medical image segmentation, where it uses two multi-scale feature representations and a Double-Level Fusion (DLF) module to fuse global and local features. Experiments showed that HiFormer outperforms other CNN-based, Transformer-based, and hybrid methods in computational complexity and accuracy. HiFormer provides an end-to-end training strategy that integrates global contextual representations from Swin Transformer and local representative features from the CNN module in the encoder, followed by a decoder that outputs the final segmentation map.

TransDeepLab [77] is a DeepLabV3+ architecture based on pure Transformer for medical image segmentation. It uses a hierarchical Swin-Transformer with shifted windows to model the Atrous Spatial Pyramid Pooling (ASPP) module. The encoder module splits the input image into patches and applies Swin-Transformer blocks to encode local semantic and long-range contextual representation. A pyramid of Swin-Transformer blocks with varying window sizes is designed for ASPP to capture multi-scale information. The extracted multi-scale features are then fused into the decoder module using a Cross-Contextual attention mechanism. Finally, in the decoding path, the extracted multi-scale features are up-sampled and concatenated with the low-level features from the encoder to refine the feature representation.

In summary, the hybrid architectures mentioned above either replace CNN with a Transformer in the encoder (e.g. Swin UNETR [75]) or stack a CNN with a Transformer sequentially to form a new encoder (e.g. TransUNet [69]). Replacing CNN with a Transformer in the encoder gives the ability to model long distance dependency in the network. However, it results in a lack of detailed texture feature extraction due to the removal of CNN in the encoder. Stacking CNN with a Transformer to form a new encoder fails to account for the complementary relationship between the global modeling capability of self-attention and the local modeling capability of convolution. Instead, they treat the convolution operation and

self-attention as two separate and unrelated operations [154, 155].

To overcome these drawbacks, the encoder in CS-Unet uses CNN and Transformer in parallel to obtain rich feature information from training images. CNN is used to extract low-level features and Swin-T is used to extract global contextual features, which are then fused using skip connections to the decoder at different stages/layers. Moreover, to reduce feature loss in the transmission process and to increase the contextual information extracted by the Swin-T encoder, the Multi-Layer Perceptron (MLP) in two successive Swin-T blocks was replaced by Residual Multi-Layer Perceptron (ResMLP).

4.4 Methodology

In this section, we give details on how CS-Unet is implemented, how the dataset MicroLite is created, and how the encoders of CS-Unet are pre-trained.

4.4.1 CS-UNet Architecture

CNN does not capture long-range spatial relation due to its intrinsic locality. Transformer is able to overcome this limitation but it is limited in capturing low-level features. Since both local and global information are essential for dense prediction tasks such as segmentation in challenging contexts, hybrid models with both CNN and Transformer encoders are expected to provide better performance for image segmentation. CS-Unet, as shown in Figure 4.2, is such a hybrid model that consists of CNN encoders, Transformer encoders, bottlenecks, Transformer decoders, and skip connections. The CNN encoders extract low-level features and the Swin-T encoders extract global contextual features. Each Swin-T encoder operates on the input image divided into non-overlapping patches, applying self-attention mechanisms to capture global dependencies. The Swin-T encoders capture long-range dependencies and contextual information from the entire image at different scales.

Inspired by the TFCN (Transformers for Fully Convolutional dense Net) [156] and Lightweight

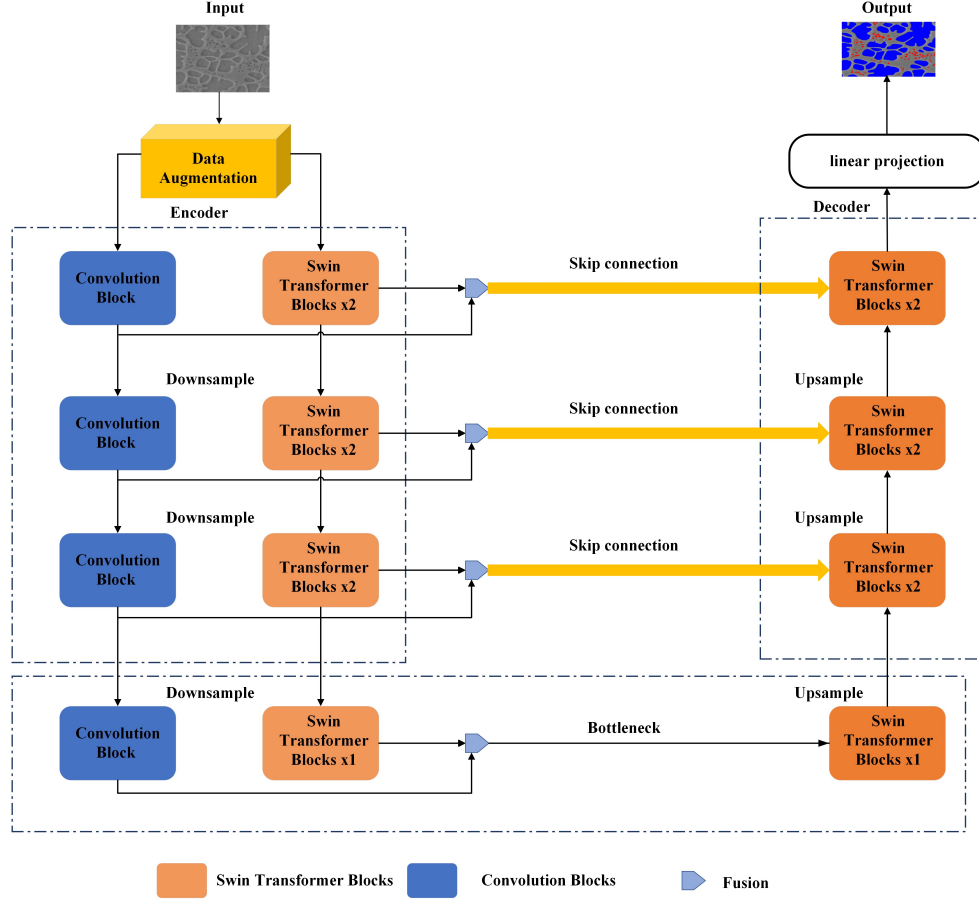


Figure 4.2: CS-UNet architecture includes CNN and Swin-T encoders, bottlenecks, skip connections, and Swin-T decoder.

Swin-Unet [8], we replaced the Multi-Layer Perceptrons (MLP) in two successive Swin-T blocks with Residual Multi-Layer Perceptrons (ResMLP) illustrated in Figure 4.3. ResMLP is used to reduce feature loss in the transmission process and to increase the contextual information extracted by the encoder. ResMLP is illustrated in Figure 4.4, which consists of two GELU [157] nonlinear layers, three Linear layers, and two dropout layers. The CNN encoder processes the input image in a series of convolution layers, gradually reducing the spatial dimensions while extracting hierarchical features. Along the way, the encoder captures low-level features in early layers and higher-level semantic features in deeper layers.

To fuse the information from both encoders, the skip connections concatenate the feature maps from the CNN encoder and the Swin-T encoder with the corresponding decoder layers. To ensure compatibility between the feature dimensions of the CNN and the Swin-T encoders,

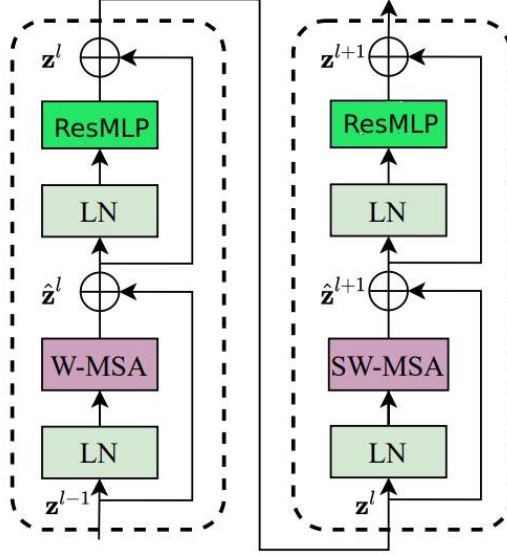


Figure 4.3: ResMLP is used to improve Swin Transformer block [8].

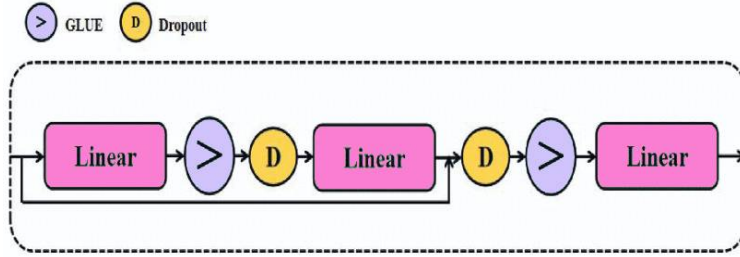


Figure 4.4: ResMLP module block [8].

we normalize the dimensions before fusing them. This is achieved by passing the features obtained from the CNN block through a linear embedding layer, which flattens and reshapes the feature map from the shape of (B, C, H, W) to $(B, C, H \times W)$, where B, C, H, W are the batch size, the number of channels, the height, and the width of the feature map, respectively. The flattened feature map is transposed to swap the last two dimensions resulting in the shape of $(B, H \times W, C)$ and is then fused with extracted features from the Swin-T encoder. By fusing the information from different encoder pathways, the skip connections enable the decoder to benefit from both the local spatial details captured by the CNN encoder and the global context captured by the Swin-T encoder.

The decoder is similar to that of Swin-Unet [59], which employs the patch-expanding layer to up-sample the extracted deep features by reshaping the feature maps of adjacent

dimensions to form a higher-resolution feature map, which effectively achieves a $2\times$ up-sampling. Additionally, it reduces the feature dimension to half of the original dimension. This allows the decoder to reconstruct the output with increased spatial resolution while reducing the feature dimension for efficient processing. The final patch-expanding layer further performs a $4\times$ up-sampling to restore the resolution of the feature maps to match the input resolution ($W\times H$). Finally, a linear projection layer is applied to these up-sampled features to generate pixel-level segmentation predictions.

Different CNN families can be used in the encoder part such as EfficientNet, ResNet, MobileNet, DenseNet, VGG, and Inception. We initialize CNN weights using MicroNet and the transformer weights using MicroLite.

4.4.2 Pre-training Dataset

The MicroLite images were collected from multiple sources including images from different materials and compounds using several measurement techniques such as light microscopy, SEM, TEM, and X-ray. MicroLite aggregates the Aversa dataset [158], UltraHigh Carbon Steel Micrograph [93], SEM images from the Materials Data Repository, and the images from some recent publications [159, 160, 161, 121, 162, 163, 164, 165]. The Aversa dataset includes over 25,000 SEM microscopy images in 10 classes, where each class consists of images in different scales (including 1, 2, 10, 20 μm and 100, 200 nm) and contrast. To properly classify these images, we used a pre-trained VGG-16 model to extract feature maps from these images and used a K-means algorithm to cluster the feature maps so that images with similar feature maps are grouped in the same class. After the pre-processing step, we obtained 53 classes. The authors of Aversa dataset manually classified a small set of the images (1038) into a hierarchical dataset, where the 10 classes are further divided into 27 subclasses [158]. Our classification of these 1038 images is largely consistent with the manually assigned subclasses. Note that we have more classes since we processed the entire Aversa dataset.

In total, MicroLite includes about 50,000 microscopy images labelled in 74 classes, which are obtained using the following pre-processing steps.

1. Remove any artifacts such as scale bars from the images.
2. Split the images into 512×512 -pixel tiles with or without overlapping depending on the size of the original images.
3. Apply data augmentation to increase the size of the dataset.
4. Aggregate the original image, the images tiles, and the augmented images to form the final dataset.

Since our current approach employs VGG-16 and K-means for feature map clustering in the pre-processing step, future work could benefit from integrating advanced clustering-based transformer methods, such as ClusterFormer [151], which can lead to more efficient and accurate results.

4.4.3 Pre-training Swin-T Encoders

We trained Swin Transformers on microscopy images to learn feature representation so that it can be transferred to tasks such as segmentation. We evaluated two types of training.

1. Fine-tune a model pre-trained on ImageNet with MicroLite (denoted by ImageNet $\rightarrow$ MicroLite).
2. Pre-train a model with MicroLite from scratch (denoted by MicroLite).

The classification tasks uses Swin-T, which is the tiny version of the Swin Transformer. Swin-T consists of two types of architectures: the original Swin-T with [2,2,6,2] transformer blocks and the intermediate network with [2,2,2,2] transformer blocks. Figure 4.5 shows the original architecture of Swin-T. We speculate that intermediate network may be enough for microscopy analysis tasks since the earlier layers learn corner edges and shapes, the

intermediate layers learn the texture or patterns, and deeper network layers in the original models learn the high-level features such as eyespots and caudal appendages. The original and intermediate Swin-T models were pre-trained on MicroLite from scratch, where the model weights are randomly initialized. The two models were also pre-trained on ImageNet and fine-tuned on MicroLite.

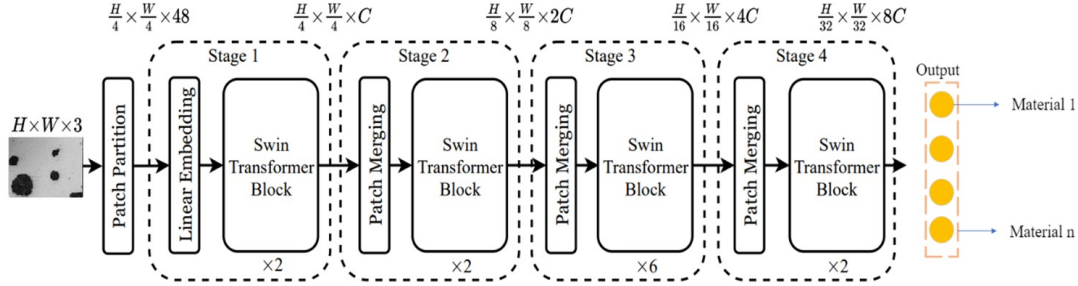


Figure 4.5: The original Swin-T architecture consists of four stages. Each stage contains two Swin transformer blocks except the stage three that contains six Swin transformer blocks [4]

The pre-training step uses an AdamW optimizer for 30 epochs with a cosine-decay learning-rate scheduler with 5 epochs of linear warm-up and batch size of 128. The initial learning rate is 10^{-3} and weight decay is 0.05. The fine-tuning step also uses an AdamW optimizer for 30 epochs with a batch size of 128 but the learning rate is reduced to 10^{-5} and the weight decay is reduced to 10^{-8} . Models were trained until there was no improvement to the validation score using an early stopping criterion with a patience of 5 epochs. Training data had been augmented using albumentations library, which included random changes to the contrast and the brightness, vertical and horizontal flips, photometric distortions, and added noise.

Swin-T models were trained by classifying microscopy images into 74 different classes. Swin-T models were either pre-trained on ImageNet (specifically the imageNet1K dataset) and fine-tuned on MicroLite, or trained with MicroLite with randomized parameters. The training stops when the validation accuracy does not improve after 5 epochs. The model accuracy is evaluated using the top-1 and top-5 accuracy. The top-1 accuracy measures the

Table 4.1: Classification accuracy of pre-trained models, where the method “ImageNet $\rightarrow$ MicroLite” indicates that the models were pre-trained on ImageNet and then fine-tuned on MicroLite.

Swin-T arch.	Pre-training method	top-1 accuracy	top-5 accuracy	# of epochs
Original	MicroLite	84.23	95.91	23
	ImageNet $\rightarrow$ MicroLite	84.63	96.353	13
Intermediate	MicroLite	84.0	96.91	19
	ImageNet $\rightarrow$ MicroLite	84.45	97.83	12

percentage of test samples for which the correct label is predicted while the top-5 accuracy measures the percentage of correct labeling in the top five predictions.

As shown in Table 4.1, the Swin-T models, when trained from scratch, take longer to converge. Specifically, the original Swin-T model takes 23 epochs, while the intermediate version takes 19 epochs. In contrast, the Swin-T models pre-trained on the ImageNet and then fine-tuned on the MicroLite converge faster. The original Swin-T model takes only 13 epochs, while the intermediate version takes 12 epochs. On average, the models initialized with ImageNet weights converged about 40.16% faster than those with the random initialization. Original Swin-T fine-tuned on MicroLite after pre-training with ImageNet achieved the top-1 accuracy of 84.63%. Overall, the Swin-T models pre-trained on ImageNet and fine-tuned on MicroLite have higher accuracy and faster convergence.

For the down-stream segmentation tasks, several models were trained for each task, which include Swin-Unet, HiFormer, and TrasDeeplapv3+. A comparative analysis was performed on these models that were pre-trained with ImageNet and microscopy images.

4.5 Evaluation of CS-UNet on Microscopy Images

In this section, we evaluate CS-UNet by comparing its performance with the results of Stucker *et al.* [10] on their 7 microscopy datasets derived from two materials: nickel-based super-alloys (Super) and Environmental Barrier Coatings (EBC). Super datasets have 3 classes: matrix, secondary, and tertiary. EBC datasets have two classes: the oxide layer and the background (non-oxide) layer. The number of images in each dataset split is shown in Table 4.2. Super-1

and EBC-1 contain the full dataset labeled for their respective materials. Super-2 and EBC-2 have only 4 images in the training set to evaluate the performance of few-shot learning. Super-3 and EBC-3 have only 1 image in the training set to evaluate performance of one-shot learning. Super-4 have test images taken under different imaging and sample conditions to evaluate the performance of out-of-distribution learning. All segmentation models in this study were trained using PyTorch [166].

Table 4.2: Number of Microscopy images in the training, validation, and test set for each experimental dataset

Experiment	# of training images	# of validation images	# of test images
Super-1	10	4	4
Super-2	4	4	4
Super-3	1	4	4
Super-4	4	4	5
EBC-1	18	3	3
EBC-2	4	3	3
EBC-3	1	3	3

EBC and Super datasets were augmented in ways similar to [10], which includes random cropping to 512×512 pixels, random changes to contrast, brightness, and gamma, and added blurring or sharpening. EBC dataset was horizontally flipped and Super dataset was randomly flipped vertically and horizontally and rotated. The training parameters such as the optimizer and the learning rate were adopted from prior research [10]. The training step used the Adam optimizer with an initial learning rate of 2×10^{-4} until the validation accuracy showed no improvement for 30 epochs. Afterwards, the training continued with a learning rate of 10^{-5} until early stopping was triggered after an additional 30 epochs without any validation improvement. Since the datasets are imbalanced, the loss function was set by the weighted sum of Balanced Cross Entropy (BCE) and dice loss with a 70% weighting towards BCE [10].

CS-UNet has a flexible architecture so that it can be trained with different encoders initialized with different pre-trained parameters. Table 4.3 shows the various combinations of pre-trained weights that were used to initialize the CS-UNet encoders. The second column

Table 4.3: The combinations of encoders in CS-UNet, where the second column shows the pre-training models for the Swin-T encoders and the third column shows the pre-training models for the CNN encoders. The last column shows the models for both types of encoders, where ‘Microscopy’ is either MicroNet or MicroLite.

Swin-T architecture	Swin-T pre-training model	CNN pre-training model	CS-UNet pre-training model
Original	ImageNet	ImageNet	ImageNet
Original	MicroLite	MicroNet	Microscopy
Original	ImageNet $\rightarrow$ MicroLite	ImageNet $\rightarrow$ MicroNet	ImageNet $\rightarrow$ Microscopy
Intermediate	MicroLite	MicroNet	Microscopy
Intermediate	ImageNet $\rightarrow$ MicroLite	ImageNet $\rightarrow$ MicroNet	ImageNet $\rightarrow$ Microscopy

shows the pre-trained weights that initialize the Swin-T encoder and the third column shows the pre-trained weights that initialize the CNN encoder. In the last column, we use the term *microscopy* to refer to the fact that the CNN encoders are initialized with weights pre-trained on Stuckner *et al.* [10]’s MicroNet and Swin-T encoders are initialized with weights pre-trained on our MicroLite dataset. Other combinations of pre-trained weights can also be used to train the CS-UNet model. For example, the Swin-T encoder could be initialized with the MicroLite weights and the CNN encoder could be initialized with the ImageNet $\rightarrow$ MicroNet weights. The flexibility of the CS-UNet architecture allows researchers to experiment with different combinations of pre-trained weights to find the best combination for their specific task.

4.5.1 Comparison between CS-UNet and CNN/Transformer-based Algorithms

Table 4.4 compares the top performance of UNet++/UNet pre-trained on MicroNet (from Figure 3–5 of *et al.* [10]), Transformer-based segmentation algorithms (including Swin-Unet, TransDeepLabV3+, and HiFormer) pre-trained on MicroLite, and CS-UNet pre-trained on MicroNet and MicroLite. The highest accuracy for each experiment is shown in bold font. CS-UNet has the best performance in most experiments except EBC-2 and EBC-3. For experiments with ample training data such as Super-1 and EBC-1, the difference between

Table 4.4: Comparison of the top performance (IoU) of UNet++/UNet pre-trained on MicroNet [10], Transformer segmentation models pre-trained on MicroLite, and CS-UNet pre-trained on MicroNet and MicroLite.

Dataset	UNet++/UNet + MicroNet	Transformer + MicroLite	CS-UNet + MicroNet + MicroLite
Super-1	96.4%	95.72%	96.43%
Super-2	94.2%	95.16%	96.06%
Super-3	93.0%	92.23%	93.5%
Super-4	78.5%	78.91%	82.13%
EBC-1	97.6%	96.59%	97.66%
EBC-2	93.3%	91.11%	92.82%
EBC-3	65.9%	82.13%	70.46%

UNet++/UNet, Transformer, and CS-UNet is small. For few-shot learning experiments such as Super-2 and EBC-2, the accuracy gain of CS-UNet is modest. For one-shot learning experiments, the result is mixed, where CS-UNet has modest improvement in Super-3 while significant gain in EBC-3. For out of context learning, CS-UNet shows significant improvement over UNet or Transformer.

Note that while CS-UNet generally outperforms CNN or Transformer-based methods, CS-UNet sometimes is worse than one of them. For example, for EBC-3 (one-shot learning), Transformer-based method is significantly better in IoU accuracy (82.13% vs 70.46%). However, CS-UNet and UNet are slightly better than Transformer for Super-3, which is also one-shot learning. For one-shot and few-shot learning cases, CNN and Transformer have uneven performances. Since CS-UNet combines both types of encoders, it tends to average out the performance.

Overall, this result suggests that CS-UNet has more consistent performance than prior algorithms. CS-UNet is similar or significantly better than UNet++/UNet in all experiments and it is better than Transformers for most experiments. Note that MicroLite is about half the size of MicroNet. Despite this, Transformer + MicroLite has comparable or better performance than that of UNet++/UNet + MicroNet.

4.5.2 Performance Impact of Pre-training Data and Encoder Configurations

Table 4.5: The configuration of top-performing Transformer-based algorithms when pre-trained on ImageNet and MicroLite.

pre-train	ImageNet			MicroLite		
<i>test data</i>	<i>Segmentation Algo.</i>	<i>Swin-T Arch.</i>	<i>IoU</i>	<i>Segmentation Algo.</i>	<i>Swin-T Arch.</i>	<i>IoU</i>
Super-1	TransDeepLabV3+	Original	95.25%	Swin-Unet	Intermediate	95.72%
Super-2	Swin-Unet	Original	94.37%	TransDeepLabV3+	Intermediate	95.16%
Super-3	Swin-Unet	Original	89.78%	TransDeepLabV3+	Intermediate	92.23%
Super-4	Swin-Unet	Original	76.42%	Swin-Unet	Original	78.91%
EBC-1	Swin-Unet	Original	96.11%	TransDeepLabV3+	Intermediate	96.59%
EBC-2	HiFormer	Original	84.21%	TransDeepLabV3+	Original	91.11%
EBC-3	TransDeepLabV3+	Original	65.77%	Swin-Unet	Original	82.13%

The middle column of Table 4.4 gives the best performance of Transformer-based algorithms including Swin-Unet, TransDeepLabV3+, and HiFormer, where their encoders have either the original or the intermediate swin-T architecture and they are pre-trained with either ImageNet or ImageLite dataset. Table 4.5 lists the configurations of the Transformer-based algorithms with the best performance. The results indicate that pre-training on MicroLite consistently provides better performance than pre-training on ImageNet. Also, intermediate architecture of Swin-T may be sufficient if pre-trained on microscopy images, who leads to less training time. While the accuracy gain of pre-training on MicroLite is modest with full training data (Super-1/EBC-1), for few-shot, one-shot, and out-of-distribution learning (Super-2/3/4, EBC-2/3), the performance improvement is more significant.

Table 4.6: The best performing encoder configurations of CS-UNet when pre-trained on ImageNet and microscopy images.

pre-train	ImageNet			MicroNet + MicroLite		
<i>test data</i>	<i>CNN Encoder</i>	<i>Swin-T Arch.</i>	<i>IoU</i>	<i>CNN Encoder</i>	<i>Swin-T Arch.</i>	<i>IoU</i>
Super-1	ResNext50_32x4d	Original	96.22%	InceptionV4	Intermediate	96.43%
Super-2	VGG16-bn	Original	96.03%	VGG16-bn	Intermediate	96.06%
Super-3	EfficientNet-b3	Original	87.01%	EfficientNet-b4	Intermediate	93.5%
Super-4	EfficientNet-b2	Original	78.89%	EfficientNet-b3	Original	82.13%
EBC-1	SE_ResNeXt50_32x4d	Original	98.0%	SE_Resnet152	Original	97.66%
EBC-2	SE_ResNet-101	Original	92.0%	SE_ResNet-101	Intermediate	92.82%
EBC-3	SE_ResNet-50	Original	61.71%	SE_ResNeXt-101_32x4d	Original	70.46%

The right column of Table 4.4 gives the best performance of CS-UNet when pre-trained

on ImageNet and microscopy images. In Table 4.6, we include the CNN and Transformer encoder combinations that have the best performance for each test when pre-trained with ImageNet and microscopy images. Similar to the Transformer-based algorithms, pre-training with microscopy images provides the best performance in most cases except EBC-1, which is the full training set. There is significant performance improvement for one-shot and out-of-distribution learning (Super-3/4 and EBC-3). This is consistent with the findings of Stuckner *et al.* [10]. However, the performance gain of few-shot learning is not as noticeable (Super-2 and EBC-2) since the performance of CS-UNet is less sensitive to pre-training data.

In Appendix E, we included some examples of segmentation results for the Super and EBC datasets to provide a visual comparison of CS-UNet when pre-trained on microscopy over natural images.

4.6 Evaluation of CS-UNet on Medical Images

While the initial application of CS-UNet is materials science images, we believe that CS-UNet can also improve the segmentation performance of medical images. Furthermore, CS-UNet pre-trained on MicroLite should have better performance for medical images such as computed tomography (CT) images since MicroLite also contains X-Ray images. To this end, we evaluated the performance of CS-UNet pre-trained on microscopy images and ImageNet on a medical image dataset called the Synapse multi-organ segmentation dataset (Synapse) [167].

Synapse dataset includes 30 patient cases with 3779 axial abdominal clinical CT images, where 18 cases are used for training and 12 cases are used for testing. The dataset contains 8 abdominal organs (aorta, gallbladder, left kidney, right kidney, liver, pancreas, spleen, and stomach). Each CT volume includes $85 \sim 198$ slices of 512×512 pixel images, with a voxel spatial resolution of $[0.54 \sim 0.54] \times [0.98 \sim 0.98] \times [2.5 \sim 5.0] \text{ mm}^3$.

Each Synapse image is reduced to 224×224 pixels. It should be noted that we follow [59]

Table 4.7: Comparison results of CS-UNet method on the Synapse dataset. Blue indicates the best result, and red displays the second-best. Where $CS-UNet^1$ (VGG16_bn, Intermediate: Swin-T), pre-trained based on MicroNet and MicroLite models, and $CS-UNet^2$ (VGG16_bn, Swin-T), pre-trained based on ImageNet model.

Methods	DSC $\uparrow$	HD $\downarrow$	Aorta	Gallbladder	Kid(L)	Kid(R)	Liver	Pancreas	Spleen	Stomach
U-Net [29]	76.85	39.70	89.07	69.72	77.77	68.60	93.43	53.98	86.67	75.58
Att-UNet [168]	77.77	36.02	89.55	68.88	77.98	71.11	93.57	58.04	87.30	75.75
TransUNet [69]	77.48	31.69	87.23	63.13	81.87	77.02	94.08	55.86	85.08	75.62
Swin-UNet [59]	79.13	21.55	85.47	66.53	83.2	79.61	94.29	56.58	90.66	76.60
TransDeepLab [77]	80.16	21.25	86.04	69.16	84.08	79.88	93.53	61.19	89.00	78.40
HiFormer [76]	80.39	14.70	86.21	65.69	85.23	79.77	94.61	59.52	90.99	81.08
CS-UNet ¹	84.20	9.89	88.42	72.86	86.84	85.59	95.44	70.51	90.11	83.81
CS-UNet ²	83.27	15.26	88.07	71.32	88.0	84.38	94.80	65.64	89.95	83.49

to choose the optimizer and learning rate for Synapse dataset. The training batch size and learning rate are 24 and 0.05, respectively. Our model is trained with the SGD optimizer with a momentum of 0.9 and weight decay of 0.0001. The average Dice-Similarity coefficient (DSC) and average 95% Hausdorff Distance (HD) are used as evaluation metrics. HD metric provides a more precise estimate of performance with respect to boundary errors. DSC values range from 0 to 1 with the larger values indicating better performance while the smaller values of HD indicate better performance.

We compared the performance of CS-UNet with 6 State-Of-The-Art (SOTA) algorithms including 2 CNN-based, 2 Transformer-based, and 2 hybrid algorithms, which are U-Net, Att-UNet, TransUNet, Swin-UNet, Transdeeplab, and HiFormer. Note that all these models are initialized with the weights pre-trained on ImageNet. Table 5.2 shows the performance of CS-UNet and SOTA methods in terms of the average DSC and average 95% HD on 8 abdominal organs. The segmentation of 2 sample images of Synapse is shown in Figure 4.6, which illustrates the better accuracy of CS-UNet when pre-trained on microscopy than on natural images.

Our results highlight the remarkable performance of CS-UNet that, when pre-trained on microscopy images, has the highest segmentation accuracy at 84.20% in average DSC and the lowest average HD at 9.89 mm. Even when pre-trained on ImageNet, CS-UNet has the second best accuracy at 83.27% in average DSC though its average HD at 15.26 mm is slightly higher than the 14.70 mm of the second best algorithm HiFormer. Lower HD

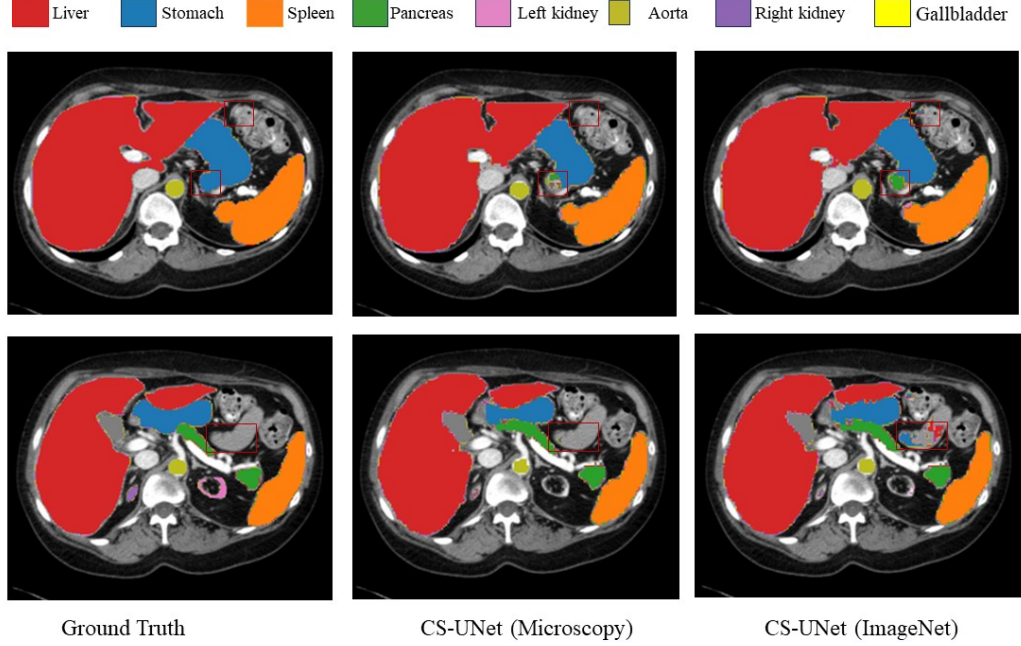


Figure 4.6: This figure compares the ground truth of 2 Synapse images (left column) with the segmentation masks of CS-UNet pre-trained on microscopy images (middle column) and CS-UNet pre-trained on ImageNet (right column). The red rectangles identify the regions where CS-UNet pre-trained on ImageNet tends to have over-segmentation problems compared to CS-UNet pre-trained on microscopy images.

Table 4.8: Comparison of CS-UNet and State-Of-The-Art algorithms based on model parameters, number of epochs to train, inference time for 1568 axial abdominal clinical CT images, and model size.

Algorithm	# of params (M)	# of training epochs	inference time (min)	model size (MB)
TransUnet [69]	105.28	150	31:09	414.412
Swin-Unet [59]	27.17	150	30:25	108.058
HiFormer [76]	25.51	400	30:56	101.161
TransDeepLab [77]	21.14	200	30:59	86.343
CS-UNet (CNN: VGG16_bn) ¹	44.96	150	30:32	177.613

indicates superior edge prediction capabilities. For individual organs, with the exception of Aorta, CS-UNet has the best and/or the second best segmentation accuracy in DSC. Even for Aorta, the accuracy of CS-UNet is also very close to the best performer with about 1% difference in accuracy. Figure 4.6 illustrates the visual differences between the segmentation ground truth of synapse images with the segmentation masks made by CS-UNet, when it is pre-trained on microscopy images and on ImageNet. The masks made by CS-UNet pre-trained ImageNet tends to have more over-segmentation problems.

¹CS-UNet is a flexible method that can use different CNN encoders. VGG16.bn is chosen here since it provided the best performance for Synapse. Appendix D shows the number of parameters of CS-UNet with

Compared to the SOTA methods, CS-UNet has larger model size than some of the models though the inference time is on the lower end. As shown in Table 5.5, our method has 44.96 million parameters, which is more than HiFormer, Swin-Unet, and Transdeeplab methods but fewer than TransUnet. Additionally, CS-UNet shows a computation time score of 30:23 minutes during the testing phase. Despite having a relatively higher number of parameters, CS-UNet exhibits less computational time compared to HiFormer, Transdeeplab, and TransUnet. Importantly, it consistently outperforms them in terms of overall performance.

4.7 Ablation study

CS-UNet combines CNN and Transformer encoders with skip connections to capture local and global features. To explore the influence of different factors on the model performance, we conducted ablation studies on the Synapse dataset to evaluate how the number of skip connections and how the CNN and Transformer encoder branches influence the performance. We use the weights pre-trained on ImageNet to initialize both the encoders and the decoder of CS-UNet.

4.7.1 Effect of the encoders in CS-UNet

As shown in Table 4.9, we separate the CNN and Transformer encoder branches. After removing the CNN branch, the mean DSC drops by about 2.5% and HD increases by about 4.48 mm, while DCS drops by about 7.6% and HD increases by about 11.74 mm if we remove the Transformer branch. It shows that the combination of CNN and Swin Transformer in CS-UNet can fully utilize the advantages of both types of models and achieve better segmentation performance than either of them alone.

different CNN encoders.

Table 4.9: Ablation study of CS-UNet encoders for Synapse dataset, separate the CNN branch and Transformer branch

Enoder	DSC$\uparrow$	HD$\downarrow$
CS-UNet (Orginal)	83.27	15.26
CS-UNet without CNN branch	80.68	19.74
CS-UNet without Transformer branch	75.61	27.00

4.7.2 Effect of the number of skip connections

The skip connections in CS-UNet connect the encoders and the decoders at 3 different stages of scale resolutions. We investigated the impact of varying the number of skip connections (0, 1, 2, and 3) on the segmentation performance of CS-UNet. Table 4.10 demonstrates that the higher number of skip connections enhances the segmentation performance of CS-UNet. Consequently, for better performance, we set the number of skip connections to 3 in this study.

Table 4.10: Ablation study on the impact of the number of skip connections.

Skip Connection	DSC $\uparrow$	HD $\downarrow$	Aorta	Gallbladder	Kid(L)	Kid(R)	Liver	Pancreas	Spleen	Stomach
0	75.81	23.72	79.801	62.24	81.57	77.66	93.70	50.83	85.00	75.64
1	78.84	18.31	84.07	62.69	84.93	80.24	94.02	58.46	86.91	79.37
2	82.38	15.72	88.06	70.17	86.89	83.86	95.01	64.52	88.34	81.64
3	83.27	15.26	88.07	71.32	88.0	84.38	94.800	65.64	89.95	83.49

4.8 Conclusion

This chapter introduced a novel U-shaped segmentation algorithm CS-UNet, which combines CNN and Transformer encoders in parallel. The encoders of CS-UNet extract low-level and high-level features from input images, which enables effective segmentation of images with long-range dependencies and high spatial resolution. Our tests on materials images showed that CS-UNet has better or comparable performance than prior state-of-the-art methods. Our tests also showed that the encoders pre-trained on microscopy images leads to better

feature representation and thus better segmentation performance than the encoders pre-trained on natural images. We further evaluated CS-UNet on a set of medical CT images, which demonstrated clear advantages over the prior state-of-the-art methods. Pre-training on microscopy images also improved the segmentation performance on the medical images due to the inclusion of X-ray images in the pre-training dataset.

While CS-UNet can preserve spatial information and process long-range dependencies, it is computationally intensive compared to CNN and Transformer-based algorithms. As future work, we will explore the training of other Transformer architectures, such as Focal Transformer and FocalNet, on large microscopy datasets. These architectures may offer further advancements in image segmentation, expanding the range of available options and potential improvements in materials analysis tasks. Future investigations may also offer deeper insights into the decision-making processes of image segmentation. As highlighted in Section 4.3, the techniques such as DNC [149], ClusterFormer [151], and CLUSTSEG [152] hold promise for achieving greater transparency and interpretability in these models.

Chapter 5

GCtx-UNet: Efficient Deep Network for Medical Image Segmentation

5.1 Abstract

Medical image segmentation is crucial for disease diagnosis and monitoring. Though effective, the current segmentation networks such as UNet struggle with capturing long-range features. More accurate models such as TransUNet, Swin-UNet, and CS-UNet have higher computation complexity. To address this problem, we propose GCtx-UNet, a lightweight segmentation architecture that can capture global and local image features with accuracy better or comparable to the state-of-the-art approaches. GCtx-UNet uses vision transformer that leverages global context self-attention modules joined with local self-attention to model long and short range spatial dependencies. GCtx-UNet is evaluated on the Synapse multi-organ abdominal CT dataset, the ACDC cardiac MRI dataset, and several polyp segmentation datasets. In terms of Dice Similarity Coefficient (DSC) and Hausdorff Distance (HD) metrics, GCtx-UNet outperformed CNN-based and Transformer-based approaches, with notable gains in the segmentation of complex and small anatomical structures. Moreover, GCtx-UNet is much more efficient than the state-of-the-art approaches with smaller model size, lower computation workload, and faster training and inference speed, making it a practical choice for clinical applications.

5.2 Introduction

Automated medical image segmentation is critical in providing valuable information for the prevention, diagnosis, progression monitoring, and prognosis of various diseases, as well as quantitative pathology assessment. The U-shaped deep-neural networks, which include encoders, decoders, and skip connections, are now the most widely used methods for medical image segmentation. Although the U-shaped networks have achieved state-of-the-art performance in numerous medical image segmentation tasks, it still has limitations. One primary limitation is the encoders’ ability to effectively extract and integrate long-range and local features. Methods based on Convolutional Neural Networks (CNNs) such as UNet [29] and UNet++ [169] excel at capturing local features, but they struggle to model long-range dependencies within data. While Transformer-based methods such as Swin-UNet [59] can model long-range pixel relations, they lack spatial induction bias in modeling local information, which leads to unsatisfactory results.

Past research explored CNN-Transformer hybrid architectures such as TransUnet [69] to capture global and local information but these models often significantly increase the number of parameters. This, in turn, translates to higher computational complexity, potentially limiting their practical applications. Recently, Hatamizadeh et al. [9] proposed a Global Context Vision Transformer (GC-ViT) that leverages global context self-attention modules and is joined with local self-attention to effectively and efficiently model both long and short-range spatial interactions. GC-ViT achieved state-of-the-art results across image classification, object detection, and semantic segmentation tasks.

In this paper, we introduce GCtx-UNet, a UNet-like segmentation network designed for medical image segmentation. GCtx-UNet effectively captures both long and short-range semantic features using GC-ViT [9] encoders and decoders with skip connections. This architecture enhances performance while requiring fewer model parameters, with higher inference speed, and lower computational complexity.

We evaluated the segmentation and runtime performance of GCtx-UNet on several medical image datasets including Synapse, ACDC, and several polyp image datasets. Our experimental results show that GCtx-UNet has better or comparable performance than the state-of-the-art segmentation algorithms including CNN-based, Transformer-based, and hybrid segmentation networks. Also, GCtx-UNet has the smallest model size and uses the least amount of training time and inference time. In addition, we pre-trained GCtx-UNet on both ImageNet and MedNet, which is a set of 200,000 medical images collected from public sources. GCtx-UNet pre-trained on in-domain images (i.e. MedNet) yielded better accuracy than GCtx-UNet pre-trained on natural images (i.e. ImageNet).

The source code for the model is available at GitHub.¹

5.3 Related work

The CNN-based methods are widely used and regarded as one of the most prominent approaches for medical image segmentation. Convolutional neural networks (CNNs), particularly encoder-decoder based architectures like UNet [29] and its derivatives, have shown exceptional efficacy in medical image segmentation. For instance, Att UNet [168] enhanced segmentation through attention gates while UNet++ [169] introduced an alternative skip connection mechanism, nested and dense, alleviating the semantic gap between levels of UNet to a certain degree. This modification yields notable performance improvements compared to UNet. However, UNet++ cannot capture semantic features at full scale. Huang et al. [170] proposed UNet3+ to maximize the use of full-scale feature maps by combining low-level details from various scales with high-level semantics. CNN-based methods have found application in diverse medical image segmentation tasks, such as retinal image segmentation [171] and skin segmentation [172], showcasing promising performance and practicality in implementation and training. Segmentation algorithms based on ResNet architecture have established its presence in medical image segmentation [173]. For example, Res-UNet [174]

¹<https://github.com/Kalrfou/GCtx-UNet>

enhanced retinal vessel segmentation with a weighted attention mechanism.

The self-attention mechanism (MSA) inherent in Transformers empowers them to perform global correlation modeling, enabling them to handle long-range dependencies effectively. Leveraging this capability, Transformers have made significant strides in both natural language processing and computer vision tasks due to their superior global modeling abilities. Several pioneering studies have introduced Transformer-based architectures for medical image segmentation. Cao et al. [59] presented Swin-UNet, integrating a Swin Transformer [4] into a U-shaped segmentation network for multi-organ segmentation. Azad et al. [77] proposed TransDeepLab for skin lesion segmentation, enhancing DeepLab with diverse window strategies. Additionally, Huang et al., [73] introduced MISSFormer to leverage global information across different scales for cardiac segmentation, while Azad et al. [175] introduced TransCeption, refining the patch merging module to capture multi-scale representations within a single stage. Combining convolution operations with a Transformer on the encoder side, Transclaw UNet [176] enables detailed segmentation and long-distance relationship learning. UNETR [74] adopts sequence-to-sequence prediction for 3D medical image segmentation. These developments underscore the transformative impact of Transformer-based approaches on medical image segmentation, charting a path towards broader adoption and deep learning advancement.

Swin Transformer [4] introduced local-window-self-attention to reduce the cost so that it grows linearly with the image size, used shifted-window-attention to capture cross-window information, and exploited multi-resolution information with hierarchical architecture. However, the shifted-window-attention struggles to capture long-range information due to small coverage area of shifted-window-attention and lacks inductive bias like ViT [3].

GC-ViT [9] is a hierarchical architecture like Swin Transformer but utilizes global-window attention instead of shifted-window attention for effectively capturing long-range information. GC-ViT also uses convolution layers for downsampling to provide the network with desirable properties such as locality bias and cross-channel interactions which are missing in

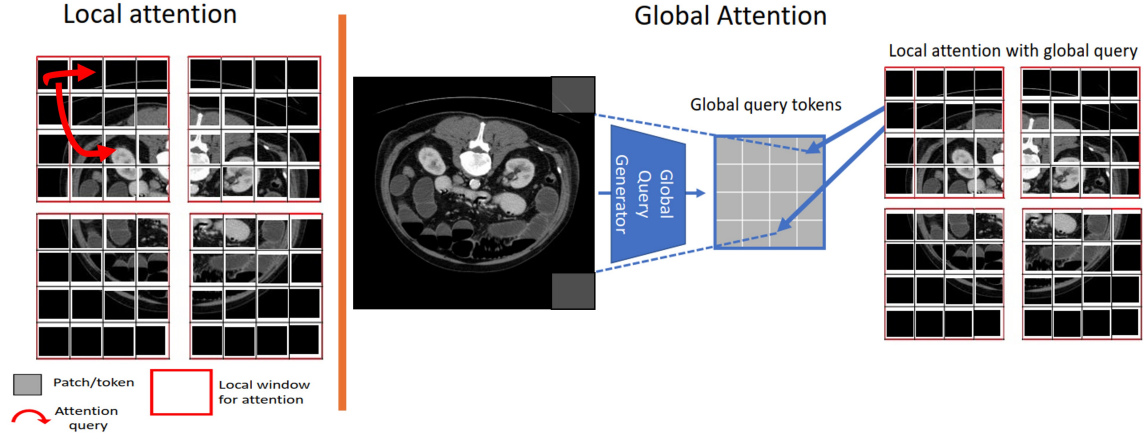


Figure 5.1: An illustration of the local and global attention mechanisms in GC-ViT [9]. Local attention is computed on feature patches within local window only (left). The global attention mechanism extracts query patches from the entire input feature map, aggregating information from all windows. The global query is interacted with local key and value tokens, hence allowing to capture long-range information.

both ViT and Swin Transformer. GC-ViT has 4 stages, each of which consists of alternating blocks of local and global Multi-head Self-Attention (MSA) layers. As shown in Figure 5.1, at each stage, global query tokens are computed by using novel fused inverted residual blocks that encompass global contextual information from different image regions. While the local self-attention modules are responsible for modeling short-range information, the global query tokens are shared across all global self-attention modules to interact with local key and value representations.

5.4 Pre-training on MedNet dataset

The majority of CNN and Transformer-based segmentation models are pre-trained on natural images such as ImageNet. However, this is suboptimal for medical image segmentation due to the semantic gap between natural and medical image modalities [177, 178]. In this work, we pre-trained the GC-ViT [9] model, specifically GCVit xxTiny, on a large medical image dataset called *MedNet* that contains more than 200,000 medical images collected from several public datasets [179] and Kaggle [180, 181, 182].

MedNet consists of different types of microscopy images such as X-ray, computed to-

mography (CT), optical coherence tomography (OCT), and MRI. Images in MedNet are divided into 65 classes. Similar to the approach of Stuckner *et al.* [10] and Alrfou *et al.* [178], the MedNet dataset is divided into training and validation sets, with each class having 100 images in the validation set, resulting in 96.75%/3.25% training/validation split. Using 100 images per class for validation is sufficient to obtain reliable accuracy metrics and to prevent overfitting during training. Although the validation sets are balanced, the training sets exhibited some class imbalance. There are a few classes, each of which contains less than 0.12% of the total images. Three classes contain 6.2% of the images. Most classes have over 2000 images representing one to two percent of the training set. MedNet includes images from various modalities such as X-ray, CT, OCT, and MRI, and encompassed a wide range of medical diseases such as Kidney Cancer, Cervical Cancer, Alzheimer’s, Covid-19, Pneumonia, Tuberculosis, Monkeypox, Breast Cancer, and Malaria.

We trained and tested GC-ViT xxTiny with the AdamW optimizer [183] for 100 epochs with an initial learning rate of 0.0001, weight decay of 0.05, and cosine decay scheduler. The training data had been augmented using the albumentations library, which included random changes to the contrast and brightness, vertical and horizontal flips, photometric distortions, and added noise.

The training process continued until the validation score showed no improvement, employing an early stopping criterion with a patience of 10 epochs. Performance was evaluated using top-1 and top-5 accuracy metrics. Top-1 accuracy measures the percentage of test samples for which the correct label is the top prediction, while top-5 accuracy measures the percentage of test samples for which the correct label appears within the top five predictions. The top-1 accuracy of the GC-ViT xx-Tiny model is 82.3%, and the top-5 accuracy is 98.2%.

5.5 GCtx-UNet Architecture

The core component of GCtx-UNet is GC-ViT block, the local and global attention mechanisms of which are illustrated in Figure 5.1. As shown in Figure 5.2, Each GC-ViT block includes a local and global Multi-head Self-Attention (MSA), Multilayer Perceptron (MLP), a Global Token Generator (GTG) and a downsampling layer. The GTG component adds global context to the computations. Local MSA can only query patches within a local window, while global MSA can query different image regions while still operating within the window. At each stage, the global query component is pre-computed. The block also introduces a CNN-based module in the downsampling layer to include inductive bias, a useful feature for images that have been missing in both ViT and Swin Transformer.

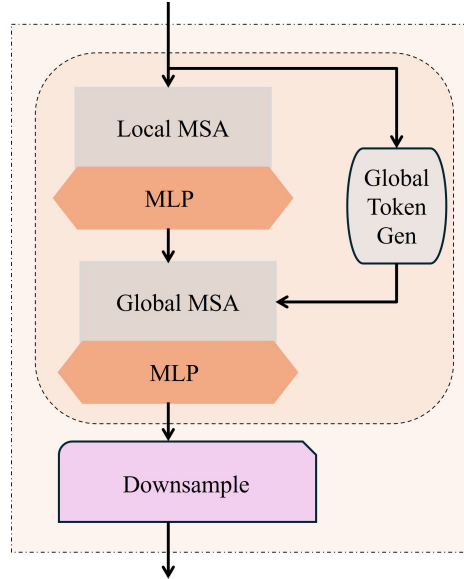


Figure 5.2: A GC-ViT block has a local and global attention, a global token generator, and a downsampling layer.

GCtx-UNet is a GC-ViT-based U-shaped Encoder-Decoder architecture with skip-connections for long and short-range semantic feature learning. As shown in Figure 5.3, the GCtx-Unet consists of encoder, bottleneck, decoder, and skip connections.

1. Both encoder and decoder used GC-ViT [9] to model long and short-range spatial

interactions, without the need for expensive operations such as computing attention masks or shifting local windows.

2. At each stage, the GC-ViT encoder and decoder consist of alternating local and global self-attention modules to extract spatial features. Both operate in local windows like Swin Transformer.
3. Skip connections concatenate the feature maps from the GC-ViT encoder with the corresponding decoder stages. The bottleneck is employed to acquire the deep feature representation, maintaining both feature dimension and resolution unchanged within this component.
4. The downsampler between stages in the encoder part and upsampler between stages in the decoder part provide desirable properties such as inductive bias and modeling of inter-channel dependencies.

Within the encoder, the initial image is partitioned into four patch blocks, which act as input for the four-stage GC-ViT module. Following the encoding process, the image dimensions are decreased to $(H/32) \times (W/32)$. In the decoder, the upsample operations are utilized to increase the image dimensions by 2 and reduce the number of channels by 2. The features from each stage of the encoder are concatenated with their corresponding stage in the decoder using skip connections. The decoder accomplishes its task through upsampling.

5.5.1 Encoder

The encoder utilizes a hierarchical GC-ViT approach to acquire feature representations at various resolutions. This is achieved by reducing spatial dimensions while simultaneously increasing embedding dimensions by a factor of 2 across 4 stages. Initially, the input image $x \in \mathbb{R}^{H \times W \times 3}$ undergoes processing through the patchify layer. This layer comprises a 3×3 convolution operation with a stride of 2, along with padding, to generate overlapping patches.

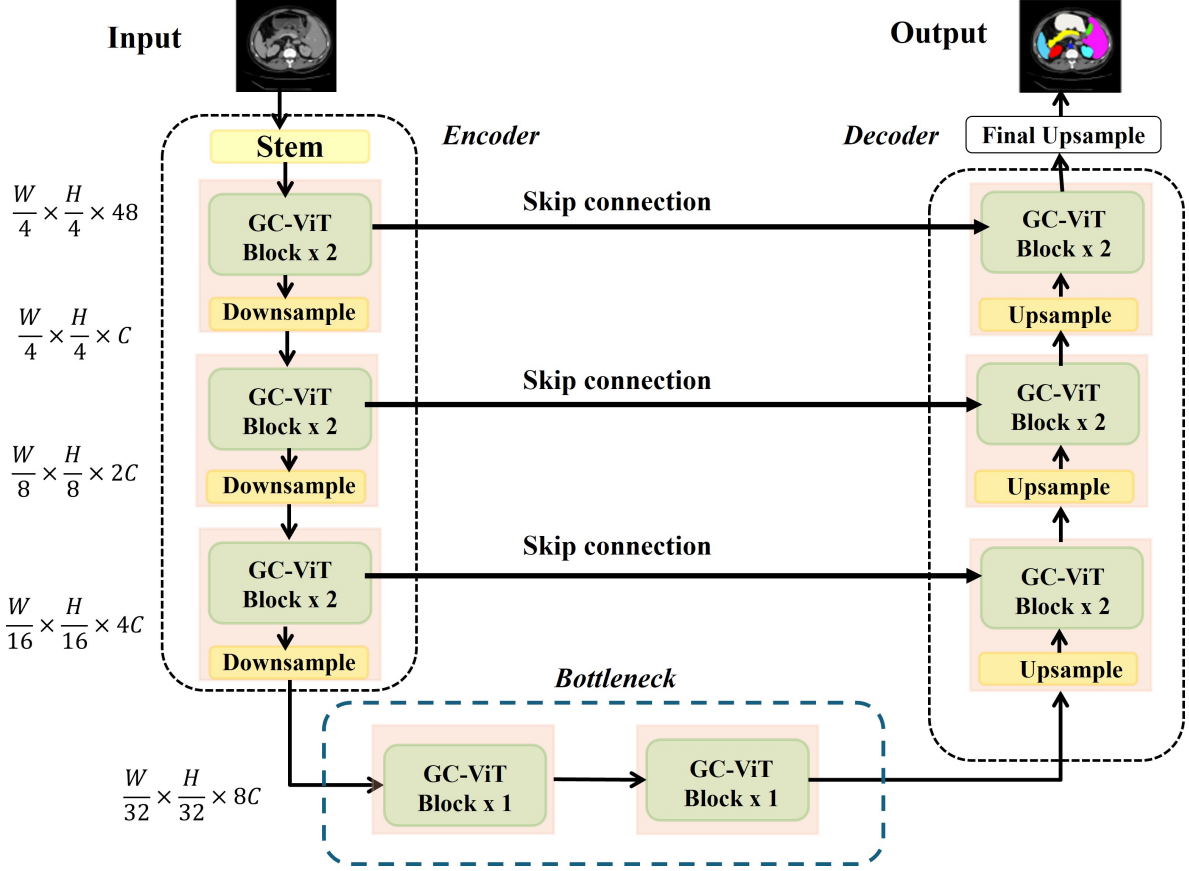


Figure 5.3: GCTx-UNet architecture includes encoders, bottlenecks, skip connections, and decoder. Encoder, bottleneck and decoder are all constructed based on GC-ViT block

Subsequently, these patches are projected into an embedding space of dimension C via another 3×3 convolution layer. After each stage in the GC-ViT backbone, the spatial resolution is decreased while the number of channels is increased through a downsampling layer. This downsampling operation helps in extracting hierarchical features at different resolutions.

5.5.2 Downsampler

In the downsampler, we incorporate the Fused-MBConv module to generate hierarchical representations by injecting inductive bias into the network and modeling inter-channel correlations. Then, the convolution layer with a kernel size of 3 and a stride of 2 was used to downsample the spatial feature resolution by 2 while doubling the number of channels. The

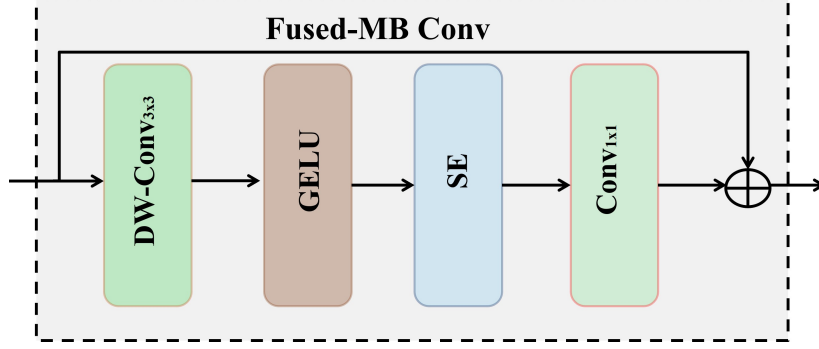


Figure 5.4: Fused-MBConv module

Fused-MBConv, as shown in Figure 5.4, includes $\text{DW-Conv}_{3 \times 3}$, GELU, SE, and $\text{Conv}_{1 \times 1}$. The Fused-MBConv operation can be defined by the following equations.

$$\hat{x} = \text{DW-Conv}_{3 \times 3}(x),$$

$$\hat{x} = \text{GELU}(\hat{x}),$$

$$\hat{x} = \text{SE}(\hat{x}),$$

$$x = \text{Conv}_{1 \times 1}(\hat{x}) + x.$$

where DW-Conv refers to Depthwise-CNN, SE refers to the Squeeze and Excitation block, and GELU represents the Gaussian Error Linear Unit function.

5.5.3 Bottleneck

Similar to Swin-UNet [59], two GC-ViT blocks are used for bottleneck construction. The bottleneck is strategically designed to facilitate the learning of deep feature representations. Within this structure, the feature dimension and resolution remain unchanged.

5.5.4 Decoder

The symmetric decoder, corresponding to the encoder, is constructed using the GC-ViT Transformer block. The decoder mirrors the encoder design, replacing the patchy block with

an unpatched block, the embedding layer with a de-embedding layer, and the downsample block with an upsample block. The decoder’s upsample block replaces the encoder’s downsample block. The upsample block restructures the feature map of adjacent dimensions into a higher-resolution feature map and reduces the feature dimension by half. This upsample block effectively increases the spatial resolution while refining and normalizing feature representations, making it suitable for decoding and reconstructing higher-resolution features in segmentation models. The skip connection fuses the features of the encoder with the deep features recovered from the up-sample, therefore mitigating the loss of spatial data produced by the downsampling.

Table 5.1: List of the methods that used in our experiments.

Architecture	Method
CNN	U-Net [29]
	Att-UNet [168]
	R50-UNet [69]
	R50-AttUNet [69]
	UNet++ [60]
	PraNet [184]
Transformer	Swin-UNet [59]
	TransDeepLab [77]
	MISSForme [73]
Hybrid CNN and Transformer	TransUNet[69]
	HiFormer [76]
	R50-ViT [3]
	CS-UNet [178]
	GPA-TUNet [185]

5.6 Results

We compared the performance of GCtx-Unet with different State-Of-The-Art (SOTA) algorithms including CNN-based,Transformer-based, and hybrid algorithms. Table 5.1 presents the list of methods that used in our experiments.

5.7 Experimental Evaluation

We compared the performance of GCtx-Unet with state-of-the-art algorithms including CNN-based, Transformer-based, and hybrid algorithms. Table 5.1 summarizes the list of methods that used in our experiments. We evaluated the performance of GCtx-Unet pre-trained on MedNet and ImageNet on three different types of medical image datasets, including the Synapse multi-organ segmentation dataset (Synapse), Automated Cardiac Diagnosis Challenge (ACDC) [186], and the Polyp datasets which includes CVC-ClinicDB [187], Kvasir-SEG [188], EndoScene [189], ColonDB [190], and ETIS-LaribDB [191].

Synapse Synapse multi-organ segmentation dataset (Synapse) includes 30 patient cases with 3779 axial abdominal clinical CT images, where 18 cases are used for training and 12 cases are used for testing. The dataset contains 8 abdominal organs (aorta, gallbladder, left kidney, right kidney, liver, pancreas, spleen, and stomach). Each CT volume includes $85 \sim 198$ slices of 512×512 pixel images, with a voxel spatial resolution of $[0.54 \sim 0.54] \times [0.98 \sim 0.98] \times [2.5 \sim 5.0] \text{ mm}^3$.

ACDC Automated Cardiac Diagnosis Challenge dataset (ACDC) [186] compiles MRI scan results of various patients from the MICCAI 2017 dataset. The ACDC dataset contains 100 cardiac MRI scans, each containing three organs: the right ventricle (RV), the myocardium (Myo), and the left ventricle (LV). Following TransUNet [69], we partitioned the dataset into 70 training cases, 10 validation cases, and 20 test cases.

Polyp We used 5 polyp datasets with early colorectal cancer diagnosis images. The CVC-ClinicDB [187] and Kvasir-SEG [188] datasets are used for binary segmentation. The CVC-ClinicDB dataset contains 612 RGB colonoscopy images with labeled polyps from MICCAI 2015 with a pixel resolution of 288×384 . The Kvasir-SEG dataset contains 1000 polyp images with a pixel resolution ranging from 332×487 to 1920×1072 and their corresponding ground truth. Following the setting in PraNet [184], we used 900

images from the CVC-ClinicDB dataset and 548 images from the Kvasir dataset for training. The remaining 64 images from CVC-ClinicDB and 100 images from Kvasir were used as test sets. To evaluate the generalization performance, we tested the model on three unseen datasets: CVC-300, CVC-ColonDB, and ETIS-LaribDB.

5.7.1 Implementation

The GCtx-UNet is implemented with PyTorch library and the model is trained on an Nvidia GeForce GTX TITAN X with 12 GB of memory. The input image size was reduced to 224×224 pixels. Our model is trained with batch size 24, learning rate 0.0001, and AdamW optimizer with momentum 0.9 and weight decay 0.0001. For evaluation metrics, we used the average Dice-Similarity Coefficient (DSC) across the Synapse, ADCD, and Polyp datasets. Additionally, the average 95% Hausdorff Distance (HD) was used on the Synapse dataset. HD metric provides a more precise estimate of performance with respect to boundary errors. DSC values range from 0 to 1 with the larger values indicating better performance while the smaller values of HD indicate better performance.

5.7.2 Synapse

We compare GCtx-UNet with current state-of-the-art methods on the Synapse dataset, including U-Net, Att-UNet, TransUnet, SwinUnet, MISSFormer, TransDeepLab, HiFormer, GPA-TUNet, and CS-UNet. Table 5.2 summarizes the performance comparison in DSC and HD.

GCtx-UNet pre-trained on MedNet has the second-best average DSC (82.39%) and the third-best average HD (15.94 mm). When pre-trained on ImageNet, GCtx-UNet is the fourth-best in both average DSC (81.95%) and average HD (16.8 mm). Note that these metrics are not too far from the best average DSC (83.27%) and average HD (14.7 mm), which are from CS-UNet and HiFormer, which have much higher computation complexity. Also note that GCtx-UNet significantly outperforms CNN-based methods UNet and Att-

UNet. For instance, UNet has an average DSC of 76.85% and an average HD of 39.70mm and Att-UNet has an average DSC of 77.77% and an average HD of 36.02mm. GCtx-UNet has better performance than Transformer-based methods like Transdeeplab, MISSFormer, and Swin-UNet, which has an average DSC ranging from 79.13 to 81.96% and an average HD ranging from 18.20 to 21.25 mm. GCtx-UNet’s performance is also competitive with the hybrid models like TransUNet, GPA-TUNet, HiFormer, and CS-UNet, which have average DSC ranging from 77.48 to 83.27% and average HD ranging from 14.70 to 31.69 mm.

Table 5.2: Comparison of GCtx-UNet and state-of-the-art algorithms on Synapse (the columns are average DSC in %, average HD in mm, and DSC in % for each organ). Blue indicates the best result and red displays the second-best. GCtx-UNet¹ is pre-trained on ImageNet while GCtx-UNet² is pre-trained on MedNet. All other models are pre-trained on ImageNet.

Algorithm	DSC↑	HD↓	Aorta	Gallbladder	Kid(L)	Kid(R)	Liver	Pancreas	Spleen	Stomach
U-Net [29]	76.85	39.70	89.07	69.72	77.77	68.60	93.43	53.98	86.67	75.58
Att-UNet [168]	77.77	36.02	89.55	68.88	77.98	71.11	93.57	58.04	87.30	75.75
Swin-UNet [59]	79.13	21.55	85.47	66.53	83.2	79.61	94.29	56.58	90.66	76.60
TransDeepLab [77]	80.16	21.25	86.04	69.16	84.08	79.88	93.53	61.19	89.00	78.40
MISSFormer [73]	81.96	18.20	86.99	68.65	85.21	82.00	94.41	65.67	91.92	80.81
TransUNet [69]	77.48	31.69	87.23	63.13	81.87	77.02	94.08	55.86	85.08	75.62
GPA-TUNet [185]	80.37	20.55	88.74	65.63	83.51	80.37	94.84	63.89	87.58	78.40
HiFormer [76]	80.39	14.70	86.21	65.69	85.23	79.77	94.61	59.52	90.99	81.08
CS-UNet [178]	83.27	15.26	88.07	71.32	88.00	84.38	94.80	65.64	89.95	83.81
GCtx-UNet ¹	81.95	16.80	86.96	66.26	87.75	83.86	94.53	61.06	91.42	83.74
GCtx-UNet ²	82.39	15.94	86.30	69.32	86.11	81.89	94.64	64.88	91.81	84.15

Organ-wise performance Figure 5.5 shows qualitative comparison on two Synapse images, where GCtx-UNet has superior performance than Swin-UNet and CS-UNet. GCtx-UNet segmented most of the organs correctly, with a few misclassifications in the Gallbladder area. In comparison, Swin-UNet over-segmented the spleen (some areas belonging to the spleen were misclassified as the left kidney) and CS-UNet over-segmented the pancreas. We speculate that this improvement is due to the use of GC-ViT, which introduces a parameter-efficient downsampling module with modified Fused MB-Conv blocks. These modifications address the lack of inductive bias in ViTs, enabling GCtx-UNet to accurately capture relatively large regions and perform well with organs close to each other.

The segmentation of larger organs with clear boundaries such as kidney, pancreas, and

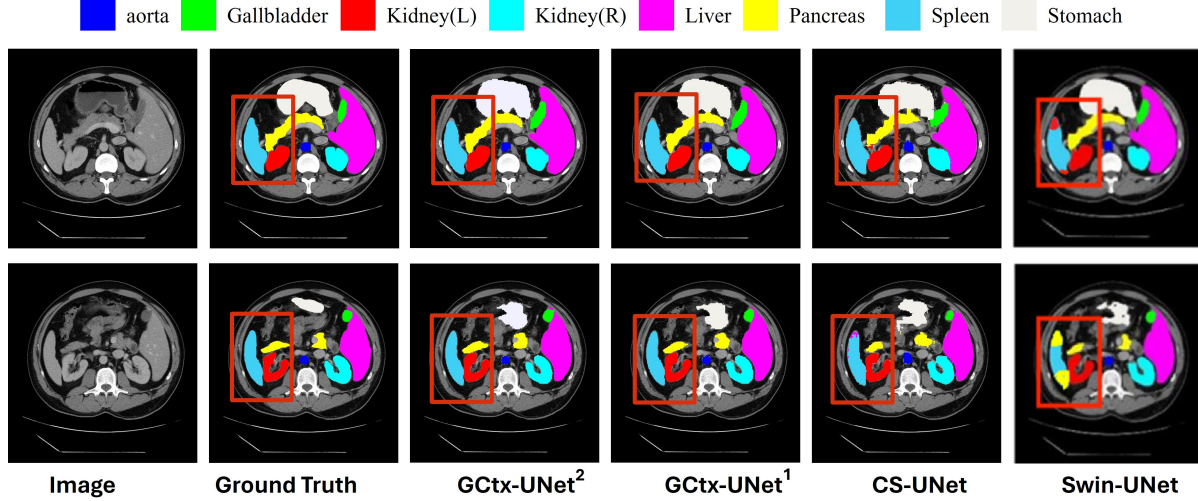


Figure 5.5: Comparison of GCtx-UNet with ground truth, CS-UNet, and Swin-UNet on two sample images in Synapse dataset. Note that GCtx-UNet¹ is pre-trained on ImageNet and GCtx-UNet² is pre-trained on MedNet. The red rectangles identify the regions where Swin-UNet tends to have over-segmentation problems compared to GCtx-UNet and CS-UNet.

spleen require the network to capture global features. We speculate that this is why Transformer-based models are more accurate compared to CNN-based models. The segmentation of smaller organ like aorta benefits more from the detection of local features. This is probably why CNN-based models have more accurate results than Transformed-based models. The segmentation of larger organs with complex boundaries, such as liver and stomach, requires to capture local and global features. That is probably why hybrid models have more accurate result.

5.7.3 ACDC

We compare the performance of GCtx-UNet on the ACDC dataset with some state-of-the-art methods including CNN-based methods (R50-UNet and R50-Atten-UNet), Transformer-based methods (R50-ViT, Swin-UNet, MISSFormer), and hybrid methods (TransUNet, GPATUNet, and CS-UNet). The results are shown in Table 5.3. GCtx-UNet is better than all other methods in terms of average DSC, where GCtx-UNet pre-trained on MedNet has the best performance. In particular, GCtx-UNet is better in the Right Ventricle (RV) and Left Ventricle (LV). GCtx-UNet pre-trained on MedNet is also better than GCtx-UNet pre-

trained on ImageNet.

Table 5.3: Comparison of GCtx-UNet with the state-of-the-art methods on ACDC dataset in DSC. Blue denotes the best results and red denotes the second best. GCtx-UNet¹ is pre-trained on ImageNet while GCtx-UNet² is pre-trained on MedNet.

Algorithm	DSC(%) $\uparrow$	Right Ventricle	Myocardium	Left Ventricle
R50-UNet [69]	87.55	87.10	80.63	94.92
R50-Atten-UNet [69]	86.75	87.58	79.20	93.47
R50-ViT [69]	87.57	86.07	81.88	94.75
Swin-UNet [59]	90.00	88.55	85.62	95.83
MISSFormer [73]	90.86	89.55	88.04	94.99
TransUNet [69]	89.71	88.86	84.53	95.73
GPA-TUNet [185]	90.37	89.44	87.98	93.68
CS-UNet [178]	90.38	88.28	86.50	96.35
GCtx-UNet ¹	90.98	89.63	86.77	96.55
GCtx-UNet ²	91.23	89.88	87.25	96.57

Figure 5.6 includes 3 example images in the ACDC dataset for a qualitative comparison between GCtx-UNet and CS-UNet. GCtx-UNet pre-trained on MedNet was able to segment right ventricle and left ventricle more accurately than CS-UNet and GCtx-UNet pre-trained on ImageNet.

5.7.4 Polyp datasets

We evaluated the performance of GCtx-UNet on several polyp datasets by first training it on 2 seen datasets (CVC-ClinicDB and Kvasir) and then use the trained models on 3 unseen datasets (CVC-ColonDB, ETIS-LaribDB, and CVC-300) to evaluate the generalizability of the models. Table 5.4 compares the performance of GCtx-UNet with state-of-the-art CNN-based algorithms (UNet, UNet++, PraNet) and hybrid method (CS-UNet). While GCtx-UNet pre-trained on MedNet has the best DSC metric for the Kvasir dataset, it is behind CS-UNet and PraNet on the CVC-ClinicDB dataset though the difference is relatively small.

Generalizability As shown in Table 5.4 (column 3–5), GCtx-UNet is more generalizable to unseen datasets (CVC-ColonDB, ETIS-LaribDB, and CVC-300). Compared to other approaches, GCtx-UNet models have the best and second best DSC value in CVC-ColonDB

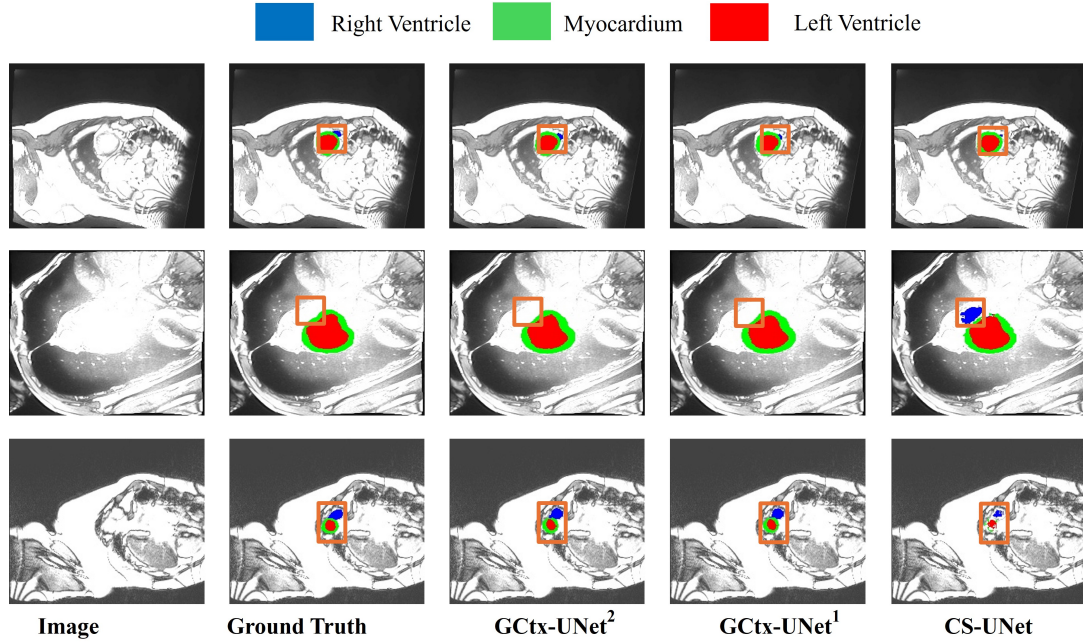


Figure 5.6: Comparison of GCtx-UNet with ground truth and CS-UNet on 3 example images in ACDC dataset. Note that GCtx-UNet¹ is pre-trained on ImageNet and GCtx-UNet² is pre-trained on MedNet. The orange rectangle box identifies the regions where GCtx-UNet¹ and CS-UNet have over or under segmentation problems compared to GCtx-UNet².

and ETIS-LaribDB datasets and the second best DSC value in CVC-300 dataset. The overall performance of GCtx-UNet is rather remarkable.

Table 5.4: Comparison of GCtx-UNet and State-Of-The-Art algorithms on Pylop datasets (the columns are average DSC in %). Blue indicates the best result and red displays the second-best. GCtx-UNet¹ is pre-trained on ImageNet while GCtx-UNet² is pre-trained on MedNet.

Methods	CVC-ClinicDB	Kvasir	CVC-300	CVC-ColonDB	ETIS-LaribDB
UNet[29]	82.3	81.8	71.0	51.2	39.8
UNet++ [60]	79.4	82.1	70.7	48.3	40.1
PraNet [184]	89.9	89.8	87.1	70.9	62.8
CS-UNet [178]	90.67	90.00	85.59	72.00	64.50
GCtx-UNet ¹	89.48	89.26	86.52	74.95	65.97
GCtx-UNet ²	89.44	90.02	86.20	78.08	72.03

Figure 5.7 presents the qualitative segmentation results of various methods, including GCtx-UNet that trained on MedNet and ImageNet. Five samples, one from each dataset, are selected to highlight ambiguous boundaries and small polyps, facilitating a differentiated comparison of segmentation performance. GCtx-UNet, pre-trained on MedNet, shows a

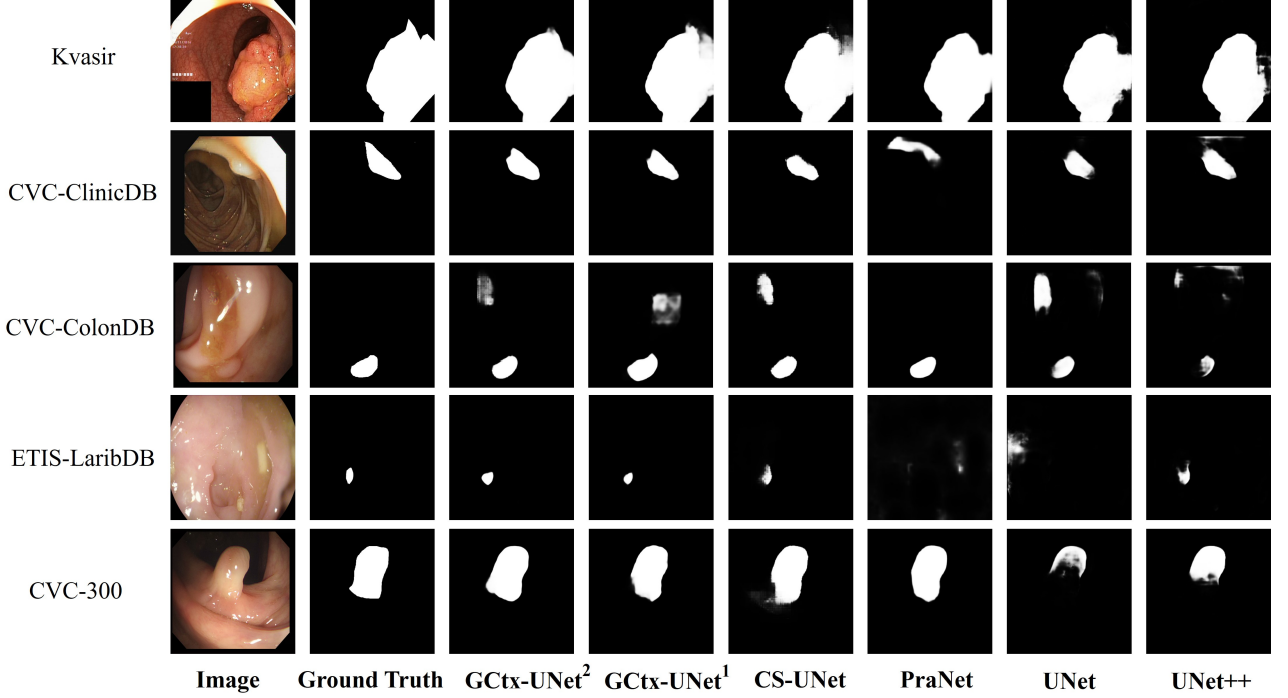


Figure 5.7: Comparison of GCtx-UNet with CS-UNet, PraNet, UNet, and UNet++ on 5 example images in the polyp datasets. GCtx-UNet¹ is pre-trained on ImageNet and GCtx-UNet² is pre-trained on MedNet.

significant reduction in false positives and false negatives. This improvement is attributed to its enhanced ability to distinguish between the obscure boundaries of polyp regions and normal regions.

5.8 Computation Complexity

GCtx-UNet not only has good segmentation performance but also lower computation complexity than the state-of-the-art segmentation methods. We compared the computation complexity of GCtx-UNet with that of Transformer-based and hybrid methods in terms of model parameter numbers, floating point operations (FLOPs) per epoch of training, inference time, and model sizes. This assessment is evaluated on the Synapse dataset. As shown in Table 5.5, GCtx-UNet has the least number of parameters at 12.34 million, while the model with nearest number of parameters, TransDeepLab, has 21.14 million parameters. Correspondingly, GCtx-UNet has the smallest model size at 49.75 MB while the second

smallest model, TransDeepLab, has 86.343 MB. GCtx-UNet also has the least number of FLOPs per epoch of training at 30.41G while the closest model Swim-UNet has 61.64G. Similarly, GCtx-UNet has the least training time per epoch and highest FPS for inference.

The low computation cost and small memory footprint of GCtx-UNet make it the best trade-off between performance and model complexity. It has better or comparable performance than most of the state-of-the-art models, which have substantially larger model size and higher computation cost. The efficient design of GCtx-UNet highlights its potential for achieving better segmentation results in clinical applications.

Table 5.5: Comparison of GCtx-UNet, Transformer-based, and hybrid algorithms based on model parameters, the floating point operations (FLOPs) per training epoch, number of epochs to train, training time per epoch (TTPE), and inference time in Frame Per Second (FPS) for 1568 axial abdominal clinical CT images, and model size. For FLOPs and TTPE, the batch size is 10. Training epochs is what is needed for the final result.

Algorithm	# of params (M)	FLOPs (G)	# of epochs	TTPE (m:s)	FPS	Model size (MB)
TransDeepLab [77]	21.14	160.00	200	1:36	20	86.343
Swin-UNet [59]	27.17	61.64	150	1:45	27	108.058
MISSFormer [73]	42.46	98.86	400	4:01	19	166.124
HiFormer [76]	25.51	80.45	400	1:27	19	101.161
CS-UNet [178]	44.96	110.00	150	2:45	26	177.613
TransUNet [69]	105.28	290.00	150	2:54	17	414.412
GCtx-UNet	12.34	30.41	150	1:16	30	49.75

5.9 Ablation Study

To investigate the impact of various factors on model performance, we conducted ablation studies using the Synapse dataset. Below, we discuss the effects of upsampling and the optimal hyperparameters for training our model.

5.9.1 Hyper-parameter Tuning

GCtx-UNet is trained with a combination of two loss functions, dice loss and cross-entropy loss, which aligns with many current segmentation methods. During the training process, we improve the performance by choosing an optimal combination of dice and cross-entropy losses

and an optimal learning rate. We conducted experiments to identify the optimal settings for the combined losses and the learning rate. Table 5.6 compares the performance of GCtx-UNet in terms of the Dice-Similarity Coefficient (DSC) and Hausdorff Distance (HD) values for various hyper-parameter values. The optimal DSC and HD are achieved when (dice loss, cross-entropy loss) = (0.3, 0.7) and learning rate is 0.0001. This configuration was used in all of our subsequent experiments.

Table 5.6: Ablation study on the impact of the training hyper-parameters to the performance of GCtx-UNet. The hyper parameters include the loss function (cross entropy loss and dice loss) and learning rate.

Optimizer	dice loss	cross entropy loss	learning rate	DSC (%)	HD (mm)
AdamW	0.6	0.4	0.00001	81.67	22.40
AdamW	0.6	0.4	0.0001	81.58	22.35
AdamW	0.7	0.3	0.00001	81.53	23.16
AdamW	0.7	0.3	0.0001	82.39	15.94

5.9.2 Upsample

Similar to Swin-UNet [59], to complement the downsampling layer in the encoder, we specifically designed an upsampling layer in the decoder to perform upsampling and feature dimension increase. To assess the effectiveness of this upsampling layer, we evaluated GCtx-UNet on the Synapse dataset to compare GCtx-UNet with bilinear interpolation (with or without SE (Squeeze and Excitation) block) to GCtx-UNet with transposed convolution (Fused-MBConv module with or without SE block) in the upsampling layer. The results in Table 5.7 indicate that GCtx-UNet with transposed convolution (Fused-MBConv module with SE block) in the upsampling layer has the best performance.

Table 5.7: Ablation study on the impact of the upsampling types to the performance of GCtx-UNet.

Upsampling Type	DSC (%)	HD (mm)
bilinear interpolation	80.31	22.64
bilinear interpolation + SE	80.89	25.76
transposed convolution (Fused-MBConv)	81.80	21.12
transposed convolution (Fused-MBConv + SE)	82.39	15.94

5.10 Conclusion

We introduced GCtx-UNet, a U-shaped network that incorporates a lightweight vision transformer to enhance medical image segmentation by effectively capturing both global and local features. The downsampling and upsampling blocks between encoder and decoder components provide inductive bias and model inter-channel dependencies effectively. GCtx-UNet has better or comparable performance than traditional CNN-based, Transformer-based, and hybrid methods on various medical image datasets. At the same time, GCtx-UNet has lower model complexity with less number of model parameters, lower model size, lower training and inference time, and lower FLOPs for training. The ability of GCtx-UNet to model long-range spatial dependencies and its competitive performance in segmenting complex and small anatomical structures make it a promising tool for clinical applications. The architecture’s design, which includes GC-ViT encoders and decoders with skip connections, contributes to its high performance while maintaining a lower computational complexity compared to state-of-the-art methods. The pre-training on a medical image dataset – MedNet and the subsequent evaluation on multiple medical imaging tasks shows the model’s robustness and generalization capabilities, positioning GCtx-UNet as a practical and powerful approach for medical image segmentation. As future work, we plan to introduce GCtx-UNet 3D model for voxel segmentation of medical images.

Chapter 6

Conclusion

In this thesis, we introduced two novel segmentation algorithms, GCtx-UNet and CS-UNet, designed to enhance the accuracy and efficiency of medical and materials science image analysis. GCtx-UNet, a UNet-like segmentation network, incorporates Global Context Vision Transformers (GC-ViT) both as encoders and decoders to effectively capture long and short-range semantic features. This architecture allows GCtx-UNet to excel in various medical image segmentation tasks, as demonstrated by its outstanding performance on the Synapse, ACDC, and polyp segmentation datasets. The results underscore GCtx-UNet’s ability to handle both global and local feature extraction, making it particularly adept at segmenting both large and small anatomical structures. Additionally, GCtx-UNet maintained a reasonable parameter count and computational time, indicating its practical applicability for clinical workflows.

CS-UNet, our other proposed algorithm, combines CNN and Transformer encoders in parallel to extract both low-level and high-level features, enabling effective segmentation of images with long-range dependencies and high spatial resolution. Our evaluations on materials images and medical CT images showed that CS-UNet offers better or comparable performance than prior state-of-the-art methods. Pre-training CS-UNet on large microscopy datasets led to improved feature representation and segmentation performance, demonstrating clear advantages over models pre-trained on natural images. This approach leverages the unique characteristics of microscopy images, which differ significantly from natural images in terms of signal-to-noise ratios and structural complexity.

Both GCtx-UNet and CS-UNet benefit significantly from being trained on large microscopy datasets. This training strategy enhances their ability to generalize across various tasks and improves their robustness in handling diverse and complex images. The use of

transfer learning from large microscopy datasets allows these models to capture high-level microscopy features that are not typically present in natural images, resulting in superior segmentation performance.

Moreover, we developed a combination of unsupervised and supervised machine learning methods for the segmentation of graphite nodules in the microstructure of ductile iron. K-means clustering, as the unsupervised method, was applied to generate the training dataset, which was then used for training a UNet-based segmentation network. Data augmentation was applied to reduce the likelihood of overfitting, and focal-loss was used to enhance prediction accuracy due to the unbalanced datasets. The automated ML method demonstrated the potential to replace human-operated procedures for quantifying the area fraction of graphite nodules in ductile iron microstructures. This approach can be easily translated to the microstructural analysis of other composite materials, leading to more accurate and robust PSP relationships essential for better material design.

In future work, we plan to enhance CS-UNet’s efficiency and performance by removing the window shifting operation from the two consecutive Swin Transformer blocks and replacing it with the GC module. The GC module can model cross-region interactions, eliminating the need for window shifting and reverse window shifting in these blocks. This change will reduce the processing time required by the Swin Transformer. Moreover, we plan to introduce the GCtx-UNet 3D model specifically designed for voxel segmentation in medical images. This extension aims to leverage the strengths of GC-UNet for more detailed and accurate 3D medical image analysis. Future investigations will also explore techniques such as Differentiable DNC, ClusterFormer, and CLUSTSEG to enhance the transparency and interpretability of image segmentation models, providing deeper insights into their decision-making processes. By pursuing these avenues, we aim to enhance the efficiency, capability, and understandability of segmentation models, addressing both current limitations and emerging needs in various image analysis domains.

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Appendix A

Appendix A: Average Performance of Segmentation Algorithms on Microscopy Images

In this section, we compare the effect of pre-training on the average performance of CNN-based, Transformer-based segmentation algorithms, and CS-UNet. After that, we compare the average performance of the three types of algorithms. Our results indicate that pre-training on microscopy images generally has positive impacts to the performance. CS-UNet outperforms UNet in all experiments while it has similar or better performance than Transformer-based algorithms.

sectionCNN-based Image Segmentation

Table A.1: The average performance of **UNet** when it is initialized with different pre-training weights. Each entry shows the mean and standard deviation of IoU value for a pre-training model. The best score per test is in bold.

Test data	ImageNet	MicroNet	ImageNet→MicroNet
Super-1	96.09%±0.20%	95.99%±0.16%	95.92%±0.19%
Super-2	95.08%±1.27%	95.28%±0.26%	95.38±1.03%
Super-3	63.30%±4.55%	74.61%±17.21%	78.69±11.24%
Super-4	71.46%±6.88%	75.1%±9.0%	77.78%±0.17
EBC-1	95.18%±0.82%	94.65%±1.44%	95.69%±1.03%
EBC-2	80.74%±10.76%	87.06%±1.55%	86.0%±5.17%
EBC-3	39.36%±7.81%	41.95%±4.89%	46.44%±4.35%

We examine the performance of UNet [29] with 3 types of pre-trained encoders. We include this result since the configurations of CS-UNet only used 19 of the 35 CNN encoders in Stuckner *et al.* [10]. As shown in Appendix B, the 19 encoders have top-5 accuracy in at least one of the segmentation tasks. This selection reduces the number of experiments

needed for a fair comparison,

The average performance of UNet when it is pre-trained with ImageNet or MicroNet is in Table A.1, which indicates that ImageNet→MicroNet model (i.e. CNN encoders initialized with ImageNet model and fine-tuned with MicroNet) achieves the best outcome in majority of the cases. The configurations of the top-performing CNN encoders are shown in Table C.1, which also indicates that pre-training on MicroNet provides better outcome in most of the cases. EfficientNet and Se_ResNet families are better than the older families such as VGG.

Table A.2: The best performing CNN encoders of UNet that are pre-trained on ImageNet and MicroNet

Dataset	ImageNet		MicroNet	
	best performing CNN encoder	IoU	best performing CNN encoder	IoU
Super-1	SE ResNeXt-50 32x4	96.2%	SE_ResNeXt-101_32x4d	95.95%
Super-2	EfficientNet-b5	95.45%	SENet-154	95.50%
Super-3	EfficientNet-b3	70.74%	EfficientNet-b3	92.5%
Super-4	EfficientNet-b2	77.34%	EfficientNet-b1	78.95%
EBC-1	SENet-154	96.23%	SENet-154	96.67%
EBC-2	DenseNet201	91.36%	InceptionResnetV2	90.97%
EBC-3	EfficientNet-b3	44.4%	EfficientNet-b3	52.84%

Unsurprisingly, our results are largely consistent with that of Stuckner *et al.* [10], since we initialized our networks with the weights pre-trained on MicroNet. Specifically, pre-training with MicroNet improves the performance of one-shot and out-of-distribution learning. Since we picked CNN encoders that have top-5 performance in at least one experiment, the IoU scores are higher than the average scores shown in Stuckner *et al.*. In fact, the performances on Super-2 (few-shot learning) are basically the same with different pre-training models.

A.1 Transformer-based Image Segmentation

Table A.3 shows the average performance of Transformer-based segmentation algorithms (Swin-Unet, HiFormer, and TransDeepLabv3+) using different configurations of pre-training and Swin-T architectures. We compared the algorithms by using the original or the intermediate Swin-T architecture and by initializing their weights with ImageNet or microscopy

Table A.3: Average performance of **Transformer**-based segmentation algorithms when initialized with different pre-trained weights. Each entry shows the mean and standard deviation of IoU value for a pre-training model. The highest score for each dataset is in bold.

<i>Test set</i>	Original Swin-T			Intermediate Swin-T	
	<i>ImageNet</i>	<i>MicroLite</i>	<i>ImageNet→MicroLite</i>	<i>MicroLite</i>	<i>ImageNet→MicroLite</i>
Super-1	94.94%±0.31%	95.0%±0.43%	94.89%±0.45%	94.72%±0.97%	94.41%±0.51%
Super-2	93.26%±0.76%	93.83%±0.62%	93.55%±0.30%	94.03%±0.83%	93.43%±0.97%
Super-3	76.24%±7.06%	79.53%±13.69%	70.65%±10.74%	78.74%±17.08%	69.31%±6.44%
Super-4	73.04%±1.64%	73.18%±1.81%	72.89%±2.93%	72.10%±2.39%	71.23%±2.32%
ECB-1	91.73%±3.40%	94.67%±1.59%	91.44%±2.87%	94.95%±1.41%	90.56%±3.01%
ECB-2	82.47%±5.90%	86.02%±4.98%	83.20%±3.34%	86.67%±2.61%	83.07%±4.22%
ECB-3	52.21%±6.76%	65.90%±13.12%	55.15%±10.42%	56.91%±5.24%	49.03%±4.67%
mean IoU	80.56	84.02	80.25	82.59	78.72

pre-training models. Our results indicate that the algorithms perform well with the Micro-Lite pre-training model and that the original Swin-T architecture is slightly better with 1-shot learning and out-of-distribution learning. Overall, pre-training with microscopy images provided better results for Transformer-based segmentation algorithms than pre-training on natural images.

A.2 CS-UNet Image Segmentation

Table A.4: Average performance of **CS-UNet** when initialized with different pre-trained weights for each experiment. Each entry in the table shows the mean value and standard deviation of the evaluation IoU metric for a particular pre-training model. The highest accuracy score for each dataset is shown in bold.

<i>Test set</i>	Original Swin-T			Intermediate Swin-T	
	<i>ImageNet</i>	<i>Microscopy</i>	<i>ImageNet→Microscopy</i>	<i>Microscopy</i>	<i>ImageNet→Microscopy</i>
Super-1	96.11%±0.11%	96.19%±0.13%	96.16%±0.14%	96.22%±0.15%	96.02%±0.34%
Super-2	95.63%±0.23%	95.63%±0.44%	95.78%±0.14%	95.67%±0.46%	95.86%±0.11%
Super-3	78.64%±9.3%	76.59%±15.78%	78.01%±13.32%	78.18%±12.13%	80.68%±12.60%
Super-4	73.74%±3.74%	77.25%±3.31%	72.87%±6.26%	76.65%±2.53%	75.07%±1.10%
ECB-1	97.09%±0.86%	95.48%±1.04%	96.4%±0.79%	94.72%±1.37%	96.21%±0.78%
ECB-2	83.57%±7.71%	86.12%±1.76%	88.58%±1.46%	86.41%±1.65%	88.08%±2.98%
ECB-3	45.88%±10.12%	44.7%±8.57%	46.08%±13.92%	46.35%±9.91%	45.16%±10.58%
mean IoU	81.5	81.71	81.98	82.03	82.44

We also compare the performance of our hybrid segmentation algorithm CS-UNet in Table A.4 when it uses the original or the intermediate Swin-T architecture and when it is initialized with weights from ImageNet or microscopy models. Since CS-UNet uses both CNN and Transformer encoders, the results are mixed where pre-training with microscopy images

does not provide better performance in all cases. The weaker performance of CNN encoders reduced the advantage of Transformer encoders when they are pre-trained on microscopy images. When we consider the mean IoU scores across all experiments, however, pre-training with microscopy images still has the better outcome.

A.3 Comparison of CS-UNet, CNN-based, and Transformer-based Algorithms

Table A.5: The average performance of CNN, Transformer, and CS-UNet on all datasets. Each entry in the table shows the mean value and standard deviation of the IoU evaluation metric for a particular method.

<i>Test set</i>	<i>UNet [29]</i>	<i>CS-UNet</i>	<i>Swin-Unet [59]</i>	<i>TransDeepLabV3+ [77]</i>	<i>HiFormer [76]</i>
Super-1	95.98% $\pm$ 0.20%	96.14% $\pm$ 0.21%	95.30% $\pm$ 0.34 %	94.91% $\pm$ 0.47%	94.30% $\pm$ 0.58%
Super-2	95.24% $\pm$ 0.96%	95.70% $\pm$ 0.33%	93.39% $\pm$ 1.1%	94.1% $\pm$ 0.61%	93.11% $\pm$ 0.5%
Super-3	72.20% $\pm$ 13.79%	77.99% $\pm$ 12.90%	83.0% $\pm$ 6.14%	81.99% $\pm$ 8.29%	64.19% $\pm$ 8.14%
Super-4	74.57% $\pm$ 7.23%	75.08% $\pm$ 4.15%	77.16% $\pm$ 0.97%	70.95% $\pm$ 1.86%	72.61% $\pm$ 1.67%
EBC-1	95.17% $\pm$ 1.21%	95.98% $\pm$ 1.28	93.49 $\pm$ 2.66%	91.59% $\pm$ 4.13%	93.96% $\pm$ 1.1%
EBC-2	84.60% $\pm$ 7.48%	86.73% $\pm$ 4.03%	83.13% $\pm$ 3.1%	83.1% $\pm$ 5.68%	86.67% $\pm$ 1.44%
EBC-3	42.58% $\pm$ 6.57%	45.69% $\pm$ 10.78%	58.62% $\pm$ 20.37%	58.7% $\pm$ 9.70%	52.84% $\pm$ 5.41%

In Table A.5, we compare the performance of UNet, CS-UNet, and Transformer-based algorithms (Swin-Unet, HiFormer, and TransDeepLabv3+) averaged over different pre-training models and Swin-T architectures. The results show that CS-UNet is better than UNet on average across all experiments. While Transformer-based segmentation algorithms may be superior in 1-shot learning or out-of-distribution learning, their performance is not always consistently better than UNet. This result indicates that our hybrid algorithm CS-UNet is more robust regardless of the pre-training models.

Appendix B

Appendix B: Selected CNN Encoders

Table B.1 shows the 19 CNN encoders that were chosen to evaluate the performance of CS-UNet and UNet.

Table B.1: Each encoder has at least 1 top-5 average IoU score for Super/EBC datasets (based on Figure 4 of [10]).

CNN Encoder	Super-1	Super-2	Super-3	Super-4	EBC-1	EBC-2	EBC-3
SE_ResNet-50 [192]							✓
SE_ResNeXt-50_32x4d [193]	✓	✓			✓	✓	✓
SE_ResNeXt-101_32x4d [193]	✓	✓					✓
SE_ResNet-152 [192]	✓				✓		
SE_ResNet-101 [192]					✓	✓	✓
SENet-154 [192]		✓	✓		✓		
ResNeXt-101_32x8d [193]	✓						
Inception-V4 [194]		✓	✓		✓		
Inception-ResNet-V2 [194]			✓				
DenseNet201 [53]						✓	
DenseNet161 [53]						✓	
VGG-16_bn [25]	✓		✓			✓	
VGG-13_bn [25]			✓				
MobileNet-V2 [45]							✓
EfficientNet-b1 [195]				✓			
EfficientNet-b2 [195]				✓			
EfficientNet-b3 [195]				✓			
EfficientNet-b4 [195]				✓			
EfficientNet-b5 [195]		✓		✓			

Appendix C

Appendix C: UNet Performance

Table C.1 shows the CNN encoders that produced the best performance of UNet for each dataset when pre-trained on ImageNet or MicroNet. EfficientNet and Se_ResNet families have better performance than the older encoder families such as VGG.

Table C.1: The best performing CNN encoders of UNet that are pre-trained on ImageNet and MicroNet.

Dataset	ImageNet		MicroNet	
	best performing CNN encoder	IoU	best performing CNN encoder	IoU
Super-1	SE_ResNeXt-50_32x4	96.2%	SE_ResNeXt-101_32x4d	95.95%
Super-2	EfficientNet-b5	95.45%	SENet-154	95.50%
Super-3	EfficientNet-b3	70.74%	EfficientNet-b3	92.5%
Super-4	EfficientNet-b2	77.34%	EfficientNet-b1	78.95%
EBC-1	SENet-154	96.23%	SENet-154	96.67%
EBC-2	DenseNet201	91.36%	InceptionResnetV2	90.97%
EBC-3	EfficientNet-b3	44.4%	EfficientNet-b3	52.84%

Appendix D

Appendix D: CS-UNet Complexity

CS-UNet is a flexible method that can use different CNN families in the CNN encoder branch. As a result, the total number of parameters of CS-UNet can vary depending on the specific CNN encoder branch chosen. Table D.1 summarizes the parameters for different CNN branches used in CS-UNet for this study.

Table D.1: The number of parameters of CS-UNet based on different CNN families in CNN encoder branch.

CNN encoder branch	# of Params (M) of CS-UNet
DenseNet201 [53]	50.98
DenseNet161 [53]	59.99
EfficientNet-b0 [195]	33.68
EfficientNet-b1 [195]	36.18
EfficientNet-b2 [195]	37.43
EfficientNet-b3 [195]	40.48
EfficientNet-b4 [195]	47.44
EfficientNet-b5 [195]	58.34
Inception-V4 [194]	73.11
Inception-ResNet-V2 [194]	86.29
SENet-154 [192]	145.83
SE_ResNet-50 [192]	58.83
SE_ResNet-152 [192]	97.56
SE_ResNeXt-50_32x4d [193]	58.3
SE_ResNeXt-101_32x4d [193]	79.7
SE_ResNet-101 [192]	80.07
ResNeXt-101_32x8d [193]	119.53
MobileNet-V2 [45]	33.36
VGG-13_bn [25]	39.64
VGG-16_bn [25]	44.96

Appendix E

Appendix E: Segmentation Examples of Super and EBC Datasets

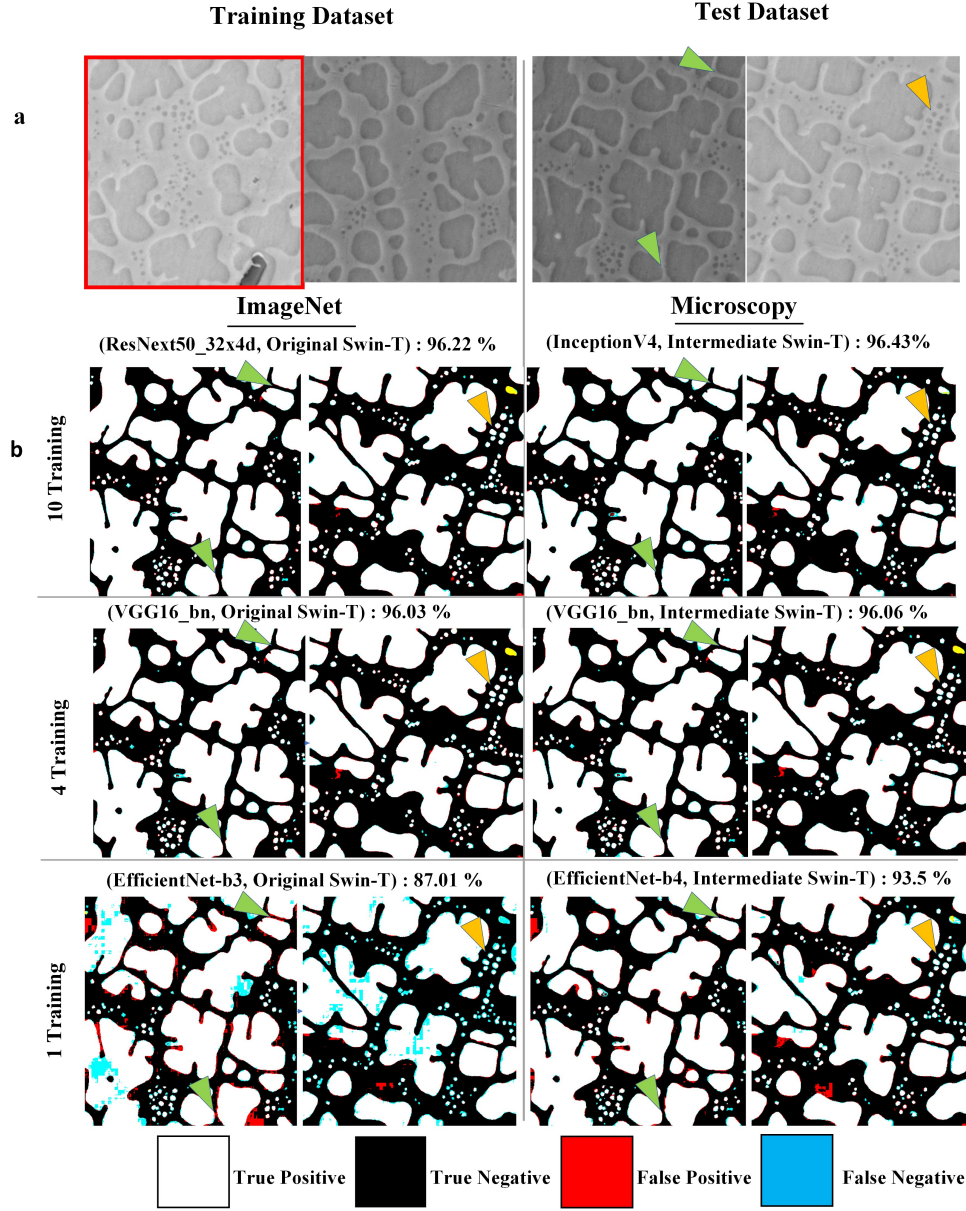


Figure E.1: (a) examples of the training and test images of the Super datasets [10], where the training image of Super-3 is outlined in red. (b) the best segmentation masks of the test images when CS-UNet is trained with Super-1, Super-2, and Super-3 dataset, where the left is pre-trained with ImageNet and the right is pre-trained with microscopy images. The CNN/Transformer encoders and the IoU score of the best model are above the segmentation mask of each experiment. The green triangle points to the area where the ImageNet (but not the microscopy) model incorrectly segmented the secondary precipitates. The yellow triangle points to the area where the ImageNet (but not the microscopy) model incorrectly segmented the tertiary precipitates.

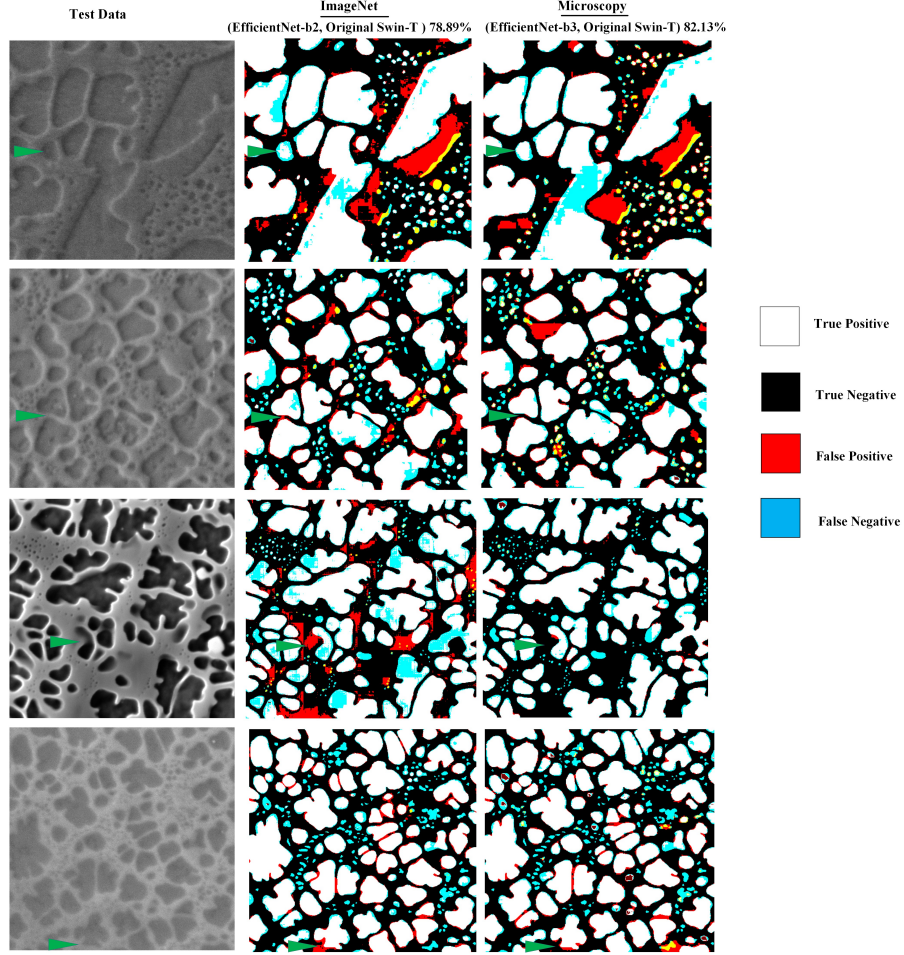


Figure E.2: The test images of Super-4 have different imaging conditions than those of the training images. The left column shows the test images of Super-4 [10]. The accuracy mask colors are the same as those in Figure E.1. The middle column shows the IoU accuracy masks for the best ImageNet model. The right column shows the same for the best microscopy model. Each row shows the test image and accuracy masks of the same image. The green arrow shows an example where the model was over-segmented and where the microscopy model accurately segmented the secondary precipitate. The yellow color indicates incorrect identification of a tertiary precipitate as a secondary precipitate. The maroon color indicates where the model improperly identified a secondary precipitate as a tertiary precipitate. The cyan color indicates where the model over-segmented the secondary precipitates.

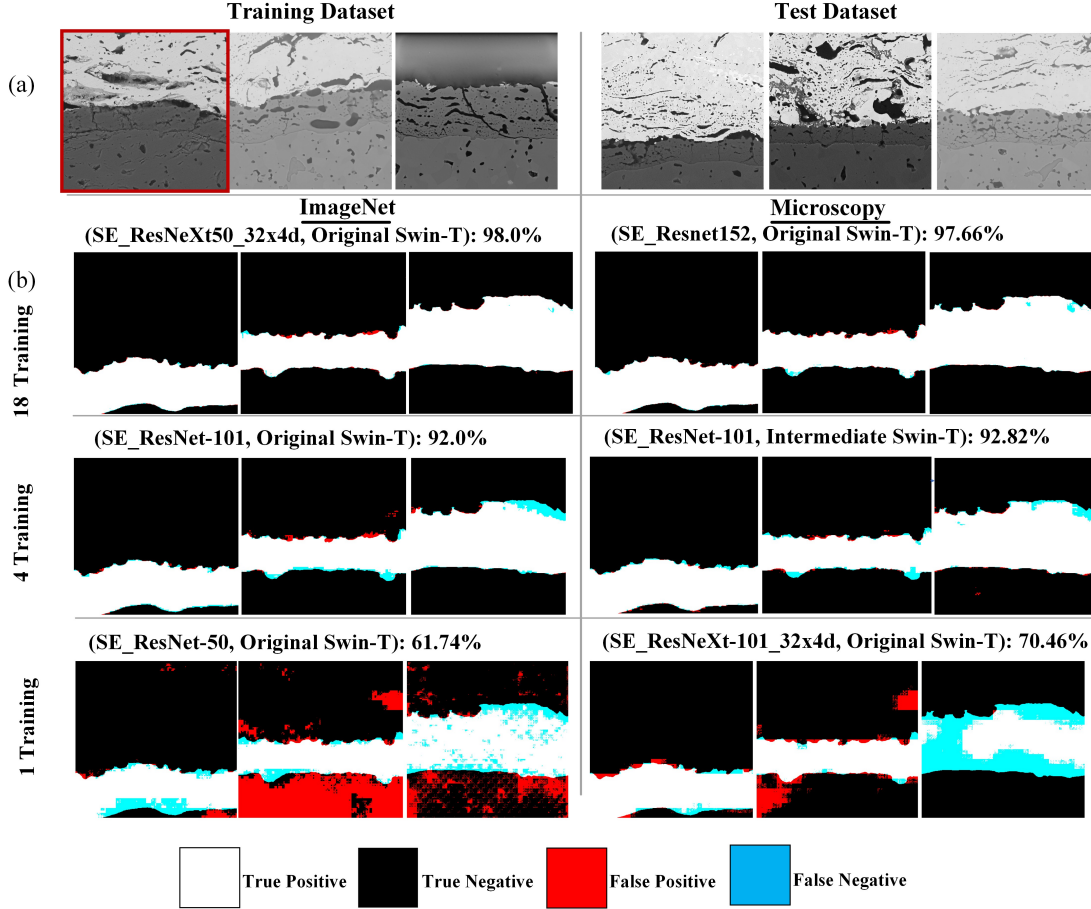


Figure E.3: (a) shows examples of the train and test images in the EBC datasets, where the left column represents the training set and the right column represents the test set. The single training image for EBC-3 is outlined in red. (b) shows the best segmentation masks of CS-UNet for each EBC dataset and for each pre-training dataset. The CNN/Transformer encoders and the IoU score is above the segmentation mask of each test. For EBC-1/2 (the second/third rows), the segmentation masks of CS-UNet with either pre-training datasets are able to distinguish between the substrate and the thermally grown oxide layer. For EBC-3 (the last row), CS-UNet pre-trained with ImageNet is not able to distinguish between the substrate and the thermally grown oxide layer, which made it difficult to accurately measure oxide thickness after simple morphological operations.

Appendix F

Appendix F: MircoLie Dataset

The MicroLite dataset was collected from multiple sources including images from different materials and compounds using several measurement techniques such as light microscopy, SEM, TEM, and X-ray. MicroLite consolidates the Aversa dataset [158], UltraHigh Carbon Steel Micrograph [93], SEM images from the Materials Data Repository, and the images from some recent publications [159, 160, 161, 121, 162, 163, 164, 165].

MicroLite consists of over 50,000 images with 74 classes. Table F.1 represents various classes of nanostructures categorized for the MiroLite dataset, captured using scanning electron microscopy (SEM).

Table F.1: Classes used in the MiroLite dataset. Each category and subcategory represents distinct types of nanostructures captured by scanning electron microscopy (SEM).

Category	Level 1 Subcategory
Tips	Zoom out
Tips	Zoom in
Tips	Tip on cantilever
Nanowires	Entangled nanowires
Nanowires	Few
Nanowires	Individual
Nanowires	Forest
Nanowires	Forest
Fibres	
Biological	
MEMS devices electrodes	Electrode
MEMS devices electrodes	Close up line
MEMS devices electrodes	3d edges
MEMS devices electrodes	Waveguide
MEMS devices electrodes	Microfluidic
Patterned surface	Line array
Patterned surface	Square array
Patterned surface	Circle array
Patterned surface	3d edge patterned surface
Patterned surface	3d line
Patterned surface	Triangle array
Patterned surface	Ring spiral
Patterned surface	Zigzag
Patterned surface	Zoom in pillar
Patterned surface	Individual pillar
Patterned surface	Triangular pillar
Particles	Dispersed
Particles	Small clusters
Particles	Individual particle
Particles	Other shape
Powder	Zoom out
Powder	Zoom in
Films and coated surfaces	Smooth film
Films and coated surfaces	Particle film
Films and coated surfaces	Other film
Porous Sponge	